MATHEMATICS

MATRICES

CONTENTS

MATRICES

JEE EXERCISES

EXERCISE–J-I ..... Page –02

EXERCISE–J-II ..... Page –21

JEE MAIN..... Page –46

JEE ADV. ..... Page –81

CHALLENGER EXERCISES

EXERCISE–C-I ..... Page –111

EXERCISE–C-II ..... Page –123

ALGEBRAIC OPERATIONS ON MATRICES

1. Let $A+2B=\begin{bmatrix}1 & 2 & 0\\ 6 & -3 & 3\\ -5 & 3 & 1\end{bmatrix}$ and $2A-B=\begin{bmatrix}2 & -1 & 5\\ 2 & -1 & 6\\ 0 & 1 & 2\end{bmatrix}$ then $\mathrm{Tr}(A)-\mathrm{Tr}(B)$ has the value equal to :

माना $A+2B=\begin{bmatrix}1 & 2 & 0\\ 6 & -3 & 3\\ -5 & 3 & 1\end{bmatrix}$ तथा $2A-B=\begin{bmatrix}2 & -1 & 5\\ 2 & -1 & 6\\ 0 & 1 & 2\end{bmatrix}$ तो $\mathrm{Tr}(A)-\mathrm{Tr}(B)$ का मान किसके बराबर होगा :

(A) 0 (B) 1 (C) 2 (D) कोई नहीं

Ans. C

Sol. $t_r(A)+2t_r(B)=-1$ (from the given matrix) and $2t_r(A)-t_r(B)=3$ (from the given matrix)

Let $t_r(A)=x$ and $t_r(B)=y$

$x+2y=-1$

$2x-y=3$

solving $x=1$ and $y=-1$

Hence $t_r(A)-t_r(B)=x-y=2$

2. If A and B are two matrices such that , $AB=B$ and $BA=A$, then $A^2+B^2=$

यदि A एवं B दो मैट्रिक्स इस प्रकार है, की $AB=B$ एवं $BA=A$ तो $A^2+B^2=$

(A) 2 AB (B) 2 BA (C) A + B (D) AB

Ans. C

Sol. $A^2=ABA=BA=A$

$B^2=BAB=AB=B$

$\Rightarrow A^2+B^2=A+B$

3. Let $f(x)=x^2-5x+6$ and $A=\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}$, then f(A) is equal to :

यदि $f(x)=x^2-5x+6$ एवं $A=\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}$ तो f(A) किस के तुल्य है :

(A) $\begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 3 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & -1 & -3 \\ 0 & -1 & -10 \\ -5 & 4 & 3 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & -1 & -3 \\ 0 & -1 & -10 \\ -5 & 4 & 0 \end{bmatrix}$

Ans. A

Sol. $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

$f(A) = A^2 - 5A + 6$

$A^2 = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$

$f(A) = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

$= \begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$

4. If $A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, then $A^2 = B^2 = C^2 =$

यदि $A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ एवं $C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, तो $A^2 = B^2 = C^2 =$

(A) I^2 (B) I (C) – I (D) 2 I

Ans. C

Sol. $A^2 = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$

5. $\text{If } A=\begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 2\\ 2 & 3 & 1\end{bmatrix}, B=\begin{bmatrix}-5 & 7 & 1\\ 1 & -5 & 7\\ 7 & 1 & -5\end{bmatrix} \text{ then } AB=$

$\text{यदि } A=\begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 2\\ 2 & 3 & 1\end{bmatrix}, B=\begin{bmatrix}-5 & 7 & 1\\ 1 & -5 & 7\\ 7 & 1 & -5\end{bmatrix} \text{ तो } AB=$

(A) I_3 (B) $2\ I_3$ (C) $4\ I_3$ (D) $18\ I_3$

Ans. D

Sol. $AB=\begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 2\\ 2 & 3 & 1\end{bmatrix}\begin{bmatrix}-5 & 7 & 1\\ 1 & -5 & 7\\ 7 & 1 & -5\end{bmatrix}$

$=\begin{bmatrix}18 & 0 & 0\\ 0 & 18 & 0\\ 0 & 0 & 18\end{bmatrix}=18\ I_3$

6. $\text{If } A=\begin{bmatrix}a & b\\ b & a\end{bmatrix} \text{ and } A^2=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}, \text{ then :}$

$\text{यदि } A=\begin{bmatrix}a & b\\ b & a\end{bmatrix} \text{ एवं } A^2=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix} \text{ तो :}$

(A) $\alpha = 2ab, \beta = a^2 + b^2$ (B) $\alpha = a^2 + b^2, \beta = ab$

(C) $\alpha = a^2 + b^2, \beta = 2ab$ (D) $\alpha = a^2 + b^2, \beta = a^2 - b^2$

Ans. C

Sol. $A^2=\begin{bmatrix}a & b\\ b & a\end{bmatrix}=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}$

$=\begin{bmatrix}a^2+b^2 & 2ab\\ 2ab & a^2+b^2\end{bmatrix}=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}$

$\Rightarrow \alpha = a^2 + b^2, \beta = 2ab$

7. If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true :

(A) $AB = BA$ (B) Either of A or B is a zero matrix

(C) Either of A or B is an identity matrix (D) $A = B$

यदि A तथा B, $n \times n$ कोटि के ऐसे वर्ग आव्यूह है कि $A^2 - B^2 = (A - B)(A + B)$, तो निम्न में से कौनसा सदैव सत्य है :

(A) $AB = BA$ (B) । अथवा B में से कोई एक शून्य आव्यूह है

(C) । अथवा B में से कोई एक तत्समक आव्यूह है (D) $A = B$

Ans. A

Sol. $A^2 - B^2 = (A - B)(A + B)$

$(A - B)(A + B) = A^2 - B^2 - BA + AB = A^2 + B^2$

$\Rightarrow AB - BA = 0$

$\Rightarrow AB = BA$

8. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, $a, b \in N$. Then :

(A) there exist more than one but finite number of B's such that $AB = BA$

(B) there exist exactly one B such that $AB = BA$

(C) there exist infinitely many B's such that $AB = BA$

(D) there cannot exist any B such that $AB = BA$

माना $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ तथा $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, $a, b \in N$ है, तो :

(A) एक से अधिक परन्तु सीमित संख्या में B इस प्रकार होंगे कि $AB = BA$

(B) केवल एक ही B इस प्रकार होगा कि $AB = BA$

(C) अनन्त B इस प्रकार होगें कि $AB = BA$

(D) कोई भी B ऐसा नही होगा कि $AB = BA$

Ans. C

Sol. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

$B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ $a, b \in N$

$AB = \begin{pmatrix} a & 2b \\ 3a & 4b \end{pmatrix}$

$$BA = \begin{pmatrix} a & 2b \\ 3b & 4b \end{pmatrix}$$

For AB = BA

$b = a \rightarrow$ their are infinite

Natural number for which a = 6

so, Infinite matrix B possible

9. Let $A = \begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$ If $|A^2| = 25$, then $|\alpha|$ equals :

माना $A = \begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$ यदि $|A^2| = 25$, तब $|\alpha|$ का मान है :

(A) 5^2 (B) 1 (C) $\frac{1}{5}$ (D) 5

Ans. C

Sol. $|A^2| = 25$

$|A|^2 = 25$

$|A| = \pm 5$

$$\begin{vmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{vmatrix} = \pm 5$$

$25\alpha = \pm 5$

$\alpha = \pm \frac{1}{5}$

10. If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, then $\alpha =$

यदि $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ तथा $|A^3| = 125$, तो $\alpha =$

(A) ± 3 (B) ± 2 (C) ± 5 (D) 0

Ans. A

Sol. $A=\begin{bmatrix}\alpha & 2\\ 2 & \alpha\end{bmatrix}$

$|A| = \alpha^2 - 4$

$(\alpha^2 - 4)^3 = 125$

$\alpha^2 - 4 = 5 \Rightarrow \pm 3$

11. A and B are two given matrices such that the order of A is 3×4 , if A' B and BA' are both defined then :

(A) order of B' is 3×4 (B) order of B'A is 4×4

(C) order of B'A is 3×3 (D) B'A is undefined

A तथा B दो आव्यूह दिये गये हैं आव्यूह A की कोटि 3×4 , है, यदि A′ B तथा BA′ दोनों ही परिभाषित है तब :

(A) ठ′ की कोटि 3×4 है (B) B′A की कोटि 4×4

(C) ठ′A की कोटि 3×3 है (D) ठ′A अपरिभाषित है

Ans. B

Sol. $A = 3 \times 4$; $A' = 4\times3$

As A′ B is defined $\Rightarrow$ let order of $B = 3 \times n$

now $BA' = (3 \times n) \times (4 \times 3) \Rightarrow n = 4$

$\therefore$ order of B is 3×4

$\therefore$ order of $B' = 4 \times 3$

order of $B'A = (4\times3) \times (3 \times 4) = 4 \times 4$ Ans

ORTHOGONAL MATRIX

12. If $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$, A^{-1} is given by :

यदि $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$ तो A^{-1} का मान होगा :

(A) –A (B) A^T (C) $-A^T$ (D) A

Ans. B

Sol. For Adj A interchange the diagonal elements and change the sign of off diagonal elements.

We have $A^{-1}=\frac{1}{\cos^2\theta+\sin^2\theta}\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}=A^T \Rightarrow$ A is orthogonal matrix

13. For a given matrix $A=\begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix}$ which of the following statement holds good?

(A) $A = A^{-1} \ \forall \ \theta \in R$ (B) A is symmetric, for $\theta = (2n+1)\frac{\pi}{2}$, $n \in I$

(C) A is an orthogonal matrix for $\theta \in R$ (D) A is a skew symmetric, for $\theta = n\pi$; $n \in I$

आव्यूह $A=\begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix}$ के लिए निम्न में से कौन सा कथन सत्य है :

(A) $A = A^{-1} \ \forall \ \theta \in R$ (B) A सममित है $\theta = (2n+1)\frac{\pi}{2}$ के लिए $n \in I$

(C) । एक लाम्बिक आव्यूह है $\theta \in R$ के लिए (D) A विषम सममित है $\theta = n\pi$ के लिए $n \in I$

Ans. C

Sol. Obv. A is orthogonal as $a_{11}^2 + a_{12}^2 = 1 = a_{21}^2 + a_{22}^2$

for skew symmetric matrix $a_{ii} = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$

for symmetric matrix, $A = A^T \Rightarrow \sin\theta = 0 \Rightarrow \theta = n\pi$

Also $\text{adj}A=\begin{pmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{pmatrix}$ and $|A| = 1$ hence $A = A^{-1}$ is possible if $\sin\theta = 0$

14. If A is an orthogonal matrix, then A^{-1} equals :

(A) A (B) A′ (C) A^2 (D) None of these

यदि A एक लाम्बिक मैट्रिक्स है, तो A^{-1} किस के तुल्य होगा :

(A) A (B) A′ (C) A^2 (D) इनमें से कोई नहीं

Ans. B

Sol. A is an orthogonal matrix

i.e $AA^T = I$

$\Rightarrow A^{-1}AA^T = A^{-1}I$

$\therefore A^{-1} = A^T$

15. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to : **[JEE(Main)-2015]**

यदि $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ एक ऐसा आव्यूह है जो आव्यूह समीकरण $AA^T = 9I$, को संतुष्ट करता है, जहाँ I, 3×3 का तत्समक आव्यूह

g§r ksfr ; qXr (a, b) का मान है : **[JEE(Main)-2015]**

(A) (2, –1) (B) (–2, 1) (C) (2, 1) (D) (–2, –1)

Ans. D

Sol. All rows of A represents mutually perpendicular vectors of magnitude 3

a must be 2 or – 2

b must be 1 or – 1

$4 - 2 + 2b = 0$ $2 + 2 + 2a = 0$

$b = -1$ $a = -2$

ADJOINT OF A SQUARE MATRIX

16. If A is a square matrix of order n then, adj (adj A) is equal to :

यदि A एक वर्ग आव्यूह है, जिसकी कोटि n है तो adj (adj A) बराबर होगा–

(A) $|A|^n A$ (B) $|A|^{n-1} A$ (C) $|A|^{n-2} A$ (D) $|A|^{n-3} A$

Ans. C

Sol. adj (adj A) = $|A|^{n-2} A$

17. If $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ and $A \text{ adj } A = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ the value of 'k' is :

यदि $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ एवं $A \text{ adj } A = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ है। तो 'k' का मान होगा :

(A) $\sin x \cos x$ (B) 1 (C) 2 (D) 3

Ans. B

Sol. We know that,

$A(\text{adj } A) = |A|I$

Given, $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$

$|A| = \cos^2 x + \sin^2 x = 1$

$\therefore A(\text{adj } A) = 1 . I = kI$

$k = 1.$

18. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $5a + b$ is equal to : **[JEE(Main)-2016]**

यदि $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ तथा $A \text{ adj } A = AA^T$ हैं, तो $5a + b$ बराबर है : **[JEE(Main)-2016]**

(A) 4 (B) 13 (C) –1 (D) 5

Ans. D

Sol. $A \text{ adj } A = AA^T$

$\text{adj } A = A^T$

$\begin{bmatrix} 2 & b \\ -3 & 5a \end{bmatrix} = \begin{bmatrix} 5a & 3 \\ -b & 2 \end{bmatrix}$

$2 = 5a, b = 3$

19. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then adj $(3A^2 + 12A)$ is equal to : **[JEE(Main)-2017]**

यदि $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$ है, तो adj $(3A^2 + 12A)$ बराबर है : **[JEE(Main)-2017]**

(A) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$ (B) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$ (C) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (D) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

Ans. D

Sol. $(3A^2 + 12A) = 3A (A + 4I)$

$$= \begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$$

$$\text{adj}\,(3A^2 + 12A) = \begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$$

INVERSE OF A SQUARE MATRIX

20. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$. If $A^{-1} = xA + yI_2$, (I_2 is Identity matrix of 2 x 2) then the value of x and y can be :

(A) $x = -\frac{1}{13}, y = \frac{7}{13}$ (B) $x = \frac{1}{13}, y = \frac{7}{13}$ (C) $x = \frac{1}{13}, y = -\frac{7}{13}$ (D) None of these

यदि $A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$ एवं $A^{-1} = xA + yI_2$, (जहाँ I_2 ,2 x 2 ईकाई मैट्रिक्स है) हो तो x एवं y का मान क्या होगा :

(A) $x = -\frac{1}{13}, y = \frac{7}{13}$ (B) $x = \frac{1}{13}, y = \frac{7}{13}$ (C) $x = \frac{1}{13}, y = -\frac{7}{13}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $|A - \lambda I| = \begin{vmatrix} 2-\lambda & 3 \\ -1 & 5-\lambda \end{vmatrix}$

$(2 - \lambda)(5 - \lambda) + 3 = \lambda^2 - 7\lambda + 13 = 0$

$\Rightarrow A^2 - 7A + 13I = 0$

$$I = \frac{-1}{13}A^2 + \frac{7}{13}A$$

$$\Rightarrow A^2 = \frac{-1}{13}A + \frac{7}{13}I$$

21. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}\left(A^2 + cA + dI\right)$, then the value of c and d are :

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ तथा $A^{-1} = \frac{1}{6}\left(A^2 + cA + dI\right)$ है, तो c तथा d का मान है :

(A) –6, –11 (B) 6, 11 (C) –6, 11 (D) 6, – 11

Ans. C

Sol. $|A-\lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 4 \\ 0 & -2 & 4-\lambda \end{vmatrix}$

$= (1-\lambda)((1-\lambda)((4-\lambda)+2)$

$= (1-\lambda)(\lambda^2 - 5\lambda + 6)$

$= \lambda^2 - 5\lambda + 6 - 11\lambda + 6 = 0$

$\Rightarrow \lambda^{-1} = \frac{1}{6}\lambda^2 + \lambda + \frac{11}{6}I$

$c = -6$ $d = 11$

22. A is an involutary matrix given by $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ then the inverse of $\frac{A}{2}$ will be :

A एक अर्न्तवलनीय आव्यूह है जहाँ $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ तो $\frac{A}{2}$ का व्युत्क्रम निम्न में किसके बराबर होगा :

(A) 2A (B) $\frac{A^{-1}}{2}$ (C) $\frac{A}{2}$ (D) A^2

Ans. A

Sol. A is involutary $\Rightarrow$ $A^2 = I$ $\Rightarrow$ $A = A^{-1}$

now $A^2 = \left(\frac{A}{2}\right)(2A) = I$ $\Rightarrow$ $2A = \left(\frac{A}{2}\right)^{-1}$

23. Which of the following statements is incorrect for a square matrix A. ($|A| \neq 0$) :

(A) If A is a diagonal matrix, A^{-1} will also be a diagonal matrix

(B) If A is a symmetric matrix, A^{-1} will also be a symmetric matrix

(C) If $A^{-1} = A \Rightarrow$ A is an idempotent matrix

(D) If $A^{-1} = A \Rightarrow$ A is an involutary matrix

एक वर्ग आव्यूह A के लिये ($|A| \neq 0$) कौनसा कथन असत्य हैं :

(A) यदि A एक विकर्ण आव्यूह है, तो A^{-1} भी विकर्ण आव्यूह होगा

(B) यदि A एक सममित आव्यूह है, तो A^{-1} भी सममित आव्यूह होगा

(C) यदि $A^{-1} = A$ तो A एक वर्गसम आव्यूह होगा

(D) यदि $A^{-1} = A$ तो A एक अन्तर्वलनीय आव्यूह होगा

Ans. C

Sol. $A^2 = I \rightarrow$ Involutary Matrix

$A^2 = A \rightarrow$ Idempotent Matrix

24. Let the matrix A and B be defined as $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$ then the value of $\text{Det.}(2A^9B^{-1})$, is :

यदि दो आव्यूह $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$ दिया गया है, तो $\text{Det.}(2A^9B^{-1})$ का मान होगा :

(A) 2 (B) 1 (C) – 1 (D) – 2

Ans. D

Sol. $\text{Det.}(2A^9B^{-1}) = \dfrac{2^2(\text{Det.}A)^9}{\text{Det.}B} = \dfrac{2^2(-1)^9}{2} = -2$ Ans.

25. If A and B are invertible matrices, which one of the following statements is not correct :

यदि A तथा B व्युत्क्रमणीय आव्यूह है तो उनमें से कौनसा कथन सत्य नही होगा :

(A) $\text{Adj. } A = |A| A^{-1}$ (B) $\det(A^{-1}) = |\det(A)|^{-1}$

(C) $(A + B)^{-1} = B^{-1} + A^{-1}$ (D) $(AB)^{-1} = B^{-1}A^{-1}$

Ans. C

Sol. $AA^{-1} = I \Rightarrow |AA^{-1}| = |I| = 1$

hence $|A|\,|A^{-1}| = 1 \Rightarrow |A^{-1}| = \dfrac{1}{|A|} \Rightarrow$ (B) is correct

26. If $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the matrix A is equal to :

यदि $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ हो तो मैट्रिक्स A निम्न में से किस के तुल्य है :

(A) $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Ans. C

Sol. $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$

$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A = 1\begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix}$

$A = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

27. If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, then the value of $\left| A' \quad A^{-1} \right|$ is :

यदि $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, तो $\left| A' \quad A^{-1} \right|$ का मान होगा :

(A) $\cos 4x$ (B) $\sec^2 x$ (C) $-\cos 4x$ (D) 1

Ans. D

Sol. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$

$\left| A' \quad A^{-1} \right| = 1$

28. Which of the following is incorrect?

निम्न में से कौनसा विकल्प असत्य है?

(A) $\text{adj}(\text{adj}\, A) = A$ (B) $(A^T)^T = A$ (C) $(A^{-1})^T = (A^T)^{-1}$ (D) $(A - I)(I + A) = 0 \Leftrightarrow A^2 = I$

Ans. A

Sol. $(A^T)^T = A$

29. Let $A=\begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$. The only correct statement about the matrix A is :

(A) A is a zero matrix
(B) $A = (-1)\, I$, where I is a unit matrix
(C) A^{-1} does not exist
(D) $A^2 = I$

माना कि $A=\begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ तो मैट्रिक्स A के संदर्भ में सही कथन कौनसा है :

(A) । शून्य मैट्रिक्स है
(B) । ञ ; 1द्धप्एजहाँ I इकाई मैट्रिक्स है
(C) $।^{-1}$ विद्यमान नहीं है
(D) $A^2 = I$

Ans. D

Sol. $A=\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

$$A^2=\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore A^2 = I$

30. Let $A=\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ and $(10)B=\begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$. If B is the inverse of matrix A, then α is :

माना $A=\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ तथा $(10)B=\begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$ यदि B मैट्रिक A का प्रतिलोम हो, तो α का मान है :

(A) – 2 (B) – 1 (C) 2 (D) 5

Ans. D

Sol. $B = A^{-1} \Rightarrow \quad AB = I$

hence product of 2nd row and 3rd column = 0

$$\frac{1}{10}\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$4 + \alpha - 9 = 0 \Rightarrow \quad a = 5$ Ans.

31. For a non singular square matrix A of order n, which one of the following statement is true :

(A) adj (KA) = K(adjA) (B) $|\text{adj } A| = |A|$

(C) (adj KA) = K^{n-1} adj A (D) $A.(\text{adj } A) = A^{-1}$

n कोटि के एक व्युत्क्रमणीय आव्यूह A है, तो इनमें से कौनसा कथन सत्य होगा :

(A) adj (KA) = K(adjA) (B) $|\text{adj } A| = |A|$

(C) (adj KA) = K^{n-1} adj A (D) $A.(\text{adj } A) = A^{-1}$

Ans. C

Sol. $|\text{adj } A| = |A|^{n-1}$ .....(1)

$A \text{ lso } A \cdot \text{adj } A = |A| I_n$

$A \to kA$

$(KA)\,\text{adj}(KA) = |KA| I_n$

$(KA)\,\text{adj}(KA) = K^n |A| I_n$

$A \cdot \text{adj } (KA) = K^{n-1}(\text{adj } A) A$

$\text{adj } (KA) = K^{n-1}(\text{adj } A)$

32. If $A=\begin{bmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & a & 1\end{bmatrix}, A^{-1}=\begin{bmatrix}1/2 & -1/2 & 1/2\\ -4 & 3 & c\\ 5/2 & -3/2 & 1/2\end{bmatrix}$, then :

यदि $A=\begin{bmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & a & 1\end{bmatrix}, A^{-1}=\begin{bmatrix}1/2 & -1/2 & 1/2\\ -4 & 3 & c\\ 5/2 & -3/2 & 1/2\end{bmatrix}$, तो :

(A) a = 1, c = – 1 (B) $a = 2, c = -\frac{1}{2}$ (C) a = – 1, c = 1 (D) $a = \frac{1}{2}, c = \frac{1}{2}$

Ans. A

Sol. $AA^{-1} = I \quad \Rightarrow \quad R_2C_3 = 0$

$$\frac{1}{2} + 2c + \frac{3}{2} = 0$$

$2c = -2 \quad \Rightarrow \quad c = -1$

also $R_3C_2 = 0$

$$-\frac{3}{2} + 3a - \frac{3}{2} = 0$$

$3a = 3 \Rightarrow \quad a = 1$

hence $a = 1; c = -1$

33. Let P and Q be 3×3 matrices with $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of $(P^2 + Q^2)$ is equal to : **[AIEEE-2012]**

माना P तथा Q, 3×3 आव्यूह है तथा $P \neq Q$ है। यदि $P^3 = Q^3$ तथा $P^2Q = Q^2P$ है, तो सारणिक $(P^2 + Q^2)$ बराबर है : **[AIEEE-2012]**

(A) –1 (B) –2 (C) 1 (D) 0

Ans. D

Sol. $(P^2 + Q^2) P = P^3 + Q^2P$ _____(1)

$(P^2 + Q^2) Q = P^2Q + Q^3$ _____(2)

Equation (1) – Equation (2)

$(P^2 + Q^2) (P - Q) = P^3 - Q^3 + Q^2P - P^2Q$

$(P^2 + Q^2) (P - Q) = 0 \quad \because (P \neq Q)$

$P^2 + Q^2 = 0$

so $|P^2 + Q^2| = 0$

34. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then $u_1 + u_2$ is equal **to** : **[AIEEE-2012]**

माना $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$ है। यदि u_1 तथा u_2 ऐसे स्तंभ आव्यूह हैं, कि $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ तथा $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ है, तो $u_1 + u_2$ बराबर है : **[AIEEE-2012]**

(A) $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ (B) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ (C) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ (D) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

Ans. A

Sol. $A^{-1} = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix}$

and $A^{-1} A U_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ _____(1)

$A^{-1} A U_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$ _____(2)

Eq. (1) + (2)

$U_1 + U_2 = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$

35. Let A be a square matrix all of whose entries are integers. Then which one of the following is true? **[AIEEE-2008]**

(A) If $\det A \neq \pm 1$, then A^{-1} exists and all its entries are non-integers

(B) If $\det A = \pm 1$, then A^{-1} exists and all its entries are integers

(C) If $\det A = \pm 1$, then A^{-1} need not exist

(D) If $\det A = \pm 1$, then A^{-1} exists but all its entries are not necessarily integers

माना A एक वर्ग आव्यूह है जिसके सभी अवयव पूर्णांक है। तो निम्न में से कौनसा सत्य है? **[AIEEE-2008]**

(A) यदि $\det A \neq \pm 1$, तो A^{-1} का अस्तित्व है तथा इसके सभी अवयव पूर्णांक नहीं है

(B) यदि $\det A = \pm 1$, तो A^{-1} का अस्तित्व है तथा इसके सभी अवयव पूर्णांक हैं

(C) यदि $\det A = \pm 1$, तो A^{-1} के अस्तित्व का होना आवश्यक नहीं है

(D) यदि $\det A = \pm 1$, तो A^{-1} का अस्तित्व है परन्तु इसके सभी अवयव पूर्णांक होने आवश्यक नहीं है

Ans. B

Sol. As $\det (A) = \pm 1$, A^{-1} exists and $A^{-1} = \frac{1}{\det (A)} (\text{adj } A) = \pm (\text{adj } A)$

All entries in adj (A) are integers.

$\therefore$ A^{-1} has integer entries.

Hindi. det (A) = ± 1 के अनुसार A^{-1} विद्यमान है तथा $A^{-1} = \frac{1}{\det(A)}(\text{adj } A) = \pm (\text{adj } A)$

कर; । द्धमें सभी प्रविष्टियाँ पूर्णांक है।

$\therefore$ A^{-1} में प्रविष्टियाँ पूर्णांक है।

36. Let A be a 2 × 2 matrix with real entries. Let I be the 2 × 2 identity matrix. Denote by tr (A), the sum of diagonal entries of A, Assume that $A^2 = I$. **[AIEEE-2008]**

Statement- 1: If $A \neq I$ and $A \neq -I$, then det A = – 1

Statement -2 : If $A \neq I$ and $A \neq -I$, then tr (A) ≠ 0

(A) Statement-1 is true, Statement -2 is true; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is true, Statement -2 is true; Statement-2 is not a correct explanation for Statement-1

(C) Statement-1 is true, Statement -2 is false

(D) Statement-1 is false, Statement-2 is true

माना A एक 2 × 2 आव्यूह है जिसके अवयव वास्तविक संख्याएँ हैं। माना I एक 2 × 2 का तत्समक आव्यूह है। विकर्ण पर स्थित A के v o; okads; ksxQy d ksstr(A) द्वारा निर्दिष्ट कीजिए। मान लीजिए $A^2 = I$. **[AIEEE-2008]**

Statement- 1 : यदि $A \neq I$ तथा $A \neq -I$, तो det A = – 1

Statement -2 : यदि $A \neq I$ तथा $A \neq -I$, तो tr (A) ≠ 0

(A) कथन-1 सत्य है, कथन-2 सत्य हैं, कथन-2 , कथन-1 की ठीक व्याख्या है

(B) कथन-1 सत्य है, कथन-2 सत्य है, कथन-2, कथन-1 की ठीक व्याख्या नहीं है

(C) कथन-1 सत्य है, कथन-2 असत्य है

(D) कथन-1 असत्य है, कथन-2 सत्य है

Ans. C

Sol. $A^2 = I$

$|A^2| = |I|$

$|A^2| = 1$

$|A| = \pm 1$

statement–1 :

If $A \neq I, A \neq -I$

but $|A| = \pm 1$

so this statement is true

statement–2 :

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$|A| = -1$ $\quad$ $\text{tr}(A) = 0$

but $A \neq I, A \neq -I$

so statement-2 is false

SYSTEM OF EQUATION BY MATRIX METHOD

37. The equation $\begin{bmatrix}1 & 2 & 2\\ 1 & 3 & 4\\ 3 & 4 & k\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$ has a solution for (x, y, z) besides (0, 0, 0). The value of k equals :

समीकरण $\begin{bmatrix}1 & 2 & 2\\ 1 & 3 & 4\\ 3 & 4 & k\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$ का (x, y, z) के लिये (0, 0, 0) के अतिरिक्त और भी हल हो तो k का मान होगा :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $\begin{vmatrix}1 & 2 & 2\\ 1 & 3 & 4\\ 3 & 4 & k\end{vmatrix}=0$

$1(3k-16)-2(k-12)+2(4-9)=0$

$3k-16-2k+24-10=0$

$k=2$ Ans.

38. The number of 3×3 matrices A whose entries either 0 or 1 and for which the system $A\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$ has exactly two distinct solutions, is : **[JEE-2010]**

3×3 कोटि के आव्यूहों A की संख्या, जिसके सभी अवयव 0 या 1 है तथा जिनके लिए निकाय $A\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$ के ठीक दो भिन्न हल

fo| eku gksgksxh: **[JEE-2010]**

(A) 0 (B) 2^9-1 (C) 168 (D) 2

Ans. A

Sol. $a_1x+b_1y+c_1z=1$

$a_2x+b_2y+c_2z=0$

$a_3x+b_3y+c_3z=0$

No three planes can meet at two distinct points. So number of matrices is 0

Exercise J-2

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON BASIC DEFINITIONS RELATED TO MATRIX

1. A is a square matrix of order n
l = maximum number of distinct entries if A is a triangular matrix
m = maximum number of distinct entries if A is a diagonal matrix
p = minimum number of zeroes if A is a triangular matrix
If $l + 5 = p + 2m$, find the order of the matrix :

। एक n कोटि का वर्ग आव्यूह है
l = भिन्न अवयवों की अधिकतम संख्या है यदि A त्रिभुजीय आव्यूह है
m = भिन्न अवयवों की अधिकतम संख्या है यदि A विकर्ण आव्यूह है
p = शून्यों की न्यूनतम संख्या है यदि A त्रिभुजीय आव्यूह है
; fn $l + 5 = p + 2m$ हो, तो आव्यूह की कोटि ज्ञात कीजिए :

Ans. 4

Sol. $\ell = 1 + 2 + 3 + ... + n \Rightarrow \ell = \frac{n(n+1)}{2} + 1$;

$m = n + 1 \qquad p = \frac{n(n-1)}{2}$

ALGEBRAIC OPERATIONS ON MATRICES

2. If, $E = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ calculate the matrix product EF and FE and show that $E^2F + FE^2 = E$.

यदि $E = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ तथा $F = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ है, तो आव्यूह गुणन EF तथा FE ज्ञात कीजिये तथा सिद्ध कीजिये $E^2F + FE^2 = E$.

Ans. $EF = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $FE = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Sol. $\Rightarrow EF = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow FE = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

3. Find the value of x and y that satisfy the equations.

$$\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$$

समीकरण $\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$ को संतुष्ट करने वाले x और y के मान ज्ञात कीजिये।

Ans. $x = \frac{3}{2}, y = 2$

Sol. $3y - 2x = 3$

$2y + 4x = 0$

$3y - 2x = 3$

$y + 2x = 5$

$4y = 8 \Rightarrow y = 2\ x = \frac{3}{2}$

4. If, $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the maxtrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

यदि $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ है तो सिद्ध कीजिए कि आव्यूह A बहुपद $f(x) = x^3 - 6x^2 + 7x + 2$ का मूल है।

Ans.

Sol. $|A=\lambda|=\begin{vmatrix}1-\lambda & 0 & 2\\ 0 & 2-\lambda & 1\\ 2 & 0 & 3-\lambda\end{vmatrix}$

$\Rightarrow \lambda^3-6\lambda^2+7\lambda+2=0$

5. Consider the two matrices A and B where $A=\begin{bmatrix}1 & 2\\ 4 & 3\end{bmatrix}; B=\begin{bmatrix}5\\ -3\end{bmatrix}$. If n(A) denotes the number of elements in A such that n(XY) = 0, when the two matrices X and Y are not conformable for multiplication. If C = (AB)(B'A); D = (B'A)(AB) then, find the value of $\left(\frac{n(C)\left(|D|^2+n(D)\right)}{n(A)-n(B)}\right)$.

दो आव्यूह A तथा B जहाँ $A=\begin{bmatrix}1 & 2\\ 4 & 3\end{bmatrix}; B=\begin{bmatrix}5\\ -3\end{bmatrix}$ पर विचार कीजिये। यदि n(A) आव्यूह A में अवयवों की संख्या को व्यक्त करता gSt ksbl i zd kj gSfd n(XY) = 0 जबकि दोनों आव्यूह X तथा Y का गुणनफल संभव नहीं है। यदि C = (AB)(B'A); D = (B'A)(AB) व ; ठश। ठ द्ध हैं, तो $\left(\frac{n(C)\left(|D|^2+n(D)\right)}{n(A)-n(B)}\right)$ का मान ज्ञात कीजिये।

Ans. 650

Sol. n(A) = 4 ; n(B) = 2 ; n(C) = 4; n(D) = 1; |D| = 18

$\frac{n(C)(|D|^2+n(D))}{n(A)-n(B)}=\frac{4(18^2+1)}{2}=650$

SPECIAL TYPE OF MATRICES (NILPOTENT, IDEMPOTENT, INVOLUTARY, PERIODIC)

6. Define $A=\begin{bmatrix}0 & 1\\ 3 & 0\end{bmatrix}$. Find a vertical vector V such that $(A^8+A^6+A^4+A^2+I)V=\begin{bmatrix}0\\ 11\end{bmatrix}$ (where I is the 2×2 identity matrix).

। ; $\begin{bmatrix}0 & 1\\ 3 & 0\end{bmatrix}$ है, तो उर्ध्वाधर सदिश V का मान ज्ञात कीजिये जो इस प्रकार है कि $(A^8+A^6+A^4+A^2+I)V=\begin{bmatrix}0\\ 11\end{bmatrix}$ हो। (जहाँ I, (2×2) कोटि का तत्समक आव्यूह है)

Ans. $V = V = \begin{bmatrix} 0 \\ \frac{1}{11} \end{bmatrix}$

Sol. $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$; $A^2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = 3I$

$$(1 + 3 + 3^2 + 3^3 + 3^4)\, I \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$$(121) \quad \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 1/11 \end{bmatrix}$$

7. If $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is an idempotent matrix. Find the value of f(a), where $f(x) = x - x^2$, when $bc = 1/4$. Hence otherwise evaluate a.

यदि $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ एक वर्गसम (idempotent) आव्यूह है, तो f(a) का मान ज्ञात कीजिए , जहाँ $f(x) = x - x^2$ है, जबकि $bc = 1/4$ है, तो a का मान ज्ञात कीजिए।

Ans. $f(a) = 1/4$, $a = 1/2$

Sol. $a^2 + bc = a$

$$\Rightarrow a^2 + \frac{1}{4} = a$$

$$\Rightarrow a = \frac{1}{2}$$

8. If the matrix A is involutary, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = O$.

यदि आव्यूह A अन्तर्वलनीय (involutary) है, तो सिद्ध कीजिये $\frac{1}{2}(I + A)$ तथा $\frac{1}{2}(I - A)$ वर्गसम (idempotent) है तथा $\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = O$ है।

Ans.

Sol. $\frac{1}{4}(Z+A)(I+A)=\frac{1}{4}(I+A+A+A^2)=\frac{1}{2}(I+A)$

$\frac{1}{4}(I-A)(I-A)=\frac{1}{4}\left(I-A-A-A^2=\frac{1}{2}\right)(I-A)$

9. Show that the matrix $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$ can be decomposed as a sum of a unit and a nilpotent marix. Hence evaluate the matrix $\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}^{2007}$.

सिद्ध कीजिये आव्यूह $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$ को इकाई आव्यूह तथा शून्यभावी (nilpotent) आव्यूह के योग के रूप में लिखा जा सकता है तथा $\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}^{2007}$ का मान भी ज्ञात कीजिये।

Ans. $\begin{bmatrix}1 & 0\\ 4014 & 1\end{bmatrix}$

Sol. $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}+\begin{bmatrix}0 & 0\\ 2 & 0\end{bmatrix}$

$A^{2007}=I+20078=\begin{bmatrix}0 & 0\\ 2 & 0\end{bmatrix}+\begin{bmatrix}0 & 0\\ 4014 & 0\end{bmatrix}$

10. Let X be the solution set of the equation $A^x=I$, where $A=\begin{bmatrix}0 & 1 & -1\\ 4 & -3 & 4\\ 3 & -3 & 4\end{bmatrix}$ and I is the corresponding unit matrix and $x\subseteq N$ then find the minimum value of $\sum(\cos^x\theta+\sin^x\theta)$, $\theta\in R$.

माना X समीकरण $A^x = I$ के हलों का समूह है, जहाँ $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ तथा I संगत इकाई आव्यूह है तथा $x \subseteq N$ है, तो

$\sum(\cos^x \theta + \sin^x \theta)$, $\theta \in R$ का न्यूनतम मान ज्ञात कीजिये।

Ans. 2

Sol. $A^2 = I \Rightarrow \quad x = 2, 4, 6, ...$

$\sum(\cos^x \theta + \sin^x \theta)$

$= (\cos^2 \theta + \sin^2 \theta) + (\cos^4 \theta + \sin^4 \theta) + ...$

$= (\cot^2 \theta + \tan^2 \theta) \geq 2$

11. If A is an idempotent non zero matrix and I is an identity matrix of the same order, find the value of n, $n \in N$, such that $(A + I)^n = I + 127 A$.

यदि A अशून्य वर्गसम (idempotent) आव्यूह है तथा I समान कोटि का इकाई आव्यूह है तो n ($n \in N$) का मान ज्ञात कीजिए जो इस प्रकार है कि $(A + I)^n = I + 127 A$ है।

Ans. n = 7

Sol. ${}^nC_1 + {}^nC_2 + ...+{}^nC_n + 127$

$\Rightarrow 2n - 1 = 127$

$n = 7$

SYMMETRIC AND SKEW SYMMETRIC MATRICES

12. Given matrices $A = \begin{bmatrix} 1 & x & 1 \\ x & 2 & y \\ 1 & y & 3 \end{bmatrix}; B = \begin{bmatrix} 3 & -3 & z \\ -3 & 2 & -3 \\ z & -3 & 1 \end{bmatrix}$ Obtain x, y and z if the matrix AB is symmetric.

आव्यूह $A = \begin{bmatrix} 1 & x & 1 \\ x & 2 & y \\ 1 & y & 3 \end{bmatrix}; B = \begin{bmatrix} 3 & -3 & z \\ -3 & 2 & -3 \\ z & -3 & 1 \end{bmatrix}$ दिये गये है x, y तथा z ज्ञात कीजिये यदि आव्यूह AB सममित है।

Ans. $\left(-\frac{4\sqrt{2}}{3}, \frac{2}{3}, 2\sqrt{2}\right), \left(\frac{4\sqrt{2}}{3}, \frac{2}{3}, -2\sqrt{2}\right), (3, 3, -1)$

Sol. $AB = \begin{bmatrix} 1 & x & 1 \\ x & 2 & y \\ 1 & y & 3 \end{bmatrix} \begin{bmatrix} 3 & -3 & z \\ -3 & 2 & -3 \\ z & -3 & 1 \end{bmatrix}$

$$\begin{bmatrix} 3-3x+3 & -3+2x-3 & z-3x+1 \\ 3x-b+y3 & -3x+6-3y & z-3y+3 \\ 3-3y+3z & -3+2y-9 & z-3y+3 \end{bmatrix}$$

$-3+2x-3 = 3x-6+yz \Rightarrow x+yz = 0$

$3-3y+3z = z-3x+1 \Rightarrow 3x-3y+2z+2 = 0$

$-3+2y-9 = z-3y+3 \Rightarrow 3-5y =$

13. $A = \begin{pmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{pmatrix}$ is Symmetric and $B = \begin{pmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{pmatrix}$ is Skew Symmetric, then find AB.

Is AB a symmetric, Skew Symmetric or neither of them. Justify your answer.

$A = \begin{pmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{pmatrix}$ सममित है तथा $B = \begin{pmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{pmatrix}$ विषम सममित आव्यूह है तो AB का मान ज्ञात कीजिये। क्या AB

सममित है, विषम सममित है या इनमें से कोई नहीं कारण बताइये ?

Ans. AB is neither symmetric nor skew symmetric

Sol. $AB \pm BA$

14. Express the matrix $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & -6 \\ -1 & 0 & 4 \end{bmatrix}$ as a sum of a lower triangular matrix and an upper triangular matrix with zero in its leading diagonal. Also Express the matrix as a sum of a symmetric and a skew symmetric matrix.

आव्यूह $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & -6 \\ -1 & 0 & 4 \end{bmatrix}$ को निम्न त्रिभुजीय आव्यूह तथा ऊपरी आव्यूह के योग के रूप में लिखिये। जिसके मुख्य विकर्ण के सभी अवयव शून्य है। आव्यूह को सममित तथा विषम सममित आव्यूहों के योगफल के रूप में भी व्यक्त कीजिये।

Ans. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 5 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -3 \\ 2 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & -3 \\ -3 & 3 & 0 \end{bmatrix}$

Sol.

15. Let $A = [a_{ij}]_{3\times3}$ be a matrix. If $A + A^T = \begin{bmatrix} 6 & 4 & 4 \\ a_{21}+a_{12} & 10 & a_{23}+a_{32} \\ a_{31}+a_{13} & 4 & 8 \end{bmatrix}$ where a_{12}, a_{23} and a_{31} are the positive root of the equation $x^3 - 6x^2 + px - 8 = 0$, $p \in R$, then the value of det(A) is

[Note : A^T denotes the transpose of matrix A.]

माना $A = [a_{ij}]_{3\times3}$ एक आव्यूह है। यदि $A + A^T = \begin{bmatrix} 6 & 4 & 4 \\ a_{21}+a_{12} & 10 & a_{23}+a_{32} \\ a_{31}+a_{13} & 4 & 8 \end{bmatrix}$ जहाँ a_{12}, a_{23} तथा a_{31} समीकरण $x^3 - 6x^2 + px - 8 = 0$, $p \in R$ के धनात्मक मूल हो, तो det(A) का मान होगा –

[नोट : A^T, आव्यूह के परिवर्त को दर्शाता है।]

Ans 28

Sol. Clearly $A + A^T$ is a symmetric matrix

Also, equation $x^3 - 6x^2 + 8x - 8 = 0$ has roots a_{12}, a_{23}, a_{31}

$\therefore a_{12} + a_{23} + a_{31} = 8$

A.M. = GM. $\Rightarrow a_{12} = a_{23} = a_{31} = 2$

$$\left.\begin{aligned} a_{21} + a_{12} &= 4 \\ \therefore a_{31} + a_{13} &= 4 \\ a_{23} + a_{32} &= 4 \end{aligned}\right\} \Rightarrow a_{21} = a_{13} = a_{23} = 2$$

and for diagonal elements.

$2a_{11} = 6$, $2a_{22} = 10$, $2a_{33} = 8$

$\Rightarrow a_{11} = 3$, $a_{22} = 5$, $a_{33} = 4$

$$\therefore A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 5 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

$\Rightarrow |A| = 3(20 - 4) - 2(8 - 4) + 2(4 - 10)$

$= 48 - 8 - 12 = 28$

ORTHOGONAL MATRIX

16. If M is a 3×3 matrix, where $M^TM = I$ and det (M) = 1, then prove that det (M – I) = 0.

यदि M कोटि 3×3 का आव्यूह है, जहाँ $M^TM = I$ और det (M) = 1 हो तो सिद्ध कीजिये कि det (M – I) = 0 है।

Ans.

Sol. $\det(M-I) = \det(M-I)^T$

$= \det\left(M^T - I\right)$

$= \det\left(M^T - M^T M\right)$

$= \det\left(M^T\left(I-M\right)\right)$

$= \det M^T . \det\left(I-M\right)$

$= \det M . \det\left(-\left(M-I\right)\right)$

$= (-1)^3 \det\left(M-I\right)$

$\det\left(M-I\right) = -\det\left(M-I\right)$

$\therefore \det\left(M-I\right) = 0$

ADJOINT OF A SQUARE MATRIX

17. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ then the trace of the matrix Adj(Adj A) is _____

यदि $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ है तो tr. [Adj(Adj A)] का मान होगा _____

Ans. 1

Sol. $|M-I| = |M - M M^T|$

$|M-I| = |M|\,|I-M|$

$\Rightarrow |M-I| = |I-M|$

$\Rightarrow \quad |M-I| = (-1)^3 |M-I|$

$\Rightarrow |M-I| = 0$

18. If $A=\begin{bmatrix} u & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, where $(u>0)$ and det (adj(adj A)) = $(23)^4$, then the value of u is .....

यदि $A=\begin{bmatrix} u & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, है जहां $(u>0)$ तथा det (adj(adj A)) = $(23)^4$ हो तो u का मान होगा?

Ans. 4

Sol. $\left|\text{adj}(\text{adj}A)\right| = |A|^{(n-1)^2}$

$= |A|^{(3-1)^2}$

$= |A|^4$

$$|A| = \begin{vmatrix} u & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix} = 3u+11$$

$|A| = 23$

$3u + 11 = 23$

$3u = 12$

$u = \frac{12}{3} = 4$

INVERSE OF A SQUARE MATRIX

19. Let A be a 3 × 3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$ then find the value of $(x + y + z)$ where I is a unit matrix of order 3.

माना A एक (3 × 3) कोटि का आव्यूह इस प्रकार है कि $a_{11} = a_{33} = 2$ तथा सभी अन्य अवयव $a_{ij} = 1$ है। माना $A^{-1} = xA^2 + yA + zI$ है, तो $(x + y + z)$ का मान ज्ञात कीजिये। जहाँ I एक 3 कोटि का इकाई आव्यूह है।

Ans. 1

Sol. $A=\begin{bmatrix}2 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 2\end{bmatrix}$

$A^{-1}=xA^2+yA+zI$

$\Rightarrow xA^3+yA^2+zA-I=0$

is characteristic equation of A

$$\begin{vmatrix}2-\lambda & 1 & 1\\ 1 & 1-\lambda & 1\\ 1 & 1 & 2-\lambda\end{vmatrix}=0$$

$\lambda^2-5\lambda^2+5\lambda-1=0$

$\Rightarrow x+y+z=1$

20. For the matrix $A=\begin{bmatrix}4 & -4 & 5\\ -2 & 3 & -3\\ 3 & -3 & 4\end{bmatrix}$ find A^{-2}.

आव्यूह $A=\begin{bmatrix}4 & -4 & 5\\ -2 & 3 & -3\\ 3 & -3 & 4\end{bmatrix}$ के लिए A^{-2} का मान ज्ञात कीजिये।

Ans. $\begin{bmatrix}17 & 4 & -19\\ -10 & 0 & 13\\ -21 & -3 & 25\end{bmatrix}$

Sol. $A=\begin{bmatrix}4 & -4 & 5\\ -2 & 3 & -3\\ 3 & -3 & 4\end{bmatrix}$

$A^2=\begin{bmatrix}4 & -4 & 5\\ -2 & 3 & -3\\ 3 & -3 & 4\end{bmatrix}\times\begin{bmatrix}4 & -4 & 5\\ -2 & 3 & -3\\ 3 & -3 & 4\end{bmatrix}$

$$A^2 = \begin{bmatrix} 39 & -43 & 52 \\ -23 & 26 & -31 \\ 30 & -33 & 40 \end{bmatrix}$$

$$A^{-2} = \left(A^2\right)^{-1} = \frac{\text{adj}\left(A^2\right)}{\left|A^2\right|}$$

$$A^2 = 39(27) + 43(10) + 52(-21)$$

$$= 1$$

$$\text{adj}\left(A^2\right) = \begin{bmatrix} 17 & -10 & -21 \\ 4 & 0 & 3 \\ -19 & 13 & 25 \end{bmatrix}^T$$

$$\therefore A^{-2} = \begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & 3 & 25 \end{bmatrix}$$

21. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then show that $F(x).\ F(y) = F(x+y)$

Hence prove that $[\ F(x)\ ]^{-1} = F(-x)$.

यदि $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ है, तो सिद्ध कीजिये $F(x).\ F(y) = F(x+y)$

और सिद्ध कीजिये $[\ F(x)\ ]^{-1} = F(-x)$.

Ans.

Sol. $$\Rightarrow \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|f(x)| = \begin{vmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \cos x \times \cos x + \sin x \times \sin x = 1$$

$$\text{adf}(f(x)) = \begin{vmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \cos(-x) & -\sin(-x) & 0 \\ \sin(-x) & \cos(-x) & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$= f(-x)$

22. Let M be a 3×3 matrix satisfying $M\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, M\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $M\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$. Then the sum of the diagonal entries of M is. **[JEE-2011]**

माना कि M एक कोटि 3×3 की आव्यूह है जो $M\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, M\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ तथा $M\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$ को संतुष्ट करता है, तब M के विकर्ण के अवयवों का योग है। **[JEE-2011]**

Ans. 9

Sol. Let $M = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

then $a_{12} = -1$, $a_{11} - a_{12} = 1 \Rightarrow a_{11} = 0$, $a_{11} + a_{12} + a_{13} = 0 \Rightarrow a_{13} = 1$

$a_{22} = 2$, $a_{21} - a_{22} = 1 \Rightarrow a_{21} = 3$, $a_{21} + a_{22} + a_{23} = 0 \Rightarrow a_{23} = -5$

$a_{32} = 3$, $a_{31} - a_{32} = -1 \Rightarrow a_{31} = 2$, $a_{31} + a_{32} + a_{33} = 12 \Rightarrow a_{33} = 7$

Hence sum of diagonal of M is $= a_{11} + a_{22} + a_{33} = 0 + 2 + 7 = 9$

SYSTEM OF EQUATION BY MATRIX METHOD

23. Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. Find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

दिया गया है कि $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$ है, तो P का मान, जो इस प्रकार है कि $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ ज्ञात कीजिये।

Ans. $\begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$

Sol. Here $|A| = -1$, $|B| = -1$

Now $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

So, $B^{-1}BPAA^{-1} = B^{-1}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}A^{-1}$

$|P|$

$= (-1)\frac{(-1)}{I}\begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & -3 \\ 0 & -1 & 1 \\ -2 & -1 & 2 \end{bmatrix}$

$$P = \begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$$

Use matrix to solve the following system of equations (Que. no. 24 to 27).

निम्न समीकरण निकायों का हल कीजियें ।(प्रश्न संख्या 24 से 27)

24. $x + y + z = 3$
$x + 2y + 3z = 4$
$x + 4y + 9z = 6$

Ans. $x = 2, y = 1, z = 0$

Sol. Given system in matrix equation is,

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

$A = B$

Argumentes matrix [A, B]

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 8 & 3 \end{bmatrix} \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 0 \end{bmatrix} R_3 \to R_3 - 3R_2$$

Writing it in echelon form,

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} . \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

$x + y + 2 = 3$ _____(1)

$y + 2z = 1$ _____(2)

$2z = 0 \quad \Rightarrow z = 0$

$z = 0$ in _____(2)

$\Rightarrow y = 1$

$z = 0, y = 1$ in _____(1)

$x = 2, y \geq 1, 2 = 0$

$x = k = 2;\ y, n = 1, z > n = 0$

$(k \times m) - n \Rightarrow (2 \times 1) - 0 \quad \Rightarrow 2$

$\therefore 2$

25. $x + y + z = 6$
$x - y + z = 2$
$2x + y - z = 1$

Ans. $x = 1, y = 2, z = 3;$

Sol. We have,

$x + y + z = 6, x - y + z = 2, 2x + y - z = 1$

$$\text{Thus } A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}$$

$$X = \begin{vmatrix} x \\ y \\ z \end{vmatrix}, \ B = \begin{vmatrix} 6 \\ 2 \\ 1 \end{vmatrix}$$

Now solve A to get $A = 6$

$$\text{Now } x = \frac{A_X}{A} = \frac{\begin{vmatrix} 6 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}}$$

$$y = \frac{A_y}{A} = \frac{\begin{vmatrix} 1 & 6 & 1 \\ 1 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}} = \frac{12}{6} = 2$$

$$z = \frac{A_z}{A} = \frac{\begin{vmatrix} 1 & 1 & 6 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}} = \frac{18}{6} = 3$$

Thus $x = 1, y = 2, z = 3$

26. $x + y + z = 3$
$x + 2y + 3z = 4$
$2x + 3y + 4z = 7$

Ans. $x = 2+k, y = 1 - 2k, z = k$ where $k \in R$;

Sol. Given system of equations can be written as $AX = B$

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$

$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$

Here, $|A| = 0$

Now, we will find (adjA)B

$$\text{adj}\,A = C^T = \begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ 1 & -2 & 1 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 2 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

$$\text{Now, (adjA)B} = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 2 & -2 \\ -1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

$$\Rightarrow (\text{adjA})\,B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (\text{adjA})\,B = 0$$

Hence, the system of equations has infinitely many solutions.

Let $z = k$ where $k \in R$

Then

$x + y = 3 - k$

$x + 2y = 4 - 3k$

Solving these eqns, we get

$y = 1 - 2k$; $x = 2 + k$

27. $x + y + z = 3$
$x + 2y + 3z = 4$
$2x + 3y + 4z = 9$

Ans. inconsistent, hence no solution

Sol. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{vmatrix} = 1(8-9) - 1(4-6) + 1(3-4)$

$= -1 + 2 - 1 = 0$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question ns. 28 to 30

Let A and B are two matrices of same order i.e. where $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & K & 5 \\ 4 & 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$

माना A तथा B दो समान कोटि के मैट्रिक्स है जहाँ $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & K & 5 \\ 4 & 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$

28. If A is singular matrix then tr (A + B) is equal to :

यदि A एक अव्युत्क्रमणीय मैट्रिक्स है, तो tr (A + B) का मान है :

(A) 5 (B) 3 (C) 4 (D) 6

Ans. B

Sol. $|A| = 0$

$k - 10 + 3(2 - 20) + 2(4 - 4k) = 0$

$k - 10 - 54 + 8 - 8k = 0$

$7k = -56$

$k = -8$

$fr(A + B) = -6 + 9 = 3$

29. If K = 2 then tr (AB) + tr (BA) is equal to :

यदि K = 2 है, तो tr (AB) + tr (BA) का मान है :

(A) 66 (B) 42 (C) 84 (D) 63

Ans. C

Sol. $AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 2 & 5 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$

$$= \begin{bmatrix} -4 & - & - \\ - & 21 & - \\ - & - & 25 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}\begin{bmatrix} 1 & -3 & 2 \\ 2 & 2 & 5 \\ 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 16 & - & - \\ - & 0 & - \\ - & - & 26 \end{bmatrix}$$

$tr(AB) + tr(BA) = 42 + 42 = 84$

30. If $C = A - B$ and $tr(C) = 0$ then K is equal to :

यदि $C = A - B$ तथा $tr(C) = 0$ है, तो K का मान है :

(A) 5 (B) -5 (C) 7 (D) -7

Ans. C

Sol. $C = \begin{bmatrix} -1 & -4 & -1 \\ -2 & k-2 & 1 \\ 1 & -1 & -4 \end{bmatrix}$

$tr(C) = 0$

$-7 + k = 0$

$k = 7$

Paragraph for Question ns. 31 to 33

Find the rank of the following matrices

निम्न आव्यूह की कोटि (Rank) ज्ञात कीजिए।

31. $\begin{bmatrix} 2 & 3 & 5 \\ 1 & 1 & 6 \\ 1 & 4 & 3 \end{bmatrix}$:

(A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol. $\begin{vmatrix} 2 & 3 & 5 \\ 1 & 1 & 6 \\ 1 & 4 & 3 \end{vmatrix} = -42 + 9 + 15 \neq 0$

Hence rank = 3

32. $\begin{bmatrix} 1 & 3 & -2 \\ 4 & -1 & 0 \\ 2 & -7 & 4 \end{bmatrix}$

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $\begin{bmatrix} 1 & 3 & -2 \\ 4 & -1 & 0 \\ 2 & -7 & 4 \end{bmatrix}$ $R_3 \to R_3 + 2R_1 - R_2$

$\begin{bmatrix} 1 & 3 & -2 \\ 4 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Hence rank = 2

33. $\begin{bmatrix} 2 & -1 & 3 & 5 \\ 1 & 4 & 0 & 2 \\ 0 & -9 & 3 & 1 \\ 3 & 3 & 3 & 7 \end{bmatrix}$

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $R_4 \to R_4 - R_1 - R_2$

$\begin{bmatrix} 2 & -1 & 3 & 5 \\ 1 & 4 & 0 & 2 \\ 0 & -9 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$R_3 \to R_3 - R_1$

$$\begin{bmatrix} 2 & -1 & 3 & 5 \\ 1 & 4 & 0 & 2 \\ -2 & -8 & 0 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_2 \to R_2 + 2R_3$

$$\begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 0 & 0 & 0 \\ -2 & -8 & 0 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence rank = 2

Paragraph for Question nos. 34 to 36

$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$, U_1, U_2 and U_3 are columns matrices satisfying. $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$; $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$, $AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and U is 3×3 matrix whose columns are U_1, U_2, U_3 then answer the following questions

$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$, U_1, U_2 तथा U_3 तथा स्तम्भ आव्यूह $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$; $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$, $AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ तथा U 3×3 का आव्यूह है जिसके स्तम्भ U_1, U_2, U_3 हो, तो निम्न प्रश्नों के उतर दीजिए।

34. The value of $|U|$ is :
$|U|$ का मान होगा :
(A) 3 (B) – 3 (C) 3/2 (D) 2

Ans. A

Sol. Let $U_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ then $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ 2a+b \\ 3a+2b+c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow$ $a = 1, b = -2, c = 1$

$\therefore \quad U_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ Similarly, $U_2 = \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix}, U_3 = \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix}$

$\therefore \quad U = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \Rightarrow \quad |U| = 3$

35. The sum of elements of U^{-1} is :

U^{-1} के अवयवों का योग है :

(A) – 1 (B) 0 (C) 1 (D) 3

Ans. B

Sol. $U^{-1} = \frac{1}{3}\begin{bmatrix} -1 & -2 & 0 \\ -7 & -5 & -3 \\ 9 & 6 & 3 \end{bmatrix} \Rightarrow$ Sum of the elements (अवयवों का योगफल) = 0

36. The value of $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is :

$\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ का मान होगा :

(A) 5 (B) 5/2 (C) 4 (D) 3/2

Ans. A

Sol. $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 7 \\ -8 \\ -5 \end{bmatrix} = [5]$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

37. If AB = A and BA = B, then :

यदि AB = A और BA = B, तब :

(A) $_A{}^2B = A^2$ (B) $_B{}^2A = B^2$ (C) ABA = A (D) BAB = B

Ans. ABCD

Sol. We have $A^2B = A(AB) = AA = A^2$, $B^2A = B(BA) = BB = B^2$,
ABA = A(BA) = AB = A, and BAB = B(AB) = BA = B

38. A square matrix A with elements from the set of real numbers is said to be orthogonal if $A' = A^{-1}$. If A is an orthogonal matrix, then :

(A) A′ is orthogonal (B) A^{-1} is orthogonal (C) Aδφ A = A′ (D) $|A^{-1}| = 1$

एक वर्ग आव्यूह A जिसके अवयव वास्तविक संख्याओं के समुच्चय से बने हैं, लाम्बिक होगा यदि $A' = A^{-1}$. यदि A लाम्बिक आव्यूह है, r ls

(A) । ′ लाम्बिक होगा (B) $।^{-1}$ लाम्बिक होगा (C) Aδφ A = A′ (D) $|A^{-1}| = 1$

Ans. AB

Sol. $A^{-1}A = I \Leftrightarrow A'A = I \Leftrightarrow (A')'A' = I \Rightarrow A'$ is orthogonal

Also, $(A^{-1})' = (A')^{-1} = (A^{-1})^{-1} = A = A$

$\Rightarrow$ $A^{-1}(A^{-1})' = I$

$\therefore$ A^{-1} is orthogonal

Next, $AA' = I \Rightarrow |AA'| = 1 \Rightarrow |A|\,|A'| = 1$

$\Rightarrow$ $|A|^2 = 1 \Rightarrow |A| = \pm 1$.

As Aδφ A = |A| A^{-1} = |A| A′, we cannot say, Adj. A = A′

Also, if |A| = 1, then $|A^{-1}| \neq -1$

39. If A and B are two 3 × 3 matrices such that their product AB is a null matrix then :

(A) det. A ≠ 0 ⇒ B must be a null matrix

(B) det. B ≠ 0 ⇒ A must be a null matrix

(C) If none of A and B are null matrices then atleast one of the two matrices must be singular

(D) If neither det. A nor det. B is zero then the given statement is not possible

यदि A और B दो (3 × 3) कोटि के आव्यूह इस प्रकार है कि उनका गुणनफल AB शून्य आव्यूह है तो :

(A) कमजम । ≠ 0 ⇒ B शून्य आव्यूह होना चाहिए

(B) कमजम ठ ≠ 0 ⇒ A शून्य आव्यूह होना चाहिए

(C) यदि A तथा B शून्य आव्यूह है तो दोनों आव्यूह में से कम से कम एक अव्युत्क्रमणीय होगा

(D) यदि न तो det. A ना हीं det. B शून्य है तो दिया गया कथन सम्भव नहीं है

Ans. ABCD

Sol. AB = O

$\therefore$ $|AB| = 0 \quad \Rightarrow \quad |A|\,|B| = 0$

$\therefore$ det A ≠ 0

$\therefore$ A^{-1} exist

$\therefore$ $A^{-1}(AB) = A^{-1}(0) = 0$

IB = 0

Exercise J-M

1. Let $A=\begin{bmatrix}2 & 3\\ a & 0\end{bmatrix}, a\in R$ be written as P + Q where P is a symmetric matrix and Q is skew symmetric matrix. If det(Q) = 9, then the modulus of the sum of all possible values of determinant of P is equal to :

[JEE(Main)Jul.2021]

माना $A=\begin{bmatrix}2 & 3\\ a & 0\end{bmatrix}, a\in R$ को P + Q के रूप में लिखा गया है, जहाँ P एक सममित आव्यूह है तथा Q एक विषम सममित आव्यूह है। ; fn det(Q) = 9 है, तो det(P) के सभी संभव मानों के योगफल का मापांक बराबर है : **[JEE(Main)Jul.2021]**

(A) 18 (B) 36 (C) 45 (D) 24

Ans. (B)

Sol. $A=\frac{A+A^T}{2}+\frac{A-A^T}{2}$

$Q=\frac{A-A^T}{2}$

$=\begin{bmatrix}0 & \frac{3-a}{2}\\ \frac{a-3}{2} & 0\end{bmatrix}$

det (Q) = 9

$\frac{(a-3)^2}{4}=9$

$(a-3)^2=36$

$a=-3, a=9$

$P=\frac{A+A^T}{2}$

$=\begin{bmatrix}0 & \frac{3+a}{2}\\ \frac{3+a}{2} & 0\end{bmatrix}$

$\det(P)=-\left(\frac{3+a}{2}\right)^2$

If a = – 3 det (P) = 9

If a = 9 det (P) = – 36

sum = 0 – 36 = – 36

| sum | = 36

2. Let $A = [a_{ij}]$ be a real matrix of order 3×3, such that $a_{i1} + a_{i2} + a_{i3} = 1$, for i = 1,2,3. Then, the sum of all the entries of the matrix A^3 is equal to : **[JEE Main Jul. 2021]**

माना $A = [a_{ij}]$, कोटि 3×3 का एक वास्तविक आव्यूह इस प्रकार है कि प्रत्येक i = 1, 2, 3 के लिए $a_{i1} + a_{i2} + a_{i3} = 1$ है। तो आव्यूह A^3 की सभी प्रविष्टियों का योग बराबर है : **[JEE Main Jul. 2021]**

(A) 9 (B) 1 (C) 2 (D) 3

Ans. (D)

Sol. $$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Let $X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$AX = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow AX = X$

Replace X by AX

$A(AX) = AX$

$A^2X = AX = X$

$A^2X = X$

Replace X by AX

$A^3X = AX = X$

Let $A^3 = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$

Now

$$A^3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \\ z_1 + z_2 + z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

hence sum of elements = 3

3. If $P=\begin{bmatrix}1 & 0\\ 1/2 & 1\end{bmatrix}$, then P^{50} is : **[JEE Main Jul. 2021]**

यदि $P=\begin{bmatrix}1 & 0\\ 1/2 & 1\end{bmatrix}$ है, तो P^{50} है : **[JEE Main Jul. 2021]**

(A) $\begin{bmatrix}1 & 0\\ 25 & 1\end{bmatrix}$ (B) $\begin{bmatrix}1 & 50\\ 0 & 1\end{bmatrix}$ (C) $\begin{bmatrix}1 & 0\\ 50 & 1\end{bmatrix}$ (D) $\begin{bmatrix}1 & 25\\ 0 & 1\end{bmatrix}$

Ans. (A)

Sol. $P=\begin{bmatrix}1 & 0\\ 1/2 & 1\end{bmatrix}$

$P^2=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

$P^3=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1/2 & 1\end{bmatrix}$

$p^3=\begin{bmatrix}1 & 0\\ 3/2 & 1\end{bmatrix}$

$P^4=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

$P^4=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$

$P^{50}=\begin{bmatrix}1 & 0\\ 25 & 1\end{bmatrix}$

4. Let $A=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$. If $A^{-1}=\alpha I+\beta A$, $\alpha, \beta \in R$, I is a 2×2 idently matrix, then $4(\alpha-\beta)$ is equal to :

[JEE Main Jul. 2021]

माना $A=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$ है। यदि $A^{-1}=\alpha I+\beta A$, $\alpha, \beta \in R$ हैं तथा I एक 2×2 का तत्समक आव्यूह है, तो $4(\alpha-\beta)$ बराबर है :

[JEE Main Jul. 2021]

(A) 2 (B) 4 (C) 8/3 (D) 5

Ans. (B)

Sol. $A=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$

$\text{tr}(A)=5$

$|A|=6$

$A^2-5A+6I=0$ (Characteristic equation)

$A^{-1}=\alpha I+\beta A$

$AA^{-1}=\alpha AI+\beta AA$

$\beta A^2+\alpha A-I=0$

$$\frac{1}{\beta}=\frac{-5}{\alpha}=\frac{6}{-1}\Rightarrow \alpha=\frac{5}{6},\beta=\frac{-1}{6}\Rightarrow 4(\alpha-\beta)=4$$

5. Let $A=\begin{pmatrix}1 & -1 & 0\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix}$ and $B=7A^{20}-20A^7+2I$, where I is an identity matrix of order 3×3. If $B=[b_{ij}]$, then b_{13} is equal to _____. **[JEE Main Jul. 2021]**

माना $A=\begin{pmatrix}1 & -1 & 0\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix}$ तथा $B=7A^{20}-20A^7+2I$ है, जहाँ I, 3×3 कोटि का तत्समक आव्यूह है। यदि $B=[b_{ij}]$, तो b_{13} बराबर है _____ **[JEE Main Jul. 2021]**

Ans. 910

Sol. $$A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}+\begin{bmatrix}0 & -1 & 0\\ 0 & 0 & -1\\ 0 & 0 & 0\end{bmatrix}$$

$A=I+C$

$$C^2=\begin{bmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix};C^3=O$$

$B=7A^{20}-20A^7+2I$

$=(I+C)^{20}-20(I+C)^7+2I$

$=7(I+20C+190C^2)-20(I+7C+21C^2)+2I$

$=-9I+910C^2$

$$= \begin{bmatrix} -9 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 910 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & 0 & 910 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{bmatrix}$$

6. Let $A = \{a_{ij}\}$ be a 3×3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j, \\ 2 & \text{if } i = j, \\ (-1)^{i+j} & \text{if } i > j, \end{cases}$ then $\det(3\ \text{Adj}(2\ A^{-1}))$ is equal to _____.

[JEE Main Jul. 2021]

माना $A = \{a_{ij}\}$ एक 3×3 आव्यूह है, जबकि $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j, \\ 2 & \text{if } i = j, \\ (-1)^{i+j} & \text{if } i > j, \end{cases}$ तो $\det(3\ \text{Adj}(2\ A^{-1}))$ बराबर है _____

[JEE Main Jul. 2021]

Ans. (108)

Sol. $$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$|A| = 4$

$$\left|3\text{adj}\left(2A^{-1}\right)\right| = 3^3 \left|\text{Adj}\left(2A^{-1}\right)\right|$$

$$= 3^3 \left|2A^{-1}\right|^2$$

$$= 3^3 \times 2^6 \times \left|A^{-1}\right|^2$$

$$= 3^3 \times 2^6 \times \frac{1}{|A|^2} = 108$$

7. Let $A=\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$. Then the number of 3×3 matrices B with entries from the set $\{1, 2, 3, 4, 5\}$ and satisfying $AB = BA$ is. **[JEE Main Jul. 2021]**

माना $A=\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$ एक 3×3 आव्यूह है। तो 3×3 आव्यूहों B, जिनकी प्रविष्टियाँ, समुच्चय, $\{1, 2, 3, 4, 5\}$ से हैं तथा जो $AB = BA$ को संतुष्ट करते हैं, की संख्या _____ **[JEE Main Jul. 2021]**

Ans. 3125

Sol. $A=\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$

Let $B=\begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{bmatrix}$

given $AB = BA$

$$\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{bmatrix}=\begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$\begin{bmatrix}b_1 & b_2 & b_3\\ a_1 & a_2 & a_3\\ c_1 & c_2 & c_3\end{bmatrix}=\begin{bmatrix}a_2 & a_1 & a_3\\ b_2 & b_1 & b_3\\ c_2 & c_1 & c_3\end{bmatrix}$$

By comparing both sides

$b_1 = a_2$ $\quad$ $c_1 = c_2$

$b_2 = a_1$ $\quad$ $c_3 = c_3$

$b_3 = a_3$

Now b_1, b_2, b_3 $\quad$ c_1, c_2, c_3 can be choosen by $= 5 \times 5 = 25$ ways

a_1, a_2, a_3 can be choosen by

$= 5 \times 5 \times 5 = 125$ way

total number of ways $= 125 \times 25 = 3125$ ways

8. Let $M=\left\{A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}: a,b,c,d \in \{\pm 3, \pm 2, \pm 1, 0\}\right\}$. Define $f: M \to Z$, as $f(A) = \det(A)$, for all $A \in M$, where Z is set of all integers. Then the number of $A \in M$ such that $f(A) = 15$ is equal to _____ .

[JEE Main Jul. 2021]

माना $M=\left\{A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}: a,b,c,d \in \{\pm 3, \pm 2, \pm 1, 0\}\right\}$ है। $f: M \to Z$ (Z ≡ सभी पूर्णांकों का समूह);

$f(A) = \det(A)$, सभी $A \in M$, द्वारा परिभाषित है। तो उन $A \in M$ की संख्या जिनके लिए $f(A) = 15$ है, है _____

[JEE Main Jul. 2021]

Ans. 16

Sol. $f(A) = \det(A) = 15$

$ad - bc = 15$

case – 1

$ad = 9$ and $bc = -6$

$ad = (3,3)$ or $(-3,-3)$ $bc = (2,-3),(-2,3),(-3,2),(3,-2)$

No. of ways = $(2) \times (4) = 8$

case – 2

$ad = 6$ and $bc = -9$

$ad = (2, 3)\ (3, 2)\ (-2, -3)\ (-3, -2)$ and $bc = (3, -3)\ (-3, 3)$

No. of ways = $4 \times 2 = 8$

Total no. of ways = $8 + 8 = 16$

9. If $A=\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + \ldots\ldots\ldots + A^{20}$, then the sum of all the elements of the matrix M is equal to _____.

[JEE Main Jul. 2021]

यदि $A=\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ तथा $M = A + A^2 + A^3 + \ldots\ldots\ldots + A^{20}$ हैं, तो आव्यूह M के सभी अवयवों का योगफल बराबर है _____

[JEE Main Jul. 2021]

Ans. 2020

Sol. $A=\begin{bmatrix}1 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix}$

$$A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}+\begin{bmatrix}0 & 1 & 1\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}$$

$A = I + B$ (say)

$$B^2=\begin{bmatrix}0 & 1 & 1\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}0 & 1 & 1\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}=\begin{bmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$B^3=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\ \therefore\ B^4 = B^5 = B^6 = \ldots\ldots\ldots\ldots = O$$

$M = A + A^2 + A^3 + \ldots\ldots\ldots + A^{20}$

$= (I + B) + (I + B)^2 + (I + B)^3 + \ldots\ldots\ldots + (I + B)^{20}$

$= 20\ I + (1 + {}^2C_1 + {}^3C_1 + \ldots\ldots\ldots {}^{20}C_1 +)\ B + (1 + {}^3C_2 + {}^4C_2 + \ldots\ldots\ldots + {}^{20}C_2)B^2$

$$= 20\ I + \left(\frac{20(20+1)}{2}\right)B + \left(\sum_{r=2}^{20}\frac{r(r-1)}{2}\right)B^2$$

$= 20\ I + 210\ B + 1330\ B^2$

$$=\begin{bmatrix}20 & 0 & 0\\ 0 & 20 & 0\\ 0 & 20 & 20\end{bmatrix}+\begin{bmatrix}0 & 210 & 210\\ 0 & 0 & 210\\ 0 & 0 & 0\end{bmatrix}+\begin{bmatrix}0 & 0 & 1330\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$=\begin{bmatrix}20 & 210 & 1540\\ 0 & 20 & 210\\ 0 & 0 & 20\end{bmatrix}$$

Sum of all the elements

$= 3 \times 20 + 2 \times 210 + 1540$

$= 2020$

10. $A=\begin{pmatrix}\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}$, $B=\begin{pmatrix}1 & 0\\ i & 1\end{pmatrix}$, $i=\sqrt{-1}$ and $Q=A^TBA$, then the inverse of the matrix $AQ^{2021}A^T$ is equal to :

[JEE Main Aug. 2021]

यदि $A=\begin{pmatrix}\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}$, $B=\begin{pmatrix}1 & 0\\ i & 1\end{pmatrix}$, $i=\sqrt{-1}$ तथा $Q=A^TBA$ है, तो आव्यूह $AQ^{2021}A^T$ का व्युत्क्रम बराबर है :

[JEE Main Aug. 2021]

(A) $\begin{pmatrix}\frac{1}{\sqrt{5}} & -2021\\ 2021 & \frac{1}{\sqrt{5}}\end{pmatrix}$ (2) $\begin{pmatrix}1 & 0\\ -2021i & 1\end{pmatrix}$ (3) $\begin{pmatrix}1 & 0\\ 2021i & 1\end{pmatrix}$ (4) $\begin{pmatrix}1 & -2021i\\ 0 & 1\end{pmatrix}$

Ans. (B)

Sol. $A=\begin{pmatrix}\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}$; $AA^T=I$

$Q=A^TBA$

$Q^2=A^TBAA^TBA=A^TB^2A$

||ly $Q^{2021}=A^TB^{2021}A$

Let $P=AQ^{2021}A^T$

$=AA^TB^{2021}AA^T$

$P=B^{2021}$

$B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{pmatrix}1 & 0\\ i & 1\end{pmatrix}=I+C$

$C^2=O$

$B^{2021}=(I+C)^{2021}=I+2021\,C$

$P=B^{2021}=\begin{pmatrix}1 & 0\\ 2021i & 1\end{pmatrix}$; $|P|=1$

$$P^{-1} = \frac{adj(P)}{|P|} = \begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$$

11. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Then $A^{2025} - A^{2020}$ is equal to : **[JEE Main Aug. 2021]**

माना $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ है तो $A^{2025} - A^{2020}$ बराबर है : **[JEE Main Aug. 2021]**

(A) A^6 (B) $A^5 - A$ (C) A^5 (D) $A^6 - A$

Ans. (D)

Sol. $A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $A^6 = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^{2025} = \begin{bmatrix} 1 & 0 & 0 \\ 2024 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } A^{2020} = \begin{bmatrix} 1 & 0 & 0 \\ 2019 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^{2025} - A^{2020} = \begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = A^6 - A$$

12. If the matrix $A = \begin{pmatrix} 0 & 2 \\ K & -1 \end{pmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of K is : **[JEE Main Aug. 2021]**

यदि आव्यूह $A = \begin{pmatrix} 0 & 2 \\ K & -1 \end{pmatrix}$ समीकरण $A(A^3 + 3I) = 2I$ को संतुष्ट करता है, तो K का मान है : **[JEE Main Aug. 2021]**

(A) $-\frac{1}{2}$ (2) 1 (3) -1 (4) $\frac{1}{2}$

Ans. (D)

Sol. $A(A^3 + 3I) = 2I$

$A^4 + 3A - 2I = 0$

$A = \begin{pmatrix} 0 & 2 \\ k & -1 \end{pmatrix}$

$tr(A) = -1$

$|A| = -2K$

Characteristic equation

$A^2 + A - 2KI = 0$

$A^2 = 2KI - A$

$A^4 = 4K^2I - 4KA + A^2$

$A^4 = 4K^2I - 4KA + (2KI - A)$

$A^4 = (4K^2 + 2K)I - (4K + 1)A$

$A^4 + 3A - 2I = 0$

$(4K^2 + 2K)I - (4K + 1)A + 3A - 2I = 0$

$(4K - 2)A = (4K^2 + 2K - 2)I$

$(2K - 1)A = (2K - 1)(K + 1)I$

$K = \frac{1}{2}$

13. Let $A = \begin{pmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{pmatrix}$, where [t] denotes the gretest integer less than or equal to t, If det (A) = 192, then the set of values of x is the interval : **[JEE Main Aug. 2021]**

माना $A = \begin{pmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{pmatrix}$, जहाँ [t] - महत्तम पूर्णांक $\leq$ t को दर्शाता है। यदि det (A) = 192 है, तो x के मानों का समुच्चय निम्न में से कौनसा अंतराल है? **[JEE Main Aug. 2021]**

(A) [60, 61] (B) [62, 63] (C) [68, 69] (D) [65, 66]

Ans. (B)

Sol. $|A|=\begin{vmatrix}[x]+1 & [x]+2 & [x]+3\\ [x] & [x]+3 & [x]+3\\ [x] & [x]+2 & [x]+4\end{vmatrix}=192$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\begin{vmatrix}[x]+1 & [x]+2 & [x]+3\\ -1 & 1 & 0\\ -1 & 0 & 1\end{vmatrix}=192$$

$[x]=62$

$x \in [62,63)$

14. Let A be a 3 × 3 real matrix. If det(2 Adj) (2 Adj(Adj) (2A)))) = 2^{41}, then the value of det (A^2) equals ____.

[JEE Main Aug. 2021]

माना A एक 3 × 3 वास्तविक आव्यूह है। यदि det(2 Adj) (2 Adj(Adj) (2A)))) = 2^{41} है, तो det (A^2) का मान बराबर है____

[JEE Main Aug. 2021]

Ans.

Sol. adj(2A) = 2^2 adjA

$\Rightarrow$ adj(adj (2A)) = adj(4 adj A) = 16 adj(adj A)

= 16|A| A

$\Rightarrow$ adj (32 |A| A) = $(32\ |A|)^2$ adj A

$12(32|A|)^2\ |adj\ A| = 2^3\ (32|A|)^6\ |adj\ A|$

$_2{}^3.2^{30}|A|^6\ .\ |A|^2 = 2^{41}$

$|A|^8 = 2^8 \Rightarrow |A| = \pm 2$

$|A|^2 = |A|^2 = 4$

15. The number of elements in the set, $\left\{A=\begin{pmatrix}a & b\\ 0 & d\end{pmatrix}: a,b,d \in \{-1,0,1\} \text{ and } (I-A)^3 = I-A^3\right\}$ where I is 2 × 2 identity matrix, is_____.

[JEE(Main)Aug.2021]

समुच्चय $\left\{A=\begin{pmatrix}a & b\\ 0 & d\end{pmatrix}: a,b,d \in \{-1,0,1\} \text{ तथा } (I-A)^3 = I-A^3\right\}$, I , 2 × 2 का तत्समक आव्यूह है में अवयवों की संख्या gS_____

[JEE(Main)Aug.2021]

Ans. (8)

Sol. $(I-A)^3 = I - A^3$

$(I-A)(I-A)(I-A) = I - A^3$

$(I - A - A + A^2)(I-A) = I - A^3$

$(I - 2A + A^2)(I-A) = I - A^3$

$I - 2A + A^2 - A + 2A^2 - A^3 = I - A^3$

$I - 3A + 3A^2 - A^3 = I - A^3$

$3A^2 - 3A = 0$

$A^2 - A = 0$

$A^2 - A$

$$\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}\begin{bmatrix} a & b \\ 0 & d \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$$

$$\begin{bmatrix} a^2 & ab+bd \\ 0 & d^2 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$$

$a^2 = a$ $\qquad$ $ab + bd = b$

$b(a + d) = b$

$d^2 = d$ $\qquad$ $b(a + d - 1) = 0$

$a(a - 1) = 0$ $\qquad$ $b = 0$ or $a + d = 1$

$a = 0, 1$ $\qquad$ $d = 0, 1$

Total 8 Matrices

16. Let $S = \left\{\sqrt{n} : 1 \le n \le 50 \text{ and } n \text{ is odd}\right\}$. Let $a \in S$ and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$. If $\sum_{a \in S} \det(\text{adj } A) = 100\lambda$ then λ is equal to : **[JEE(Main)June2022]**

माना $S = \{\sqrt{n} : 1 \le n \le 50$ तथा n विषम है$\}$ माना $a \in S$ तथा $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$ हैं। यदि $\sum_{a \in S} \det(\text{adj } A) = 100\lambda$ है, तो λ बराबर है : **[JEE(Main)June2022]**

(1) 218 (2) 221 (3) 663 (4) 1717

Ans. Official Answer NTA (2)

Sol. $S = \left\{1, \sqrt{3}, \sqrt{5}, \sqrt{7}, \ldots\ldots\sqrt{49}\right\} \equiv \left\{\sqrt{2r-1} : 1 \le r \le 25, r \in N\right\}$

$$A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$$

$|A| = 1 + a^2$

$|\text{adj}(A)| = |A|^2 = (1 + a^2)^2$

$$\sum_{a\in S}\det\left(\text{adj}(A)\right) = \sum_{r=1}^{2}(1+2r-1)^2 = \sum_{r=1}^{25}4r^2$$

$$= 4\sum_{r=1}^{25}\left(r^2\right)$$

$$100\lambda = \frac{4(25)(26)(51)}{6}$$

$\Rightarrow \lambda = 221$

17. Let A be a 3×3 real matrix such that $A\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix}; A\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix}$ and $A\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\2\end{pmatrix}$. If $X = (x_1, x_2, x_3)^T$ and I is an identity matrix of order 3, then the system $(A - 2I)X = \begin{pmatrix}4\\1\\1\end{pmatrix}$ has : **[JEE(Main)June2022]**

(1) no solution (2) infinitely many solutions

(3) unique solution (4) exactly two solutions

माना A एक 3×3 का वास्तविक आव्यूह है, जिसके लिए $A\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix}; A\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix}$ तथा $A\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\2\end{pmatrix}$ है।

; fn $X = (x_1, x_2, x_3)^T$ है तथा I, कोटि 3 का तत्समक आव्यूह है, तो निकाय $(A - 2I)X = \begin{pmatrix}4\\1\\1\end{pmatrix}$:

[JEE(Main)June2022]

(1) कोई हल नहीं है (2) के अनंत हल हैं

(3) का केवल एक हल है (4) के मात्र दो हल हैं

Ans. Official Answer NTA (2)

Sol. $$A = \begin{bmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{bmatrix}$$

$$A\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix} = \begin{bmatrix}1\\1\\2\end{bmatrix}$$

$\Rightarrow \quad c_1 = 1, c_2 = 1, c_3 = 2$

$$A\begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}c_1 + a_1\\c_2 + a_2\\c_3 + a_3\end{bmatrix} = \begin{bmatrix}-1\\0\\1\end{bmatrix}$$

$\Rightarrow a_1 = -2, a_2 = -1, a_3 = -1$

$$A\begin{bmatrix}1\\1\\0\end{bmatrix} = \begin{bmatrix}a_1 + b_1\\a_2 + b_2\\a_3 + b_3\end{bmatrix} = \begin{bmatrix}1\\1\\0\end{bmatrix}$$

$\Rightarrow \quad b_1 = 3, b_2 = 2, b_3 = 1$

$$\Rightarrow \quad A = \begin{bmatrix}-2 & 3 & 1\\-1 & 2 & 1\\-1 & 1 & 2\end{bmatrix}$$

$$\Rightarrow \quad A - 2I = \begin{bmatrix}-4 & 3 & 1\\-1 & 0 & 1\\-1 & 1 & 0\end{bmatrix}$$

$|A - 2I| = 0$

$$\text{Now, } \begin{bmatrix}-4 & 3 & 1\\-1 & 0 & 1\\-1 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}4\\1\\1\end{bmatrix}$$

$-4x_1 + 3x_2 + x_3 = 4$ ....(1)

$-x_1 + x_3 = 1$ .....(2)

$-x_1 + x_2 = 1$ ....(3)

$(1) - [(2) + 3(3)]$

$0 = 0 \Rightarrow$ infinite solutions

18. Let $A = \begin{bmatrix}0 & -2\\2 & 0\end{bmatrix}$. If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2 is :

[JEE(Main)June2022]

(1) a non-identity symmetric matrix

(2) a skew-symmetric matrix

(3) neither symmetric nor skew-symmetric matrix

(4) an identity matrix

माना $A=\begin{bmatrix}0 & -2\\ 2 & 0\end{bmatrix}$ है। यदि दो आव्यूह M तथा N $M=\sum_{k=1}^{10}A^{2k}$ तथा $N=\sum_{k=1}^{10}A^{2k-1}$ द्वारा दिए गए हैं, तो MN^2 :

[JEE(Main)June2022]

(1) एक सममित आव्यूह है, तो तत्समक नहीं है

(2) एक विषम–सममित आव्यूह है

(3) न तो सममित आव्यूह है न ही विषम सममित

(4) एक तत्समक आव्यूह है

Ans. Official Answer NTA (1)

Sol. $A^2=\begin{bmatrix}0 & -2\\ 2 & 0\end{bmatrix}.\begin{bmatrix}0 & -2\\ 2 & 0\end{bmatrix}=\begin{bmatrix}-4 & 0\\ 0 & -4\end{bmatrix}=-4I$. (symmetric)

and $A^3=-4A\left(\text{skew symmetric}\right)$

$\Rightarrow M=\sum_{k=1}^{10}A^{2K}=\left[(-4)+(-4)^2+(-4)^3+.....+(-4)^{10}\right]$

$=-4\lambda I$ is symmetric

$\Rightarrow N=\sum_{k=1}^{10}A^{2K-1}=A\left[1+(-4)+(-4)^2+.....+(-4)^9\right]$

$=\lambda A$ is skew symmetric

where $\lambda=\{1+(-4)+(-4)^3+.....+(-4)^9\}$

Now $MN=-4\lambda^2 A=NM$

$\Rightarrow MN^2=(MN)N=(NM)N=N(MN)=N(NM)=N^2M$

Hence $(MN^2)^T=(N^2)^T M^T=(N^T)^2 M^T=(-N)^2 M=N^2M$

$\Rightarrow MN^2$ is symmetric matrix

19. Let A be a 3×3 invertible matrix. If $|\text{adj}(24A)|=|\text{adj}(3\ \text{adj}(2A))|$, then $|A|^2$ is equal to :

[JEE(Main)June2022]

माना A एक 3×3 का व्युत्क्रमणीय आव्यूह है। यदि $|\text{adj}(24A)|=|\text{adj}(3\ \text{adj}(2A))|$ है, तो $|A|^2$ बराबर है :

[JEE(Main)June2022]

(1) 6^6 (2) 2^{12} (3) 2^6 (4) 1

Ans. Official Answer NTA (3)

Sol. $|\text{adj}(24A)|=|\text{adj}\ 3(\text{adj}\ 2A)|$

$\Rightarrow |24A|^2=|3\ \text{adj}(2A)|^2$

$\Rightarrow (24^3|A|)^2=(3^3|\text{adj}(2A)|)^2$

$=3^6(|2A|^2)^2$

$\Rightarrow 24^6|A|^2=(24^3|A|)^2=3^6\times 2^{12}|A|^4$

$\Rightarrow |A|^2=\dfrac{24^6}{3^6\times 2^{12}}=64$

20. Let A and B be two 3×3 matrices such that $AB = I$ and $|A| = \frac{1}{8}$. Then $|\text{adj}(B \text{ adj}(2A))|$ is equal to :

[JEE(Main)June2022]

माना A तथा B दो 3×3 आव्यूह हैं तथा $AB = I$ और $|A| = \frac{1}{8}$ हैं। तो $|\text{adj}(B \text{ adj}(2A))|$ बराबर है :

[JEE(Main)June2022]

(1) 16 (2) 32 (3) 64 (4) 128

Ans. Official Answer NTA (3)

Sol. $|AB| = 1$

$|A|\,|B| = 1$

$\Rightarrow |B| = 8$

$$\left|\text{adj}\left(B \text{ adj}(2A)\right)\right| = \left|B \text{ adj}(2A)\right|^2$$

$$= |B|^2 \,|\text{adj}(2A)|^2$$

$$= |B|^2 \left(|2A|^2\right)^2$$

$$= |B|^2 \,|2A|^4$$

$$= |B|^2 \,(2^3)^4 \,|A|^4$$

$$= 8^2.8^4.\left(\frac{1}{8}\right)^4$$

$$= 8^2 = 64$$

21. Let A be a matrix of order 3×3 and $\det(A) = 2$. Then $\det(\det(A) \text{ adj}(5 \text{ adj}(A^3)))$ is equal to _____.

[JEE(Main)June2022]

माना A एक क 3×3 कोटि का आव्यूह है। माना $\det(A) = 2$ है। तो $\det(\det(A) \text{ adj}(5 \text{ adj}(A^3)))$ बराबर है _____.

[JEE(Main)June2022]

(1) 512×10^6 (2) 256×10^6 (3) 1024×10^6 (4) 256×10^{11}

Ans. Official Answer NTA (1)

Sol. $|A| = 2$

$$\left||A| .\text{adj}\left(5 \text{ adj}\left(A^3\right)\right)\right| = 2^3 \left|5^2 .\text{adj}\left(\text{adj } A^3\right)\right|$$

$= 2^3 \cdot 5^6 \,|\text{adj}(\text{adj } A^3)|$

$= 2^3.5^6 \,|A^3|^4$

$= 2^3.\, 5^6 .2^{12}$

$= 2^9 \cdot 10^6$

$= 512 \times 10^6$

22. Let $A=\left[a_{ij}\right]$ be a square matrix of order 3 such that $a_{ij}=2^{j-i}$, for all i, j = 1, 2, 3. Then, the matrix $A^2+A^3+.....+A^{10}$ is equal to : **[JEE(Main)June2022]**

माना कोटि 3 का एक वर्ग आव्यूह $A=\left[a_{ij}\right]$ है, जिसमें $a_{ij}=2^{j-i}$, $\forall$ i, j = 1, 2, 3 हैं। तो आव्यूह $A^2+A^3+.....+A^{10}$ बराबर है : **[JEE(Main)June2022]**

(1) $\left(\frac{3^{10}-3}{2}\right)A$ (2) $\left(\frac{3^{10}-1}{2}\right)A$ (3) $\left(\frac{3^{10}+1}{2}\right)A$ (4) $\left(\frac{3^{10}+3}{2}\right)A$

Ans. Official Answer NTA (1)

Sol. $A=\begin{bmatrix}1 & 2 & 4\\ \frac{1}{2} & 1 & 2\\ \frac{1}{4} & \frac{1}{2} & 1\end{bmatrix}=\frac{1}{4}\begin{bmatrix}4 & 8 & 16\\ 2 & 4 & 8\\ 1 & 2 & 4\end{bmatrix}$

$$A^2=\frac{1}{16}\begin{bmatrix}4 & 8 & 16\\ 2 & 4 & 8\\ 1 & 2 & 4\end{bmatrix}\begin{bmatrix}4 & 8 & 16\\ 2 & 9 & 8\\ 1 & 2 & 4\end{bmatrix}=\frac{1}{16}\begin{bmatrix}48 & 96 & 192\\ 24 & 98 & 96\\ 12 & 24 & 48\end{bmatrix}=3\begin{bmatrix}1 & 2 & 4\\ \frac{1}{2} & 1 & 2\\ \frac{1}{4} & \frac{1}{2} & 1\end{bmatrix}$$

$A^2=3A$

$A^3=A.A^2=A(3A)=3A^2=33A=3^2A$

$A^4=3^3A+.....+A^{10}=3^9A$

$$A^2+.....+A^{10}=(3+3^2+.....+3^9)=3.\frac{3^9-1}{2}A$$

$$=\frac{3^{10}-3}{2}A$$

23. Let $A=\begin{pmatrix}2 & -1\\ 0 & 2\end{pmatrix}$. If $B=I-{}^5C_1(\text{adj}\,A)+{}^5C_2(\text{adj}\,A)^2-.....-{}^5C_5(\text{adj}\,A)^5$ then the sum of all elements of the matrix B is : **[JEE(Main)June2022]**

माना $A=\begin{pmatrix}2 & -1\\ 0 & 2\end{pmatrix}$ है। यदि $B=I-{}^5C_1(\text{adj}\,A)+{}^5C_2(\text{adj}\,A)^2-.....-{}^5C_5(\text{adj}\,A)^5$ है, तो आव्यूह B के सभी v o; okad k ; kx Qy gS: **[JEE(Main)June2022]**

(1) –5 (2) –6 (3) –7 (4) –8

Ans. Official Answer NTA (3)

Sol. Given $A=\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}$

$B = I - {}^5C_1\,(\text{adj } A) + {}^5C_2\,(\text{adj } A)^2 - \ldots\ldots\ldots - {}^5C_5\,(\text{adj } A)^5$

Then $B=\left[I-\text{adj}A\right]^5$

$$=\left[\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}-\begin{bmatrix}2 & 0\\ 1 & 2\end{bmatrix}\right]^5$$

$$B=\begin{bmatrix}-1 & 0\\ -1 & -1\end{bmatrix}^5 \equiv$$

$$B=\begin{bmatrix}-1 & 0\\ -5 & -1\end{bmatrix}^5 \Rightarrow$$

Sum if element = – 1 –5 – 1= – 7

24. Let A be a 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is _____. **[JEE(Main)June2022]**

माना A एक 3×3 आव्यूह है, जिसके अवयव समुच्चय $\{-1, 0, 1\}$ से हैं। इस प्रकार के सभी आव्यूहों A, जिनके सभी अवयवों का ; ksQy 5 है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (414)

Sol. Sol. Case-I: $1 \to 7$ times

and $-1 \to 2$ times

$$\text{number of possible matrix} = \frac{9!}{7!2!} = 36$$

Case-II: $1 \to 6$ times,

$-1 \to 1$ times

and $0 \to 2$ times

$$\text{number of possible matrix} = \frac{9!}{6!2!} = 252$$

Case-III: $1 \to 5$ times,

and $0 \to 4$ times

$$\text{number of possible matrix} = \frac{9!}{5!4!} = 126$$

Hence total number of all such matrix A = 414

25. Let $A=\begin{pmatrix}2 & -2\\ 1 & -1\end{pmatrix}$ and $B=\begin{pmatrix}-1 & 2\\ -1 & 2\end{pmatrix}$. Then the number of elements in the set $\{(n, m) : n, m \in \{1, 2, \ldots\ldots,10\}$ and $nA^n + mB^m = I\}$ is _____. **[JEE(Main)June2022]**

माना $A=\begin{pmatrix}2 & -2\\ 1 & -1\end{pmatrix}$ तथा $B=\begin{pmatrix}-1 & 2\\ -1 & 2\end{pmatrix}$ हैं। तो समुच्चय $\{(n, m) : n, m \in \{1, 2, ...,10\}$ तथा $nA^n + mB^m = I\}$ है में

vo; okLrfod la[;k gS_____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (1)

Sol. Here $A^2 = A$ and $B^2 = B$

$\Rightarrow A^n = A$ and $B^m = B$

$$n\begin{pmatrix}2 & -2\\ 1 & -1\end{pmatrix}+m\begin{pmatrix}-1 & 2\\ -1 & 2\end{pmatrix}=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

$$\left.\begin{aligned}2n-m&=1\\ -2n+2m&=0\\ n-m&=0\\ -n+2m&=1\end{aligned}\right\}\Rightarrow n=1\ \&\ m=1$$

$(n, m) = (1, 1)$

26. Let $X=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}, Y=\alpha I+\beta X+\gamma X^2$ and $Z=\alpha^2 I-\alpha\beta X+\left(\beta^2-\alpha\gamma\right)X^2, \alpha,\beta,\gamma\in R$.

If $Y^{-1}=\begin{bmatrix}1/5 & -2/5 & 1/5\\ 0 & 1/5 & -2/5\\ 0 & 0 & 1/5\end{bmatrix}$, then $(\alpha-\beta+\gamma)^2$ is equal to _____. **[JEE(Main)June2022]**

माना $X=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}, Y=\alpha I+\beta X+\gamma X^2$ तथा $Z=\alpha^2 I-\alpha\beta X+\left(\beta^2-\alpha\gamma\right)X^2, \alpha,\beta,\gamma\in R$ हैं। यदि

$Y^{-1}=\begin{bmatrix}1/5 & -2/5 & 1/5\\ 0 & 1/5 & -2/5\\ 0 & 0 & 1/5\end{bmatrix}$, है, तो $(\alpha-\beta+\gamma)^2$ बराबर है _____. **[JEE(Main)June2022]**

Ans. Official Answer NTA (100)

Sol. $X=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}$

$y = \alpha I + \beta x + \gamma x^2$

$$X^2=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}=\begin{bmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$Y^2=\begin{bmatrix}\alpha & 0 & 0\\ 0 & \alpha & 0\\ 0 & 0 & \alpha\end{bmatrix}+\begin{bmatrix}0 & \beta & 0\\ 0 & 0 & \beta\\ 0 & 0 & 0\end{bmatrix}+\begin{bmatrix}0 & 0 & \gamma\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$Y=\begin{bmatrix}\alpha & \beta & \gamma\\ 0 & \alpha & \beta\\ 0 & 0 & \alpha\end{bmatrix}$$

$\therefore$ Y. $Y^{-1}=I$

$$\begin{bmatrix}\alpha & \beta & \gamma\\ 0 & 0 & \beta\\ 0 & 0 & \alpha\end{bmatrix}.\begin{bmatrix}1/5 & -2/5 & 1/5\\ 0 & 1/5 & -2/5\\ 0 & 0 & 1/5\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$\frac{\alpha}{5}=1\Rightarrow\alpha=5$$

$$\frac{-2\alpha}{5}+\frac{\beta}{5}=0$$

$$\frac{\beta}{5}=2\Rightarrow\beta=10$$

$$\frac{\alpha}{5}-\frac{2\beta}{5}+\frac{\gamma}{5}=0$$

$$\gamma=15$$

$$(\alpha-\beta+\gamma)^2=(5-10+15)^2=10^2=100$$

27. The positive value of determinant of the matrix A, whose Adj (Adj (A)) $=\begin{pmatrix}14 & 28 & -14\\ -14 & 14 & 28\\ 28 & -14 & 14\end{pmatrix}$, is _____.

[JEE(Main)June2022]

यदि Adj (Adj (A)) $=\begin{pmatrix}14 & 28 & -14\\ -14 & 14 & 28\\ 28 & -14 & 14\end{pmatrix}$ है, तो आव्यूह A के सारणिक का धनात्मक मान है _____

[JEE(Main)June2022]

Ans. Official Answer NTA (14)

Sol. $\text{adj}(\text{adj}(A))=\begin{bmatrix}14 & 28 & -14\\ -14 & 14 & 28\\ 28 & -14 & 14\end{bmatrix}$

$$\det.\left(|A|' A\right)=\begin{vmatrix}14 & 28 & -14\\ -14 & 14 & 28\\ 28 & -14 & 14\end{vmatrix}$$

$R_1 \to R_1 + R_2 + R_3$

$$|A|^3 |A| = \begin{vmatrix}28 & 28 & 28\\ -14 & 14 & 28\\ 28 & -14 & 14\end{vmatrix}$$

$C_1 \to C_1 - C_2,\ C_2 \to C_2 - C_3$

$$|A|^4 = \begin{vmatrix}0 & 0 & 28\\ -28 & -14 & 28\\ 42 & -28 & 14\end{vmatrix}$$

$|A|^4 = 28((28)^2 + 14 \times 42)$
$|A|^4 = 28 \times 14 \times 14\ (4 + 3)$
$|A|^4 = 4 \times 7 \times 2 \times 7 \times 2 \times 7 \times 7$
$|A| = 14$

28. Let A be a matrix of order 2×2, whose entries are from the set $\{0, 1, 2, 3, 4, 5\}$. If the sume of all entries of A is a prime number p, $2 < p < 8$, then the number of such matrices A is _____. **[JEE(Main)June2022]**

माना 2×2 कोटि का एक आव्यूह A है, जिसके अवयव समुच्चय $\{0, 1, 2, 3, 4, 5\}$ में से हैं। यदि A के सभी अवयवों का योग एक vHkkT; l a[; kp, $2 < p < 8$ है, तो ऐसे आव्यूहों की संख्या है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (180)

Sol. $a + b + c + d = 3$

a, b, c, d, = {0, 1, 2, 3}

$\text{No. of ways} = {}^{3+4-1}C_{4-1} = {}^{6}C_3 = 20$

$a + b + c + d = 5$

$a, b, c, d, \in \{0, 1, 2, 3, 4, 5\}$

$\text{No. of ways} = {}^{4+5-1}C_{4-1} = {}^{8}C_3$

$= 56$

$a + b + c + d = 7$

a, b, c, d = {0, 1, 2, 3, 4, 5}

$\text{Coeff of } x^7 \text{ in } (1 + x + ... + x^5)^4$

$$x^7 \text{ in } \left(\frac{1-x^6}{1-x}\right)^4$$

$$x^7 \text{ in } \left(1-x^6\right)^4 \left(1-x\right)^{-4}$$

$$x^7 \text{ in } \left(1-x\right)^{-4} - 4 + x^1 \text{ in } \left(1-x\right)^{-4}$$

$${}^{4+7-1}C_{4-1} - 4 \times {}^{4+1-1}C_{4-1}$$

$^{10}C_3 - 4.\,^4C_3$

$120 - 16 = 104$

Hence total no. of ways = 20 + 56 + 104 = 180

29. Let $A = \begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ where $i = \sqrt{-1}$. Then, the number of elements in the set $\left\{n \in \{1,2,.....,100\} : A^n = A\right\}$ is _____. **[JEE(Main)June2022]**

माना $A = \begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ है, जहाँ$i = \sqrt{-1}$ है। तो समुच्चय $\left\{n \in \{1,2,.....,100\} : A^n = A\right\}$ में अवयवों की संख्या है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (25)

Sol. $A^2 = \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix}\begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix} = \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix}$

$A^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

So $A^5 = A$, $A^9 = A$ and So on.

$\therefore$ n = 1, 5, 9 ....., 97

Number of values = 25

30. Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number an $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is _____. **[JEE(Main)June2022]**

माना $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$ है, जहाँ α एक शून्येतर वास्तविक संख्या है तथा $N = \sum_{k=1}^{49} M^{2k}$ यदि $(I - M^2)N = -2I$ है, तो α का धन पूर्णांक मान है _____. **[JEE(Main)June2022]**

Ans. Official Answer NTA (1)

Sol. Given $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}, M^2 = \begin{bmatrix} -\alpha^2 & 0 \\ 0 & -\alpha^2 \end{bmatrix} = -\alpha^2 I$

$N = M^2 + M^4 + ...+ M^{98}$

$N = [-\alpha^2 + \alpha^4 - \alpha^6 + ...]I$

$I - M^2 = (1 + \alpha^2)I$

$(I - M^2)N = 1 + -\alpha^2(\alpha^{98} + 1) = -2$

$\alpha = 1$

31. Let A be a 2×2 matrix with $\det(A) = -1$ and $\det((A+I)(\text{Adj}(A) + I)) = 4$. Then the sum of the diagonal elements of A can be : **[JEE(Main)July2022]**

माना A एक 2×2 का आव्यूह है जिसके लिए $\det(A) = -1$ तथा $\det((A+I)(\text{Adj}(A)+I)) = 4$ हैं। तो A के विकर्ण d sv o; okad k ; kx gksl d r k gS: **[JEE(Main)July2022]**

(1) -1 (2) 2 (3) 1 (4) $-\sqrt{2}$

Ans. Official Answer NTA (2)

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; ad - bc = -1$

$|A + I|\,|\text{adj}\, A + I| = 4$

$\Rightarrow ad - bc + a + d + 1 = 2 \text{ or } -2$

$a + d = 2 \text{ or } -2$

32. Let $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the value of the A'BA is : **[JEE(Main)July2022]**

माना $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$ हैं, तो A'BA का मान है : **[JEE(Main)July2022]**

(1) 1224 (2) 1042 (3) 540 (4) 539

Ans. Official Answer NTA (4)

Sol. $A'BA = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$= [9^2 + 12^2 - 15^2 - 10^2 + 13^2 + 16^2\ 11^2 - 14^2 + 17^2] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$= [9^2 + 12^2 - 15^2 - 10^2 + 13^2 + 16^2 + 11^2 - 14^2 + 17^2]$

$= [539$

33. Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in R$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to : **[JEE(Main)July2022]**

माना $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$ है तथा माना $\alpha, \beta \in R$ इस प्रकार हैं कि $\alpha A^2 + \beta A = 2I$ है। तो $\alpha + \beta$ का मान है : **[JEE(Main)July2022]**

(1) –10 (2) –6 (3) 6 (4) 10

Ans. Official Answer NTA (4)

Sol. Characteristic equation of matric A

$|A-\lambda I|=0$

$\begin{vmatrix}1-\lambda & 2\\ -2 & -5-\lambda\end{vmatrix}=0$

$\Rightarrow \lambda^2+4\lambda=1$

$\Rightarrow A^2+4A=I$

$\Rightarrow 2A^2+8A=2I$

$\Rightarrow 2A^2+8A=2I$ ..... (1)

Given that $\alpha A^2+\beta A=2I$ ..... (2)

Comparing equation (1) & (2) we get

$\alpha=2, \quad \beta=8$

$\therefore\ \alpha+\beta=10$

34. Let $A=\begin{pmatrix}4 & -2\\ \alpha & \beta\end{pmatrix}$. If $A^2+\gamma A+18I=O$, then det (A) is equal to : **[JEE(Main)July2022]**

माना $A=\begin{pmatrix}4 & -2\\ \alpha & \beta\end{pmatrix}$ है। यदि $A^2+\gamma A+18I=O$ है, तो det (A) बराबर है : **[JEE(Main)July2022]**

(1) –18 (2) 18 (3) –50 (4) 50

Ans. Official Answer NTA (2)

Sol. The characteristic equation for A is $A|A-\lambda I|=0$

$\Rightarrow \begin{vmatrix}4-\lambda & -2\\ \alpha & \beta-\lambda\end{vmatrix}=0$

$\Rightarrow (4-\lambda)(\beta-\lambda)+2\alpha=0$

$\Rightarrow \lambda^2-(\beta+4)\lambda+4\beta+2\alpha=0$

Put $\lambda=A$

$A^2-(\beta-4)A+(4\beta+2\alpha)I=0$

On comparison

$-9(\beta+4)=\gamma$ & $4\beta+2\alpha=18$

and $|A|=4\beta+2\alpha=18$

35. Let the matrix $A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ and the matrix $B_0=A^{49}+2A^{98}$. If $B_n=\text{Adj}(B_{n-1})$ for all $n\geq 1$, then $\det(B_4)$ is equal to : **[JEE(Main)July2022]**

माना आव्यूह $A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ तथा आव्यूह $B_0=A^{49}+2A^{98}$ हैं। यदि सभी $n\geq 1$ के लिए $B_n=\text{Adj}(B_{n-1})$ है, तो $\det(B_4)$ बराबर है : **[JEE(Main)July2022]**

(1) 3^{28} (2) 3^{30} (3) 3^{32} (4) 3^{36}

Ans. Official Answer NTA (3)

Sol. $A^2=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$

$=\begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$

$a\leftrightarrow R_2$

$=\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$

$R_2\leftrightarrow R_3$

$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=I$

$B_0=A^{49}+2A^{98}$

$=A+2I$

$B_n=\text{Adj}\,(B_n-1)$

$B_4=\text{Adj}(\text{Adj}(\text{Adj}(\text{Adj}B_0))$

$=|B_0|^{(n-1)^4}$

$=|B_0|^{16}$

$$B_0 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$= 2(4 - 0) - 1(0 - 1)$

$= 9$

$B_4 (9)^{16} = (3)^{32}$

36. Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true? **[JEE(Main)July2022]**

(1) $A^4 - B^4$ is a symmetric matrix (2) $AB - BA$ is a symmetric matrix

(3) $A^5 - B^5$ is a skew-symmetric matrix (4) $AB - BA$ is a skew-symmetric matrix

माना A तथा B क्रमशः कोई भी दो 3×3 की सममित तथा विषम सममित आव्यूह हैं। तो निम्न में से कौनसा सत्य नहीं gS **[JEE(Main)July2022]**

(1) $A^4 - B^4$ एक सममित आव्यूह है (2) $AB - BA$ एक सममित आव्यूह है

(3) $A^5 - B^5$ एक विषम सममित आव्यूह है (4) $AB - BA$ एक विषम सममित आव्यूह है

Ans. Official Answer NTA (3)

Sol. Given that $A^T = A$, $B^T = -B$

(A) $C = A^4 - B^4$

$C^T = (A^4 - B^4) = (A^4)^T - (B^4)^T = A^4 - B^4 = C$

(B) $C = AB - BA$

$C^T = (AB - BA)^T = (AB)^T - (BA)^T$

$= B^T A^T - A^T B^T = -BA + AB = C$

(C) $C = B^5 - A^5$

$C^T = (B^5 - A^5)^T = (B^5)^T - (A^5)^T = -B^5 - A^5$

(D) $C = AB + BA$

$C^T = (AB + BA)^T = (AB)^T + (BA)^T$

$= -BA - AB = -C$

$\therefore$ Option C is not true.

37. Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then :

[JEE(Main)July2022]

(1) The system of linear equations AX = 0 has a unique solution

(2) The system of linear equations AX = 0 has infinitely many solution

(3) B is an invertible matrix

(4) adj(A) is an invertible matrix

माना A तथा B, 3×3 के दो अशून्य वास्तविक आव्यूह हैं जिसके लिए AB एक शून्य आव्यूह है। तो :

[JEE(Main)July2022]

(1) रैखिक समीकरण निकाय AX = 0 का अद्वितीय हल है

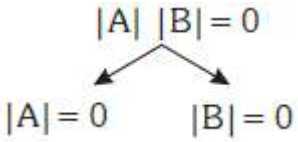
(2) रैखिक समीकरण निकाय AX = 0 के अनंत हल हैं

(3) B एक व्युत्क्रमणीय आव्यूह है

(4) adj(A) एक व्युत्क्रमणीय आव्यूह है

Ans. Official Answer NTA (2)

Sol. $AB = 0 \Rightarrow |AB| = 0$



If $|A| \neq 0, B = 0$ (not possible)

If $|B| \neq 0, A = 0$ (not possible)

Hence $|A| = |B| = 0$

$\Rightarrow AX = 0$ has infinitely many solutions

38. Which of the following matrices can NOT be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation? **[JEE(Main)July2022]**

आव्यूह $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ पर केवल एक प्रारंभिक पंक्ति संक्रिया से निम्न में से कौनसा आव्यूह प्राप्त नहीं किया जा सकता है?

[JEE(Main)July2022]

(1) $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ (3) $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$ (4) $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Ans. Official Answer NTA (3)

Sol. $A = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

(1) $R_1 \to R_1 + R_2; \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$ possible

(2) $R_1 \leftrightarrow R_2; \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ possible

(3) Option is not possible

(4) $R_2 \to R_2 + 2R_1; \begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$ possible

39. Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$, $a, b \in R$. If for some $n \in N$, $A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$ then $n + a + b$ is equal to _____.

[JEE(Main)July2022]

माना $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$, $a, b \in R$ है। यदि किसी $n \in N$ के लिए $A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$ है, तो $n + a + b$ बराबर है

_____

[JEE(Main)July2022]

Ans. Official Answer NTA (24)

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} = I + B$

$A = \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & ab \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$B^3 = 0$

$\therefore A^n = (1 + B)^n = {}^nC_0 I + {}^nC_1 B + {}^nC_2 B^2 + {}^nC_3 B^3 + ...$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & na & na \\ 0 & 0 & nb \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & \frac{n(n-1)ab}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & na & na + \frac{n(n-1)ab}{2} \\ 0 & 1 & nb \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 48 \\ 0 & 0 & 1 \end{bmatrix}$

On compaing we get na = 48, nb = 96 and

$$na + \frac{n(n+1)}{2}ab = 2160$$

$\Rightarrow$ a= 4, n = 12 and b = 8

n + a + b = 24

40. The number of matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c, d $\in \{-1, 0, 1, 2, 3, ...10\}$, such that $A = A^{-1}$, is _____.

[JEE(Main)July2022]

अव्यहों $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, जहाँ a, b, c, d $\in \{-1, 0, 1, 2, 3, ...10\}$ हैं $A = A^{-1}$ है, की संख्या है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (50)

Sol. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Given $A = A^{-1}$

$\therefore A^2 = A \cdot A^{-1} = I$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\therefore a^2 + bc = 1$ .....(1)

$ab + bd = 0$ .....(2)

$ac + cd = 0$ .....(3)

$bc + d^2 = 1$ .....(4)

(1) – (4) gives

$a^2 - d^2 = 0$

$\Rightarrow (a + d) = 0$ or $a - d = 0$

Case – I

$a + d = 0 \Rightarrow (a, d) = (-1, 1), (0, 0), (1, -1)$

(a) (a, d) = (.1, 1)

$\therefore$ from equation (1)

$1 + bc = 1 \Rightarrow bc = 0$

b = 0 C = 12 possibilities

c = 0 b = 12 possibilities

but (0, 0) is repeated

$\therefore 2 \times 12 = 24$

24 – 1 (repeated) = 23 pairs

(b) $(a, d) = (1, -1) \Rightarrow bc = 0 \rightarrow$ 23 pairs

(c) $(a, d) = (0, 0) \Rightarrow bc = 1$

$\Rightarrow (b, c) = (1, 1)$ & $(-1, -1)$, 2 pairs

Case – II

$a = d$

from (2) and (3)

$a \neq 0$ then $b = c = 0$

$a^2 = 1$

$a = \pm 1 = d$

$(a, d) = (1, 1), (-1, -1) \rightarrow 2$ pairs

$\therefore$ Total $= 23 + 23 + 2 + 2$

$= 50$ pairs

41. Let S be the set containing all 3×3 matrices with entries from $\{-1, 0, 1\}$. The total number of matrices $A \in S$ such that the sum of all the diagonal elements of $A^T A$ is 6 is _____. **[JEE(Main)July2022]**

माना 3×3 के सभी आव्यूहों जिनके अवयव $\{-1, 0, 1\}$ में से हैं, का समुच्चय S है। आव्यूह $A \in S$ जिसके लिए A^TA के विकर्ण के सभी अवयवों का योग 6 है, की कुल संख्या है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (5376)

Sol. $\text{Tr}(AA^T) = 6$

$$AA^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Now given $a^2 + d^2 + g^2 + b^2 + e^2 + h^2 + c^2 + f^2 + i^2 = 6$

$= {}^9C_3 \times 2^6$

$= 5376$

42. Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers. If $\frac{\det\left(\text{adj}\left(\text{adj}\left(\text{adj}\left(\text{adj } A\right)\right)\right)\right)}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32} \times 3^{16}$, then the number of such 3-tuples (α, β, γ) is _____.

[JEE(Main)July2022]

आव्यूह $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$ है, जहाँ α, β, γ तीन भिन्न पूर्णांक संख्याऐं हैं। यदि

$\frac{\det\left(\text{adj}\left(\text{adj}\left(\text{adj}\left(\text{adj } A\right)\right)\right)\right)}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32} \times 3^{16}$ है, तो इस प्रकार के त्रिकों (α, β, γ) की संख्या है _____.

[JEE(Main)July2022]

Ans. Official Answer NTA (42)

Sol. $A=\begin{bmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \beta+\gamma & \gamma+\alpha & \alpha+\beta\end{bmatrix}$

$R_3 \to R_3 + R_1$

$\Rightarrow |A| = |\alpha+\beta+\gamma| \begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ 1 & 1 & 1\end{vmatrix}$

$\Rightarrow |A| = (\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$

$\because |\text{adj}A| = |A|^{n-1}$

$|\text{adj}(\text{adj}A)| = |A|^{(n-1)^2}$

$|\text{adj}(\text{adj}(\text{adj}(\text{adj}A)))| = |A|^{(n-1)^4} = |A|^{2^4} = |A|^{16}$

$\therefore (\alpha+\beta+\gamma)^{16} = 2^{32}.3^{16}$

$\Rightarrow (\alpha+\beta+\gamma)^{16} = (2^2.3)^{16} = (12)^{16}$

$\Rightarrow \alpha+\beta+\gamma = 12$

$\because \alpha,\beta,\gamma \in N$

$(\alpha-1)+(\beta-1)+(\gamma-1) = 9$

number all tuples $(\alpha,\beta,\gamma) = {}^{11}C_2 = 55$

1 case for $\alpha = \beta = \gamma$

& 12 case when any two of these are equal

So, No. of distinct tuples (α,β,γ)

$= 55 - 13 = 42$

43. Let $A=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}$, $\alpha, \beta \in R$. Let α_1 be the value of α which satisfies $(A+B)^2 = A^2 + \begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$ and α_2 be the value of α which satisfies $(A + B)^2 = B^2$. Then $|\alpha_1 - \alpha_2|$ is equal to _____.

[JEE(Main)July2022]

माना $A=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}$ तथा $B=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}$, $\alpha, \beta \in R$ हैं। माना $(A+B)^2 = A^2 + \begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$ को संतुष्ट करने वाला α का eku α_1 है तथा $(A + B)^2 = B^2$ को संतुष्ट करने वाला α का मान α_2 हैं। तो $|\alpha_1 - \alpha_2|$ बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (2)

Sol. $A+B=\begin{bmatrix}\beta+1 & 0\\ 3 & \alpha\end{bmatrix}$

$$(A+B)^2=\begin{bmatrix}\beta+1 & 0\\ 3 & \alpha\end{bmatrix}\begin{bmatrix}\beta+1 & 0\\ 3 & \alpha\end{bmatrix}$$

$$=\begin{bmatrix}(\beta+1)^2 & 0\\ 3(\beta+1)+3\alpha & \alpha^2\end{bmatrix}$$

$$A^2=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}$$

$$=\begin{bmatrix}-1 & -1-\alpha\\ 2+2\alpha & \alpha^2-2\end{bmatrix}$$

$$\therefore\begin{bmatrix}1 & -\alpha+1\\ 2\alpha+4 & \alpha^2\end{bmatrix}=\begin{bmatrix}(\beta+1)^2 & 0\\ 3(\alpha+\beta+1) & \alpha^2\end{bmatrix}$$

$\alpha = 1 = \alpha_1$

$$B^2=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}$$

$$=\begin{bmatrix}\beta^2+1 & \beta\\ \beta & 1\end{bmatrix}=\begin{bmatrix}(\beta+1)^2 & 0\\ 3(\beta+1)+3\alpha & \alpha^2\end{bmatrix}$$

$\therefore\ \beta=0,\alpha=-1=\alpha_2$

$|\alpha_1-\alpha_2|=|1-(-1)|=2$

44. The number of matrices of order 3 × 3, whose entries are either 0 or 1 and sum of all the entries is a prime number, is _____. **[JEE(Main)July2022]**

कोटि 3 × 3 आव्यूहों, जिसके अवयव या तो 0 या 1 हैं तथा सभी अवयवों का योग एक अभाज्य संख्या है, की संख्या _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (282)

Sol. $A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}a_{ij}\in\{0,1\}$

$\sum a_{ij}=2,3,5,7$

Total matrix $= {}^9C_2 + {}^9C_3 + {}^9C_5 + {}^9C_7$

$= 282$

45. Let $x = \begin{bmatrix}1\\1\\1\end{bmatrix}$ and $A = \begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}$. For $k \in N$, If $X' A^k X = 33$, then k is equal to _____.

[JEE(Main)July2022]

माना $x = \begin{bmatrix}1\\1\\1\end{bmatrix}$ तथा $A = \begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}$ हैं। $k \in N$ के लिए, यदि $X' A^k X = 33$ है, तो k बराबर है_____

[JEE(Main)July2022]

Ans. Official Answer NTA (10)

Sol. $X = \begin{bmatrix}1\\1\\1\end{bmatrix}; A = \begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}$

$X^T A^K X = 33$

$$\begin{bmatrix}1 & 1 & 1\end{bmatrix}\begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}^k\begin{bmatrix}1\\1\\1\end{bmatrix} = 33$$

$$\begin{bmatrix}1 & 1 & 1\end{bmatrix}\begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix} = 33$$

$$\text{As } A^2 = \begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}\begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix} = \begin{bmatrix}1 & 0 & 6\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

$$A^4 = \begin{bmatrix}1 & 0 & 6\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 6\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 12\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

$$A^8 = \begin{bmatrix}1 & 0 & 24\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

$$A^{10}=\begin{bmatrix}1&0&6\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1&0&24\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}1&0&30\\0&1&0\\0&0&1\end{bmatrix}$$

$$\text{for } K\to \text{Even } A^K=\begin{bmatrix}1&0&3K\\0&1&0\\0&0&1\end{bmatrix}$$

$X^TA^KX=33$ (This is not correct)

$$\begin{bmatrix}1&1&1\end{bmatrix}\begin{bmatrix}1&0&3K\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}$$

$$=\begin{bmatrix}1&1&3K+1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}=[3K+3]$$

$\therefore 3K+3=33 \therefore K=10$

But it should be dropeed as 33 is not matrix

If K is odd

$X^TA^KX=33$

$X^TAAA^{K-1}X=33$

$$\begin{bmatrix}1&1&1\end{bmatrix}\begin{bmatrix}-1&2&3\\0&1&6\\0&0&-1\end{bmatrix}\begin{bmatrix}1&0&3k-3\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}=33$$

$$\begin{bmatrix}-1&3&8\end{bmatrix}\begin{bmatrix}3k-2\\1\\1\end{bmatrix}=[33]$$

$[-3k+13]=[33]$

$k=20/3$ (not possible)

Exercise J - A

1. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \le i, j \le 3$. If the determinant of P is 2, then the determinant of the matrix Q is : **[JEE-2012]**

माना कि $P = [a_{ij}]$ कोटि 3×3 का आव्यूह है और माना $Q = [b_{ij}]$, जहाँ $b_{ij} = 2^{i+j}a_{ij}$ जब $1 \le i, j \le 3$ है। यदि P के सारणिक का eku 2 है, तो आव्यूह Q के सारणिक का मान है : **[JEE-2012]**

(A) 2^{10} (B) 2^{11} (C) 2^{12} (D) 2^{13}

Ans. D

Sol. $|Q| = \begin{vmatrix} 2^2a_{11} & 2^3a_{12} & 2^4a_{13} \\ 2^3a_{21} & 2^4a_{22} & 2^5a_{23} \\ 2^4a_{31} & 2^5a_{32} & 2^6a_{33} \end{vmatrix}$

$$\Rightarrow |Q| = 2^2.2^3.2^4.\begin{vmatrix} a_{11} & 2a_{12} & 2^2a_{13} \\ a_{21} & 2a_{22} & 2^2a_{23} \\ a_{31} & 2a_{32} & 2^2a_{33} \end{vmatrix}$$

$= 2^2.2^3.2^4|P|.2^3$

$= 2^2.2^3.2^4.2.2^3 = 2^{13}$

2. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identuty matrix, then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that : **[JEE-2012]**

एक 3×3 कोटि का आव्यूह P इस प्रकार है कि $P^T = 2P + I$, जहाँ P^T आव्यूह P का परिवर्त है तथा I, 3×3 तत्समक आव्यूह है, तब , d Lr Hkkv dk, oxz(column matrix) $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ इस प्रकार है कि : **[JEE-2012]**

(A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (B) $PX = X$ (C) $PX = 2X$ (D) $PX = -X$

Ans. D

Sol. $P^T = 2P + I$

$\Rightarrow P = 2P^T + I$

$\Rightarrow P = 2(2P + I) + I$

$\Rightarrow P = 4P + 3I$

$\Rightarrow P = -I$

$\Rightarrow PX = -X$

3. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are) : **[JEE-2012]**

यदि 3×3 कोटि के आव्यूह P का सहखंडज $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$ है, तो P के सारणिक का संभावित मान होगा : **[JEE-2012]**

(A) –2 (B) –1 (C) 1 (D) 2

Ans. AD

Sol. $|\text{adjP}| = \begin{vmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{vmatrix}$

$\Rightarrow |P|^2 = 4 \Rightarrow |P| = \pm 2$

4. For 3×3 matrices M and N, which of the following statment(s) is (are) NOT correct? **[JEE-2013]**

(A) $N^TM\,N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric

(B) $M\,N - N\,M$ is skew symmetric for all symmetric matrices M and N

(C) $M\,N$ is symmetric for all symmetric matrices M and N

(D) (adj M) (adj N) = adj(M N) for all invertible matrices M and N

3×3 आव्यूहों M तथा N के लिए निम्न में से कौन प्रकथन सत्य नहीं है (हैं)? **[JEE-2013]**

(A) ड के सममित या विषम सममित होने के अनुसार $N^TM\,N$ सममित या विषम सममित है

(B) सभी सममित आव्यूहों M तथा N के लिए $M\,N - N\,M$ विषम सममित है

(C) सभी सममित आव्यूहों M तथा N के लिए $M\,N$ सममित है

(D) सभी व्युत्क्रमणीय आव्यूहों M तथा N के लिए (adj M) (adj N) = adj(M N)

Ans. CD

Sol. (A) $(N^TMN)^T = N^T\,M^T\,N$ is symmetric if M is symmetric and skew-symmetric if M is skew-symmetric.

(B) $(MN - NM)^T = (MN)^T - (NM)^T = NM - MN = -(MN - NM)$

skew symmetric

(C) $(MN)^T = N^T\,M^T = NM \neq MN$ hence NOT correct

(D standard result is
adj(MN) = (adjN) (adj M)] ≠ (adjM) (adjN)

5. Let M and N be two 3 × 3 matrices such that MN = NM. Further, if $M \neq N^2$ and $M^2 = N^4$, then :

[JEE(Adv)-2014]

(A) determinant of $(M^2 + MN^2)$ is 0

(B) there is a 3 × 3 non-zero matrix U such that $(M^2 + MN^2)$ U is the zero matrix

(C) determinant of $(M^2 + MN^2) \geq 1$

(D) for a 3 × 3 matrix U, if $(M^2 + MN^2)$ U equals the zero matrix then U is the zero matrix

मानाकि दो 3 × 3 आव्यूह M तथा N इस प्रकार है कि MN = NM है। यदि $M \neq N^2$ तथा $M^2 = N^4$ हो, तो :

[JEE(Adv)-2014]

(A) ;ड $^2 + MN^2$) के सारणिक का मान शून्य है

(B) एक ऐसा 3 × 3 शून्येतर आव्यूह U है जिसके लिये $(M^2 + MN^2)$ U शून्य आव्यूह है

(C) ;ड $^2 + MN^2$) के सारणिक का मान ≥ 1 है

(D) 3 ऊ 3 आव्यूह U जिसके लिये $(M^2 + MN^2)$ U शून्य आव्यूह है तो U भी एक शून्य आव्यूह होगा

Ans. AB

Sol. MN = NM & $M^2 - N^4 = 0$

$$(M - N^2)(M + N)^2 = 0$$

$M - N^2 = 0$	$M + N^2 = 0$	$\lvert M + N^2 \rvert = 0$
Not Possible	$M - N^2 \neq 0$	$\lvert M - N^2 \rvert = 0$

In any case $|M + N^2| = 0$

(A) $|M^2 + MN^2| = |M|\,|M + N^2|$
$= 0$
(A) is correct

(B) If $|A| = 0$ then AU = 0 will have ∞ solution.
Thus $(M^2 + MN^2)$ U = 0 will have many 'U'
(B) is correct

(C) Obvious wrong.

(D) If AX = 0 & $|A| = 0$ then X can be non zero.
(D) is wrong

6. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if : **[JEE(Adv)-2014]**

(A) the first column of M is the transpose of the second row of M

(B) the second row of M is the transpose of the first column of M

(C) M is a diagonal matrix with nonzero entries in the main diagonal

(D) the product of entries in the main diagonal of M is not the square of an integer

माना कि 2×2 सममित आव्यूह M के सभी पूर्णांक हैं। तब M व्युत्क्रमणीय है, यदि : **[JEE(Adv)-2014]**

(A) ड का पहला स्तम्भ M की दूसरी पंक्ति का परिवर्त है

(B) ड की दूसरी पंक्ति M के पहले स्तम्भ का परिवर्त है

(C) ड एक विकर्ण आव्यूह है जिसके मुख्य विकर्ण के अवयव शून्यतर है

(D) ड के मुख्य विकर्ण के अवयवों का गुणनफल किसी भी पूर्णांक का वर्ग नहीं है

Ans. CD

Sol. $M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

(A) $\begin{bmatrix} a \\ b \end{bmatrix}$ and [b c] are transpose.

So $\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ c \end{bmatrix}$ is given $\Rightarrow$ $a = b = c$

$M = \begin{bmatrix} a & a \\ a & a \end{bmatrix}$ $\Rightarrow$ $|M| = 0$ A is wrong.

(B) [b c] and $\begin{bmatrix} a \\ b \end{bmatrix}$ are transpose.

So $a = b = c$ B is wrong

(C) $M = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix}$ $\Rightarrow$ $|M| = ac \neq 0$ C is correct

(D) $M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ given $ac \neq \lambda^2$. D is correct

(C, D) are correct.

7. Let X and Y be two arbitrary, 3×3, non-zero, skew-symmetric matrices and Z be an arbitrary 3×3, non zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric? **[JEE(Adv)-2015]**

माना कि X एवं Y दो स्वेच्छ, 3×3 शून्येतर विषम सममित आव्यूह है और Z एक स्वेच्छ 3×3, शून्येत्तर सममित आव्यूह है। तब fuEufyf[kr esa lsdkSl k¼ lefer vkO;wggSa\? **[JEE(Adv)-2015]**

(A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

Ans. CD

Sol. (C) $(X^4Z^3 - Z^3X^4)^T = (X^4Z^3)^T (Z^3X^4)^T$

$= (Z^T)^3(X^T)^4 - (X^T)^4(Z^T)^3$

$= Z^3X^4 - X^4Z^3$

$= -(X^4Z^3 - Z^3X^4)$

(D) $(X^{23} + Y^{23})^T = -X^{23} - Y^{23}$ $\Rightarrow$ $X^{23} + Y^{23}$ is skew–symmetric

8. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where $k \in \mathbb{R}$, $k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then : **[JEE(Adv)-2016]**

माना कि $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, जहाँ $\alpha \in \mathbb{R}$ है। मान लीजिए कि $Q = [q_{ij}]$ एक ऐसा आव्यूह है कि $PQ = kI$, जहाँ $k \in \mathbb{R}$, $k \neq 0$ और I तीन कोटि का तत्समक आव्यूह है। यदि $q_{23} = -\frac{k}{8}$ और $\det(Q) = \frac{k^2}{2}$ हो, तब : **[JEE(Adv)-2016]**

(A) $\alpha = 0, k = 8$ (B) $4\alpha - k + 8 = 0$ (C) $\det(P\,\text{adj}(Q)) = 2^9$ (D) $\det(Q\,\text{adj}(P)) = 2^{13}$

Ans. BC

Sol. $P\left(\frac{Q}{K}\right) = I$

$\Rightarrow P^{-1} = \left(\frac{Q}{K}\right)$

$\Rightarrow \left(P^{-1}\right)_{23} = \frac{q_{23}}{K} = -\frac{1}{8}$

$\Rightarrow \frac{-(3\alpha + 24}{12\alpha + 20} = -\frac{1}{8} \Rightarrow \alpha = -1$

Hence $|P| = 12\alpha + 20 = 8$

Again $|P|\left|\frac{Q}{K}\right| = |I| \Rightarrow \frac{8|Q|}{K^3} = 1$

$\Rightarrow |Q| = \frac{K^3}{8}$

$$\Rightarrow \frac{K^3}{8} = \frac{K^2}{2} \Rightarrow K = 4 \Rightarrow |Q| = 8$$

(C) $\det(P\ adj(Q)) = \det(P) \cdot \det(adj(Q))$

$= \det(P) \cdot (\det(Q))^2 = 8 \times 8^2 = 2^9$

(D) $\det(Q\ adj(P)) = \det(Q) \cdot (\det P)^2 = 8 \times 8^2 = 2^9$

9. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then

$\frac{q_{31} + q_{32}}{q_{21}}$ equals : **[JEE(Adv)-2016]**

माना कि $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ और I तीन कोटि का तत्समक आव्यूह है। यदि $Q = [q_{ij}]$ एक आव्यूह इस प्रकार है कि $P^{50} - Q = I$ है,

r c $\frac{q_{31} + q_{32}}{q_{21}}$ का मान है : **[JEE(Adv)-2016]**

(A) 52 (B) 103 (C) 201 (D) 205

Ans. B

Sol. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} = I + A$

$$A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$A^n = 0 \quad \forall\ n \geq 3$

Given : $P^{50} = Q + I$

$$\Rightarrow I + Q = (I + A)^{50} = I + 50A + \frac{50 \times 49}{2} A^2 + 0$$

$$\Rightarrow Q = 50\begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} + \frac{50\times 49}{2}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$q_{31} = 50\times 16 + \left(\frac{50\times 49}{2}\right)16 = 200\times 102$$

$q_{32} = 50\times 4 = 200$, $q_{21} = 50\times 4 = 200$

$$\frac{q_{31}+q_{32}}{q_{21}} = 103$$

10. Which of the following is(are) NOT the square of a 3×3 matrix with real entries? **[JEE(Adv)-2017]**

निम्न में से कौनसा/कौनसे वास्तविक संख्याओं के 3×3 आव्यूह का वर्ग नहीं है/हैं? **[JEE(Adv)-2017]**

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

Ans. BD

Sol. $|A^2| = |A|^2$ will be +ve
Now check option

11. How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of M^TM is 5? **[JEE(Adv)-2017]**

ऐसे कितने 3×3 आव्यूह M हैं जिनकी प्रविष्टियाँ $\{0, 1, 2\}$ में हैं तथा M^TM की विकर्णीय प्रविष्टियों का योग 5 है? **[JEE(Adv)-2017]**

(A) 162 (B) 135 (C) 126 (D) 198

Ans. D

Sol. $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

$$a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2 = 5$$

Case -I : Five $(1's)$ and four $(0's)$ ${}^9C_5 = 126$
Case -II : One (2) and one (1) ${}^9C_2 \times 2! = 72$
$\therefore$ Total = 198

12. For a real number α, if the system $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ of linear equations, has infinitely many solutions,

then $1 + \alpha + \alpha^2 =$ **[JEE(Adv)-2017]**

वास्तविक संख्या α के लिये, यदि रैखिक समीकरण निकाय $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ के अनन्त हल है, तब $1 + \alpha + \alpha^2 =$

[JEE(Adv)-2017]

Ans. 1

Sol. $\Delta = 0$

$$\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{vmatrix} = 0 \Rightarrow \alpha = 1 \quad \text{or} \quad \alpha = -1$$

For $\alpha = 1$
System will have no solution

For $\alpha = -1$
System will have infinite solution

13. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of equations

(in real variables)

$$-x + 2y + 5z = b_1$$
$$2x - 4y + 3z = b_2$$
$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least one solution

for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$? **[JEE(Adv)-2018]**

माना कि S उन सभी स्तम्भ आव्यूहों $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ का समुच्चय है जिनके लिए $b_1, b_2, b_3 \in \mathbb{R}$ और वास्तविक चरों वाले समीकरण निकाय

$$-x + 2y + 5z = b_1$$
$$2x - 4y + 3z = b_2$$
$$x - 2y + 2z = b_3$$

का कम से कम एक हल है। तब निम्नलिखित वास्तविक चरों वाले निकायों में से किस (कौन से) निकाय (निकायों) का भी प्रत्येक

$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$ के लिए कम से कम एक हल है? **[JEE(Adv)-2018]**

(A) $x + 2y + 3z = b_1$, $4y + 5z = b_2$ and $x + 2y + 6z = b_3$

(B) $x + y + 3z = b_1$, $5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$

(C) $-x + 2y - 5z = b_1$, $2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$

(D) $x + 2y + 5z = b_1$, $2x + 3z = b_2$ and $x + 4y - 5z = b_3$

Ans. AD

Sol. $\Delta = 0$ so for at least one solutions $\Delta_1 = \Delta_2 = \Delta_3 = 0 \quad \Rightarrow \quad b_1 + 7b_2 = 13b_3$ .....(i)

οπτιον (A) $\Delta \neq 0 \Rightarrow$ unique solution $\Rightarrow$ option (A) is correct

οπτιον (Δ) $\Delta \neq 0 \Rightarrow$ unique solution $\Rightarrow$ option (B) is correct

οπτιον (X) $\Delta = 0 \Rightarrow$ equations are $x - 2y + 5z = -b_1$

$$x - 2y + 5z = \frac{b_2}{2}$$

$$x - 2y + 5z = b_2$$

There planes are parallel so they must be coincident

$\Rightarrow \quad -b_1 = \frac{b_2}{2} = b_3$ which does not satisfies equation (1) for all $b_1, b_2, b_3 \quad \Rightarrow \quad$ option (C) is incorrect.

οπτιον (B) $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 5 & 2 & 2 \\ 2 & 1 & 1 \end{vmatrix} = 0$. Also $\Delta_1 = 0$

For infinite solutions, Δ_2 and Δ_3 must be 0

$\Rightarrow \quad \begin{vmatrix} 1 & b_1 & 1 \\ 5 & b_2 & 2 \\ 2 & b_3 & 1 \end{vmatrix} = 0 \quad \Rightarrow \quad -b_1 - b_2 + 3b_3 = 0$ which does not satisfy (i) for all $b_1, b_2\, b_3$ so option(s)

wrong

14. Let P be a matrix of order 3 × 3 such that all the entries in P are from the set {–1, 0, 1}. Then, the maximum possible value of the determinant of P is _____. **[JEE(Adv)-2018]**

माना कि P, 3 × 3 कोटि का एक ऐसा आव्यूह है कि P की सभी प्रविष्टियाँ समुच्चय {–1, 0, 1} में से है। तब P के सारणिक का vf/kdre l aHkkfor eku gS_____. **[JEE(Adv)-2018]**

Ans. 4

Sol. $\Delta = 0$

$$\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{vmatrix} = 0 \Rightarrow \alpha = 1 \quad \text{or} \quad \alpha = -1$$

For $\alpha = 1$
System will have no solution

For $\alpha = -1$
System will have infinite solution

15. Let $M = \begin{bmatrix} \sin^4\theta & -1-\sin^2\theta \\ 1+\cos^2\theta & \cos^4\theta \end{bmatrix} = \alpha I + \beta M^{-1}$ where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers and I is the 2×2 identity matrix. If α^* is the minimum of the set $\{\alpha(\theta): \theta \in [0,2\pi)\}$ and β^* is the minimum of the set $\{\beta(\theta): \theta \in [0,2\pi)\}$ then the value of $\alpha^* + \beta^*$: **[JEE(Adv)-2019]**

माना कि $M = \begin{bmatrix} \sin^4\theta & -1-\sin^2\theta \\ 1+\cos^2\theta & \cos^4\theta \end{bmatrix} = \alpha I + \beta M^{-1}$ जहाँ $\alpha = \alpha(\theta)$ और $\beta = \beta(\theta)$ वास्तविक संख्याएँ है और I एक 2×2 तत्समक आव्यूह है, यदि समुच्चय $\{\alpha(\theta): \theta \in [0,2\pi)\}$ का निम्नतम α^* है और समुच्चय $\{\beta(\theta): \theta \in [0,2\pi)\}$ का निम्नतम β^* है। तो $\alpha^* + \beta^*$ का मान है : **[JEE(Adv)-2019]**

(A) $-\frac{31}{16}$ (B) $-\frac{29}{16}$ (C) $-\frac{37}{16}$ (D) $-\frac{17}{16}$

Ans. B

Sol. $M = \alpha I + \beta M^{-1}$

$M^2 = \alpha M + \beta I$

$M^2 - \alpha M - \beta I = 0$

$\alpha = tr(M) = \sin^4\theta + \cos^4\theta$

$$\alpha = 1 - \frac{1}{2}\sin^2 2\theta$$

$$(\alpha)_{min} = 1 - \frac{1}{2} = \frac{1}{2} \quad \text{when } \sin^2 2\theta = 1$$

$$-\beta = |M|$$

$\beta = -|M|$

$= -\left(\sin^4\theta\cos^4\theta + \left(1+\sin^2\theta\right)\left(1+\cos^2\theta\right)\right) = -\left(\sin^4\theta\cos^4\theta + \sin^2\theta\cos^2\theta + 2\right)$

$= -\left(\left(\sin^2\theta\cos^2\theta + \frac{1}{2}\right)^2 + \frac{7}{4}\right) = -\left(\left(\frac{\sin^2 2\theta}{4} + \frac{1}{2}\right)^2 + \frac{7}{4}\right)$

$(\beta)_{min} = \frac{-37}{16}$ when $\sin^2 2\theta = 1$

$(\alpha)_{min} + (\beta)_{min} = \frac{-29}{16}$

16. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and adj $M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ Where a and b are real numbers, which of the following options is/are correct : **[JEE(Adv)-2019]**

(A) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$

(B) If $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ then $\alpha - \beta + \gamma = 3$

(C) $a + b = 3$

(D) $\det(\text{adj}M^2) = 81$

माना कि $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ तथा adj $M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ जहाँ a और b वास्तविक संख्याएँ है। निम्न में से कौन सा (से) विकल्प l ghgS(है) : **[JEE(Adv)-2019]**

(A) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$

(B) यदि $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ तब $\alpha - \beta + \gamma = 3$

(C) $a + b = 3$

(D) $\det(\text{adj}M^2) = 81$

Ans. ABC

Sol. $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$

$$\text{adj}(M)=\begin{bmatrix}2-3b & \ldots & \ldots\\ \ldots & -3a & \ldots\\ \ldots & \ldots & \ldots\end{bmatrix}\quad\begin{matrix}2-3b=-1\\ 3b=3\\ b=1\end{matrix}\Bigg|\begin{matrix}-3a=-6\\ a=2\end{matrix}$$

$$M=\begin{bmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & 1 & 1\end{bmatrix}\Rightarrow |M|=-2$$

(A) $\text{adj}(M^{-1}) + (\text{adj } M)^{-1} = -M$

$(\text{adj } M)^{-1} + (\text{adj } M)^{-1} = -M$

$2(\text{adj } (M))^{-1} = -M$

$2(\text{adj } M)^{-1} \text{ adj } M = -M \text{ adj } M$

$2I = -|M| I$

$2I = 2I$

(B) $$\begin{bmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & 1 & 1\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\\ \gamma\end{bmatrix}=\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$$

$$\begin{bmatrix}\alpha\\ \beta\\ \gamma\end{bmatrix}=\begin{bmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & 1 & 1\end{bmatrix}^{-1}\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$$

$$M^{-1}=\frac{\text{adj}(M)}{(M)}=-\frac{1}{2}\begin{bmatrix}-1 & 1 & -1\\ 8 & -6 & 2\\ -5 & 3 & -1\end{bmatrix}$$

$$\begin{bmatrix}\alpha\\ \beta\\ \gamma\end{bmatrix}=-\frac{1}{2}\begin{bmatrix}-1 & 1 & -1\\ 8 & -6 & 2\\ -5 & 3 & -1\end{bmatrix}\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$$

$$=-\frac{1}{2}\begin{bmatrix}-2\\ 2\\ -2\end{bmatrix}\qquad\begin{bmatrix}\alpha\\ \beta\\ \gamma\end{bmatrix}=\begin{bmatrix}1\\ -1\\ 1\end{bmatrix}\Rightarrow \alpha=1,\ \beta=-1,\ \gamma=1$$

(C) $a + b = 1 + 2 = 3$

(D) $|\text{adj}(M^2)| = |M^2|^2 = |M^4| = 16$

17. Let $x \in R$ and let $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$. Then which of the following options is are/ correct? **[JEE(Adv)-2019]**

(A) For $x = 1$ there exists a unit vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ for which $R\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(B) $\det R = \det\begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in R$

(C) For $x = 0$, if $R\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6\begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$ then $a + b = 5$

(D) There exists a real number x such that $PQ = QP$

माना कि $x \in R$ और माना कि $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ और $R = PQP^{-1}$ तब निम्न में से कौन सा (से) विकल्प सही g$(हैं)? **[JEE(Adv)-2019]**

(A) $x = 1$ के लिए, एक ऐसा मात्रक सदिश (unit vector) $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ सम्भव है, जिसके लिए $R\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(B) सभी $x \in R$ के लिए, $\det R = \det\begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$

(C) $x = 0$ के लिए, यदि $R\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6\begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, तब $a + b = 5$

(D) एक ऐसी वास्तविक संख्या x सम्भव है जिसके लिए $PQ = QP$

Ans. BC

Sol. Option (2) $PQP^{-1} = R$

$PQ = RP$

$\det P \det Q = \det R \det P$

$\det R = \det Q$

$\det Q = 48 - 4x^2$

2nd option $\Rightarrow 40 - x^2 + 8$

$R = PQP^{-1}$

Option (3) $R = \frac{1}{6}\begin{bmatrix} 6x+12 & 3x+6 & 4-10x \\ 12 & 24 & 8-4x \\ 18x & 0 & 36-6x \end{bmatrix}$

if $x = 0$

$$R = \frac{1}{6}\begin{bmatrix} 2 & 1 & \frac{2}{3} \\ 0 & 4 & \frac{4}{3} \\ 0 & 0 & 6 \end{bmatrix}$$

$$R\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = \begin{bmatrix} 2+a+\frac{2}{3}b \\ 4a+\frac{4}{3}b \\ 6b \end{bmatrix} \Rightarrow a = 2 \text{ and } b = 3$$

$a + b = 5$

Option (4) $PQ = QP$ not possible

Option (1) if we put $x = 1$ then

$$R = \frac{1}{6}\begin{bmatrix} 18 & 9 & -6 \\ 12 & 24 & 4 \\ 18 & 0 & 30 \end{bmatrix}$$

$|R| \neq 0$

$\Rightarrow$ system will have unique solution

$\Rightarrow$ only $\alpha = \beta = \gamma = 0$ will satisfy the system

18. Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

and $X = \sum_{k=1}^{6} P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$ where P_k^T denotes the transpose of the matrix P_k then which of the following options is/are correct : **[JEE(Adv)-2019]**

(A) The sum of diagonal entries of X is 18 (B) If $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ then a = 30

(C) X is a symmetric matrix (D) X–30I is an invertible matrix

माना कि

$P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

और $X = \sum_{k=1}^{6} P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$ जहाँ आव्यूह (matrix) P_k के परिवर्त (transpose) को P_k^T से दर्शाया गया है। तब निम्न में से कौन

l k(से) विकल्प सही है (हैं)? **[JEE(Adv)-2019]**

(A) X के विकर्ण (diagonal) की प्रविष्टियों (entries) का योग 18 है

(B) यदि $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, तब a = 30

(C) X एक सममित (symmetric) आव्यूह है

(D) $X - 30I$ एक व्युत्क्रमणीय (invertible) आव्यूह है

Ans. ABC

Sol. $P_1 = P_1^T = P_1^{-1}$

$P_2 = P_2^T = P_2^{-1}$

$P_6 = P_6^T = P_6^{-1}$

and $A^T = A$

$(A + B)^T = A^T + B^T$

$X^T = (P_1AP_1T + ...P_6AP_6^T)^T$

$= X$ so $\times$ is symmetry

trace $\Rightarrow (2 + 0 + 1) + (2 + 0 + 1).....6$ time

$= 18$

$$\text{Let } B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$XB = P_1AP_1^TB + ...P_6A_6^TB$

$$XB = (P_1 + P_2 + P_3.....P_6) \begin{bmatrix} 6 \\ 3 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 6\times2+3\times2+6\times2 \\ 6\times2+3\times2+6\times2 \\ 6\times2+3\times2+6\times2 \end{bmatrix} = \begin{bmatrix} 30 \\ 30 \\ 30 \end{bmatrix} = 300 \Rightarrow \alpha = 30$$

$$\text{Since } x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 30 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{(x-301)B = 0 has a non trivial solution } B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow |x-30| = 0$

19. Suppose $\det \begin{bmatrix} \sum_{k=0}^{n} k & \sum_{k=0}^{n} {}^nC_k k^2 \\ \sum_{k=0}^{n} {}^nC_k k & \sum_{k=0}^{n} {}^nC_k 3^k \end{bmatrix} = 0$ holds for some positive integer n then $\sum_{k=0}^{n} \frac{{}^nC_k}{k+1}$ equals _____.

[JEE(Adv)-2019]

माना कि किसी धनात्मक पूर्णांक (positive integer) n के लिए $\begin{bmatrix} \sum_{k=0}^{n} k & \sum_{k=0}^{n} {}^nC_k k^2 \\ \sum_{k=0}^{n} {}^nC_k k & \sum_{k=0}^{n} {}^nC_k 3^k \end{bmatrix} = 0$ तब $\sum_{k=0}^{n} \frac{{}^nC_k}{k+1}$ बराबर _____.

[JEE(Adv)-2019]

Ans. 6.2

Sol. $\begin{vmatrix} \frac{n(n+1)}{2} & n(n+1)2^{n-2} \\ n2^{n-1} & 4^n \end{vmatrix} = 0$

$n = 4$

put $n = 4$ In $\sum_{k=0}^{n} \frac{{}^nC_k}{k+1}$

$\sum_{k=0}^{4} \frac{{}^4C_k}{k+1} = \frac{2^5-1}{5} = \frac{31}{5}$

$= 6.2$

20. Let M be a 3×3 invertible matrix with real entries and let I denote the 3×3 identity matrix. If $M^{-1} = \text{adj (adj M)}$, then which of the following statements is / are ALWAYS TRUE? **[JEE(Adv)-2020]**

माना कि M वास्तविक संख्याओं का एक 3×3 व्यूत्क्रमणीय आव्यूह है एवं माना कि 3×3 के तत्समक आव्यूह को I से दर्शाया जाता gSA; fn $M^{-1} = \text{adj (adj M)}$ है, तब निम्न में से कौनसा / से कथन सदैव सही है (हैं)? **[JEE(Adv)-2020]**

(A) $M = I$ (B) $\det M = 1$ (C) $M^2 = I$ (D) $(\text{adj M})^2 = I$

Ans. BCD

Sol. $M^{-1} = \text{adj (adj (M))}$

$M^{-1} = |M| \; M$

$I = |M| \; M^2$ .....(1)

$|I| = ||M| M^2|$

$I = |M|^3 \; |M|^2$

$|M|^5 = 1 \Rightarrow |M| = 1$ .....(2)

$\Rightarrow$ Put in (1)

$M^2 = I \Rightarrow M^{-1} = M$

$\frac{\text{adj(M)}}{|M|} = M$

$\text{adj (M)} = M$

$\Rightarrow (\text{adj (M)})^2 = I$

21. The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2×2 matrix such that the trace of A is 3 and the trace of A^3 is –18, then the value of the determinant of A is_____ **[JEE(Adv)-2020]**

एक वर्ग आव्यूह के अनुरेख को उसके विकर्ण की प्रविष्टियों के योगफल द्वारा परिभाषित किया जाता है। यदि A एक ऐसा 2×2 आव्यूह gSr 1 d kv uqjs k3 है एवं A^3 का अनुरेख –18 है, तब A के सारणिक का मान है_____ **[JEE(Adv)-2020]**

Ans. 5

Sol. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $a + d = 3$

$d = 3 - a$

$A = \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix}$

$|A| = a(3-a) - bc$

$|A| = 3a - a^2 - bc$

$A^2 = \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix} \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix} = \begin{bmatrix} a^2+bc & 3b \\ 3c & (3-a)^2+bc \end{bmatrix}$

$A^3 = \begin{bmatrix} a^2+bc & 3b \\ 3c & (3-a)^2+bc \end{bmatrix} \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix}$

$\text{tr}(A^3) = a^3 + abc + 3bc + 3bc + (3-a)[(3-a)^2 + bc] = -18$

$\text{tr}(A^3) = a^3 + abc + 6bc + (3-a)^3 + 3bc - abc = -18$

$\text{tr}(A^3) = a^3 + 9bc + (3-a)^3 = -18$

$\text{tr}(A^3) = a^3 + 9bc + 27 - a^3 - 9a(3-a) = -18$

$9bc + 27 - 27a + 9a^2 = -18$

$bc + 3 - 3a + a^2 = -2$

$5 = 3a - a^2 - bc$

hence

$|A| = 3a - a^2 - bc$

$|A| = 5$

22. For any 3×3 matrix M, let $|M|$ denote the determinant of M. Let $E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$ If Q is a nonsingular matrix of order 3×3, then which of the following statements is (are) TRUE? **[JEE(Adv)-2021]**

(A) F=PEP and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

किसी 3×3 आव्यूह M के लिए, माना $|M|$, M के सारणिक को निरूपति करता है। माना $E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

तथा $F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$ यदि Q कोटि 3×3 का एक व्युत्क्रमणीय है, तब निम्न में से कौनसा/कौनसे कथन सही है/हैं?

[JEE(Adv)-2021]

(A) F=PEP तथा $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $|EQ + PEQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) च$^{-1}$EP + F के विकर्ण अवयवों का योगफल $E + P^{-1}FP$ के विकर्ण अवयवों के योगफल के बराबर है

Ans. ABD

Sol. $E=\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 8 & 13 & 18\end{bmatrix}; P=\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}; F=\begin{bmatrix}1 & 3 & 2\\ 8 & 18 & 13\\ 2 & 4 & 3\end{bmatrix}$

$|Q|\neq 0,$ Q is square matrix of order 3

(A) $PE=\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 8 & 13 & 18\end{bmatrix}=\begin{bmatrix}1 & 2 & 3\\ 8 & 13 & 18\\ 2 & 3 & 4\end{bmatrix}$

$PEP=\begin{bmatrix}1 & 2 & 3\\ 8 & 13 & 18\\ 2 & 3 & 4\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}=\begin{bmatrix}1 & 3 & 2\\ 8 & 18 & 13\\ 2 & 4 & 3\end{bmatrix}=F$

$P^2=\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

$\Rightarrow$ (A) is Correct

(B) F = PEP & P^2=I (from (A))

$PF = P^2EP$

$PF = IEP$

$PF = EP$ -------(1)

LHS = $|EQ + PFQ^{-1}|$

$=|EQ + EPQ^{-1}|$

$|E||Q+PQ^{-1}|$

$=0$

RHS = $|EQ| + |PFQ^{-1}|$

$= |EQ| + |EPQ^{-1}|$

$=|E||Q| + |E|\,|PQ^{-1}|$

$= 0$

$\because |E| = 0$

(C) $|EF|=|E||F|=0$

$\Rightarrow$ (C) is Incorrect

LHS :

(D) $tr(P^{-1}EP + F)$

$= tr(P^{-1}PF + F)$ (Using (1))

$= tr(F + F)$

$= tr(2F) = 2\,tr(F) = 2(1 + 18 + 3) = 44$

RHS :

$tr(E + P^{-1}FP)$

$= tr(E + P^{-1}EP)$ (Using (1))

Now, $\because P^2 = I \Rightarrow P = P^{-1}$

$= \text{tr}(E + PEP)$

$= \text{tr}(E) + \text{tr}(PEP)$

$= \text{tr}(E) + \text{tr}(F)$

$= (1 + 3 + 18) + (1 + 18 + 3)$

$= 44$

$\Rightarrow$ (D) is correct

23. For any 3×3 matrix M, let $|M|$ denote the determinant of M. Let I be the 3×3 identity matrix. Let E and F be two 3×3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) TRUE ? **[JEE(Adv)-2021]**

किसी 3×3 आव्यूह M के लिए माना $|M|$, M के सारणिक को निरूपित करता है। माना I, 3×3 तत्समक आव्यूह है। माना E व F दो 3×3 आव्यूह है जबकि $(I - EF)$ प्रतिलोमीय है। यदि $G = (I - EF)^{-1}$, तब निम्न में से कौनसा/कौनसे कथन सही है/हैं? **[JEE(Adv)-2021]**

(A) $|FE| = |I - FE|\,|FGE|$ (B) $(I - FE)(I + FGE) = 1$

(C) $EFG = GEF$ (D) $(I - FE)(I - FGE) = 1$

Ans. ABC

Sol. $G = (I - EF)^{-1}$

$\Rightarrow I - EF = G^{-1}$

Now, $G.G^{-1} = G^{-1}.G$

$G(I - EF) = (I - EF).G$

$G - GEF = G - EFG$

$GEF = EFG$ .....(1)

(C) is correct

Now,

(A) $(I - FE)(FGE)$

$= FGE - FEFGE$ .....(2)

$I - EF = G^{-1}$

$\Rightarrow G(I - EF) = G.G^{-1}$

$\Rightarrow G - GEF = I$

$\Rightarrow G - EFG = I$ [From (1)]

$\Rightarrow G - I = EFG$ .....(3)

Put $G - I = EFG$ in (2), we get

$FGE - FEFGE$

$= FGE - F(G-I)E$

$= FGE - FGE + FE$

$= FE$

$\Rightarrow (I-FE)(FGE) = FE$

$\Rightarrow |I - FE|\,|FGE| = |FE|$

$\Rightarrow$ (A) is correct

(B) $(I-FE)(I + FGE)$

$= I + FGE - FE - FE\,FGE$

$= I + FGE - FE - F(G-I)E$ (from (3))

$= I + FGE - FE - FGE + FE$

$= I$

$\Rightarrow$ (B) is correct

(C) $(I-FE)(I - FGE)$

$= I - FGE - FE - FE\,FGE$

$= I - FGE - FE + F(G-I)E$ (from (3))

$= I - FGE - FE + FGE - FE$

$= I - 2FE$

$\Rightarrow$ (D) is incorrect

24. Let β be a real number. Consider the matrix $A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$. If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____. **[JEE(Adv)-2022]**

माना कि β एक वास्तविक संख्या है। आव्यूह $A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$ पर विचार कीजिए। यदि $A^7 - (\beta - 1)A^6 - \beta A^5$ एक अव्युत्क्रमणीय आव्यूह है, तब 9β का मान _____ है। **[JEE(Adv)-2022]**

Ans. (3)

Sol. $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, so

$|A^7 - (\beta - 1)A^6 - \beta A^5| = 0$

$|A|^5\,|(A^2 - (\beta - 1)A - \beta I)| = 0$

$|A|^5\,|A^2 - (\beta - 1)A - \beta I| = 0$

$|A|^5\,|A^2 - \beta A + A - \beta I)| = 0$

$|A|^5\,|(A + I)(A - \beta I)| = 0$

here $|A| = -1$

$$|A+I|=\begin{bmatrix}\beta+1 & 0 & 1\\ 2 & 2 & -2\\ 3 & 1 & -1\end{bmatrix}=0+1(2-6)=-4$$

$$\text{So } |A-\beta I|=\begin{bmatrix}0 & 0 & 1\\ 2 & 1-\beta & -2\\ 3 & 1 & -2-\beta\end{bmatrix}$$

$\Rightarrow 2-3+3\beta=0$
$\Rightarrow 3\beta=1$
So $9\beta=3$

25. If $M=\begin{pmatrix}\frac{5}{2} & \frac{3}{2}\\ -\frac{3}{2} & -\frac{1}{2}\end{pmatrix}$, then which of the following matrices is equal to M^{2022}? **[JEE(Adv.)2022]**

यदि $M=\begin{pmatrix}\frac{5}{2} & \frac{3}{2}\\ -\frac{3}{2} & -\frac{1}{2}\end{pmatrix}$ है, तब निम्न आव्युहों में से कौनसा M^{2022} के बराबर है ? **[JEE(Adv.)2022]**

(A) $\begin{pmatrix}3034 & 3033\\ -3033 & -3032\end{pmatrix}$ (B) $\begin{pmatrix}3034 & -3033\\ 3033 & -3032\end{pmatrix}$

(C) $\begin{pmatrix}3033 & 3032\\ -3032 & -3031\end{pmatrix}$ (D) $\begin{pmatrix}3032 & 3031\\ -3031 & -3030\end{pmatrix}$

Ans. (A)

Sol. Let $M=B+I$

$$B=\begin{bmatrix}\frac{3}{2} & \frac{3}{2}\\ -\frac{3}{2} & -\frac{3}{2}\end{bmatrix}, B^2=0 \text{ and hence } B^3, B^4.....=0$$

$$M^{2022}=(B+I)^{2022}=I+2022\ B=\begin{bmatrix}3034 & 3033\\ -3033 & -3032\end{bmatrix}$$

26. Let $M=(a_{ij})$, $i,j\in\{1,2,3\}$, be the 3×3 matrix such that $a_{ij}=1$ if $j+1$ is divisible by i, otherwise $a_{ij}=0$. Then which of the following statements is(are) true? **[JEE(Adv.)2023]**

(A) M is invertible

(B) There exists a non zero column matrix $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ such that $M\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$

(C) The set $\{X \in R^3 : MX = \mathbf{0}\} \neq \{\mathbf{0}\}$ where $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

(D) The matrix (M – 2I) is invertible, where I is the 3×3 identity matrix

Let $M = (a_{ij})$, $i, j \in \{1,2,3\}$, be the 3×3 matrix such that $a_{ij} = 1$ if $j + 1$ is divisible by i, otherwise $a_{ij} = 0$. Then which of the following statements is(are) true? **[JEE(Adv.)2023]**

माना कि 3×3 आव्यूह $M = (a_{ij})$, $i, j \in \{1,2,3\}$, इस प्रकार है कि $a_{ij} = 1$ यदि i से $j + 1$ विभाज्य (divisible) है अन्यथा $a_{ij} = 0$ है। तब निम्न में से कौनसा (से) कथन सत्य है (हैं) ? **[JEE(Adv.)2023]**

(A) M व्युत्क्रमणीय (invertible) है।

(B) एक शून्येतर (nonzero) स्तंभ आव्यूह (column matrix) $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ का अस्तित्व इस प्रकार है कि $M\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$

(C) समुच्चय $\{X \in R^3 : MX = \mathbf{0}\} \neq \{\mathbf{0}\}$, जहाँ $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

(D) आव्यूह (M – 2I) व्युत्क्रमणीय है, जहाँ I एक 3×3 तत्समक आव्यूह (identity matrix) है।

Ans. BC

Sol. $M = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$|M|=-1+1=0 \Rightarrow$ M is singular so non-invertible

(B) $M\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} -a_1 \\ -a_2 \\ -a_3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} -a_1 \\ -a_2 \\ -a_3 \end{bmatrix}$

$$\left.\begin{aligned} a_1 + a_2 + a_3 &= -a_1 \\ a_1 + a_3 &= -a_2 \\ a_2 &= -a_3 \end{aligned}\right\} \Rightarrow a_1 = 0 \text{ and } a_2 + a_3 = 0 \text{ infinite solutions exists [B] is correct.}$$

Option (D)

$$M-2I=\begin{bmatrix}1&1&1\\1&0&1\\0&1&0\end{bmatrix}-2\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}-1&1&1\\1&-2&1\\0&1&-2\end{bmatrix}$$

$|M-2\,I| = 0 \Rightarrow$ [D] is wrong

Option (C) :

$$MX=0\Rightarrow\begin{bmatrix}1&1&1\\1&0&1\\0&1&0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$x + y + z = 0$

$x + z = 0$

$y = 0$

$\therefore$ Infinite solution

[C] is correct

27. Let $R=\left\{\begin{pmatrix}a&3&b\\c&2&d\\0&5&0\end{pmatrix}: a,b,c,d\in\{0,3,5,7,11,13,17,19\}\right\}$. Then the number of invertible matrices in R is _____.

[JEE(Adv.)2023]

माना कि $R=\left\{\begin{pmatrix}a&3&b\\c&2&d\\0&5&0\end{pmatrix}: a,b,c,d\in\{0,3,5,7,11,13,17,19\}\right\}$ है। तब R में व्युत्क्रमणीय (invertible) आव्यूहों की संख्या है

_____

[JEE(Adv.)2023]

Ans. 3780

Sol. $|R| = -5\,(ad - bc)$

invertible matrices = Total – (non – invertible)

= 8^4 – (when $ad - bc = 0$)

ad = bc

<u>Case - I</u>

Non zero = non zero

(I) all are same

7 cases

(II) Two alike, two other alike

Ex : (a = b and c = d)

$= {}^7C_2 . {}^2C_1 . {}^2C_1 = 84$ cases

Case - II

zero = zero

(i) all are zero = 1 case

(ii) any three are zero $= {}^{7}C_{1} \cdot {}^{4}C_{1} = 28$

(iii) one of (a, d) and one of (b, c) is zero $= {}^{2}C_{1} \cdot {}^{2}C_{1} \cdot {}^{7}C_{1} \cdot {}^{7}C_{1}$

$=196$

So total invertible matrices

$= 8^4 - (\text{case 1}) - (\text{case 2})$

$= 4096 - (7 + 84) - (1 + 28 + 196)$

$= 3780$

Ans. 3780

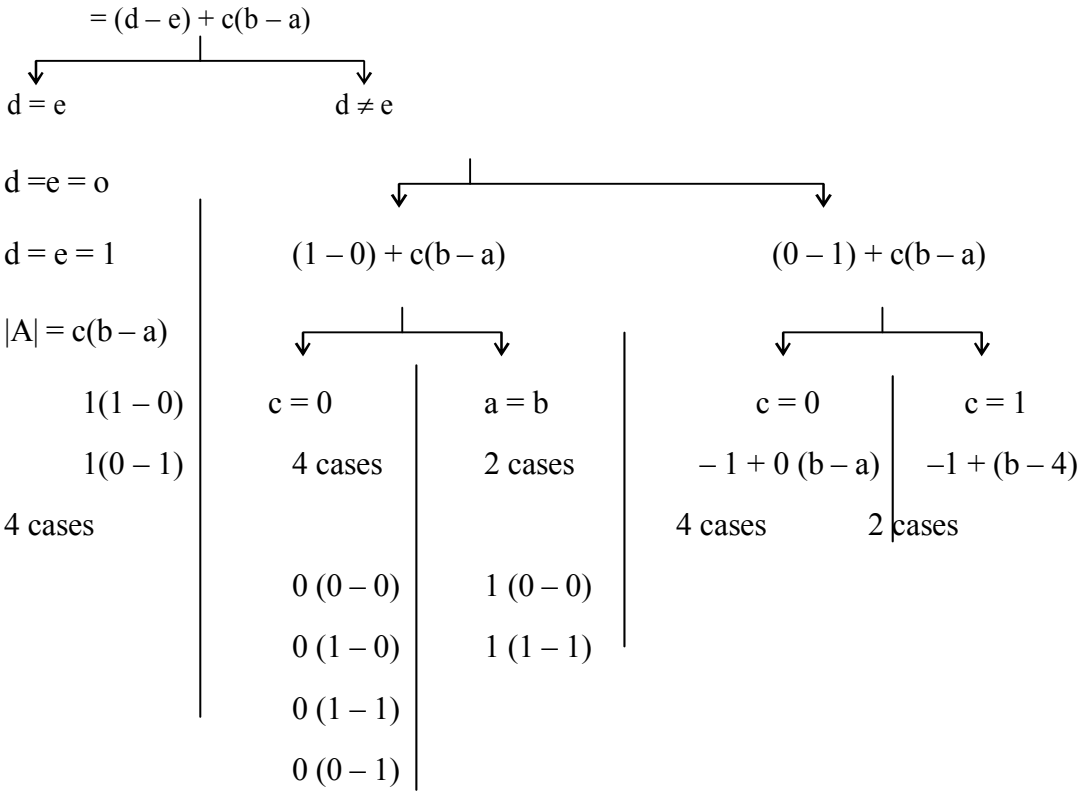
28. Let $S = \left\{ A = \begin{pmatrix} 0 & 1 & c \\ 1 & a & d \\ 1 & b & e \end{pmatrix} : a, b, c, d, e \in \{0,1\} \text{ and } |A| \in \{-1,1\} \right\}$ where $|A|$ denotes determinant of A. Then the number of elements in S is _____. **[JEE(Adv.)2024]**

माना कि $S = \left\{ A = \begin{pmatrix} 0 & 1 & c \\ 1 & a & d \\ 1 & b & e \end{pmatrix} : a, b, c, d, e \in \{0,1\} \text{ तथा } |A| \in \{-1,1\} \right\}$ जहाँ $|A|$ आव्यूह A के सारणिक को दर्शाता है। तब S में अवयवों की संख्या _____ है। **[JEE(Adv.)2024]**

Ans. 16

Sol. $|A| = \begin{vmatrix} 0 & 1 & c \\ 1 & a & d \\ 1 & b & e \end{vmatrix}$ $\quad a, b, c, d, e \in \{0,1\}$

$|A| \in \{-1,1\}$



Total 16 cases

29. Let α and β be the distinct roots of the equation $x^2 + x - 1 = 0$. Consider the set $T = \{1, \alpha, \beta\}$. For a 3×3 matrix $M = (a_{ij})_{3\times3}$, define $R_i = a_{i1} + a_{i2} + a_{i3}$ and $C_j = a_{1j} + a_{2j} + a_{3j}$ for $i = 1,2,3$ and $j = 1,2,3$.

Match each entry in **List-I** to the correct entry in **List-II**. **[JEE(Adv.)2024]**

List-I	**List-II**
(P) The number of matrices $M = (a_{ij})_{3\times3}$ with all entries in T such that $R_i = C_j = 0$ for all i, j, is	(1) 1
(Q) The number of symmetric matrices $M = (a_{ij})_{3\times3}$ with all entries in T such that $C_j = 0$ for all j, is	(2) 12
(R) Let $M = (a_{ij})_{3\times3}$ be a skew symmetric matrix such that $a_{ij} \in T$ for $i > j$. Then the number of elements in the set $\left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} : x, y, z \in R, M\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{12} \\ 0 \\ -a_{23} \end{pmatrix}\right\}$ is	(3) infinte
(S) Let $M = (a_{ij})_{3\times3}$ be a matrix with all entries in T such that $R_i = 0$ for all i. Then the absolute value of the determinant of M is	(4) 6
	(5) 0

The correct option is :

(A) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (1)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (3); (S) $\rightarrow$ (4)

माना कि α और β समीकरण $x^2 + x - 1 = 0$ के भिन्न मूल हैं। समुच्चय $T = \{1, \alpha, \beta\}$ पर विचार कीजिये। एक 3×3 आव्यूह M ¥ ; $_{ij})_{3\times3}$ के लिए, $R_i = a_{i1} + a_{i2} + a_{i3}$ और $C_j = a_{1j} + a_{2j} + a_{3j}$ परिभाषित कीजिए, जहाँ $i = 1,2,3$ और $j = 1,2,3$ है।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए। **[JEE(Adv.)2024]**

सूची-I	**सूची-II**
(P) आव्यूहों $M = (a_{ij})_{3\times3}$ जिनकी सभी प्रविष्टियाँ ज से हैं और जिनमें सभी i, j के लिए $R_i = C_j = 0$ है, की संख्या है	(1) 1
(Q) सममित आव्यूहों $M = (a_{ij})_{3\times3}$ जिनकी $M = (a_{ij})_{3\times3}$ सभी प्रविष्टियाँ T से हैं और जिनमें सभी j के लिए $C_j = 0$ है, की संख्या है	(2) 12
(R) माना कि $M = (a_{ij})_{3\times3}$ एक ऐसा विषम सममित आव्यूह है कि $i > j$ के लिए $a_{ij} \in T$ है। तब समुच्चय $\left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} : x, y, z \in R, M\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{12} \\ 0 \\ -a_{23} \end{pmatrix}\right\}$ में अवयवों की संख्या है	(3) अनन्त
(S) माना कि $M = (a_{ij})_{3\times3}$ एक ऐसा आव्यूह है कि सभी प्रविष्टियाँ T से हैं, और जिसमें सभी i के लिए $R_i = 0$ है। तब M के सारणिक का निरपेक्ष मान है	(4) 6
	(5) 0

सही विकल्प है :

(A) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (1)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (3); (S) $\rightarrow$ (4)

Ans. C

Sol. $x^2 + x - 1 = 0$

$\alpha + \beta = -1$ $\quad\quad$ $\alpha\beta = -1$

set $T = \{1, \alpha, \beta\}$ $\quad\quad$ $M = (a_{ij})_{3\times3}$

$R_i = a_{i1} + a_{i2} + a_{i3}$ $\quad\quad$ $C_j = a_{1j} + a_{2j} + a_{3j}$

(P) $R_i = C_j = 0$ for all i, j

$\alpha + \beta = -1$ $\quad T = \{1, \alpha, \beta\}$

Number of matrices

$\underline{|3} \times \underline{|2} \times \underline{|1} = 12$

Number of ways to arrange $1,\alpha,\beta$ in R_1

Number of ways to arrange $1,\alpha,\beta$ in R_2

$\left[1 \ \underline{\alpha} \ \underline{\beta}\right]$

(Q) Number of symmetric matrices = ?

$C_j = 0 \ \forall \ j$

Number of symmetric matrices

$= \underline{|3} \times 1 = 6$

$$\begin{bmatrix} 1 & \alpha & \beta \\ \alpha & \beta & 1 \\ \beta & 1 & \alpha \end{bmatrix}$$

(R) $M \rightarrow$ skew symmetric of 3×3

$|M| = 0$ $\quad a_{ij} \in T$ for $i > j$

$$M\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{12} \\ 0 \\ -a_{23} \end{pmatrix}$$

$$\begin{bmatrix} 0 & -a_{21} & -a_{31} \\ a_{21} & 0 & -a_{32} \\ a_{31} & a_{32} & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a_{12} \\ 0 \\ -a_{23} \end{bmatrix}$$

As $x, y, z \in R$ and a_{12} & $a_{23} \in R$

& $|M| = 0$

$\therefore$ System has infinite solutions.

(S) $R_i = 0 \ \forall \ i$

$$M = \begin{bmatrix} 1 & \alpha & \beta \\ \alpha & \beta & 1 \\ \beta & 1 & \alpha \end{bmatrix}$$

$C_1 \to C_1 + C_2 + C_3 \qquad |M| = \begin{vmatrix} 1+\alpha+\beta & \alpha & \beta \\ 1+\alpha+\beta & \beta & 1 \\ 1+\alpha+\beta & 1 & \alpha \end{vmatrix} = 0$ $\quad \begin{pmatrix} \because \alpha+\beta=-1 \\ \Rightarrow 1+\alpha+\beta=0 \end{pmatrix}$

$(P) \to 2, (Q) \to (4), (R) \to (3), (S) \to (5)$

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ and $I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction :

यदि $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ तथा $I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ है, तो गणितीय आगमन द्वारा ज्ञात कीजिए कि $n \geq 1$ के लिए निम्न में से कौन सा सत्य है :

(A) $A^n = nA - (n-1)I$ (B) $A^n = 2^{n-1}A - (n-1)I$

(C) $A^n = nA + (n-I)I$ (D) $A^n = 2^{n-1}A + (n-1)I$

Ans. A

Sol. Given, $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

Now, $A^2=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$; $A^2=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$; $A^3=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}$

Similarly, $A^n=\begin{bmatrix}1 & 0\\ n & 1\end{bmatrix}$ can be verified by induction.

Now, go option by option

(b) $\begin{bmatrix}1 & 0\\ n & 1\end{bmatrix} \neq \begin{bmatrix}n & 0\\ n & n\end{bmatrix}+\begin{bmatrix}n-1 & 0\\ 0 & n-1\end{bmatrix}$

(d) $nA-(n-1)I=\begin{bmatrix}n & 0\\ n & n\end{bmatrix}-\begin{bmatrix}n-1 & 0\\ 0 & n-1\end{bmatrix}=\begin{bmatrix}1 & 0\\ n & 1\end{bmatrix}$

Hence, option (d) is incorrect.

2. If $P=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}$, $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $Q = PAP^T$ and $x = P^TQ^{2005}P$, then x is equal to :

यदि $P=\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ तथा $Q = PAP^T$ तथा $x = P^TQ^{2005}P$ है, तो x का मान है :

(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 4+2005\sqrt{3} & 6015 \\ 2005 & 4-2005\sqrt{3} \end{bmatrix}$

(C) $\frac{1}{4}\begin{bmatrix} 2+\sqrt{3} & 1 \\ -1 & 2-\sqrt{3} \end{bmatrix}$ (D) $\frac{1}{4}\begin{bmatrix} 2005 & 2-\sqrt{3} \\ 2+\sqrt{3} & 2005 \end{bmatrix}$

Ans. A

Sol. $P^TP = I$

$A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow A^2=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and so on

$Q = PAP^T$

$Q^2 = (PAP^T)(PAP^T) = PA^2P^T$

$Q^{2005} = PA^{2005}P^T$

$x = P^T(PA^{2005}P^T)P \Rightarrow x = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

3. Let $A=\begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$ and suppose that det.(A) = 2 then the det.(B) equals, where $B=\begin{bmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{bmatrix}$:

माना कि $A=\begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$ तथा det.(A) = 2 तो det.(B) का मान कितना होगा जहाँ $B=\begin{bmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{bmatrix}$:

(A) det(B) = – 2 (B) det(B) = – 8 (C) det(B) = – 16 (D) det(B) = 8

Ans. C

Sol. $\det(B) = \begin{vmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{vmatrix} = (4)(2)(-1)\begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix} = -8\begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} = -8\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$

$= -8 \times 2 = -16$ Ans.

4. If A is a non singular maxrix satisfying $A = AB - BA$, then which one of the following holds true :

यदि A एक व्युत्क्रमणीय आव्यूह है जो $A = AB - BA$, के सन्तुष्ट करती है तब कौन सा कथन सत्य है :

(A) det. B = 0 (B) B = 0 (C) det. A = 1 (D) det. (B + I) = det. (B – I)

Ans. D

Sol. A is non singular $\det A \neq 0$

Given $AB - BA = A$ hence $AB = A + BA = A(I + B)$

$\det. A \cdot \det. B = \det. A \cdot \det. (I + B)$ $(\det A \neq 0)$

$\det. B = \det. (I + B)$ ....(1) (as A is non singular)

again $AB - A = BA$

$A(B - I) = BA$

$(\det. A) \cdot \det.(B - I) = \det. B \cdot \det. A \Rightarrow \det. (B - I) = \det. (B)$ ....(2)

from (1) and (2)

$\det. (B - I) = \det. (B + I)$ Ans.

5. $A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ then $B^T A^T$ is :

(A) a null matrix (B) an identity matrix

(C) scalar, but not an identity, matrix (D) such that $\text{tr}(B^TA^T) = 4$

$A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ तथा $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ तो $B^T A^T$ होगा :

(A) शून्य आव्यूह (B) इकाई आव्यूह

(C) सदिश (लेकिन इकाई नहीं) आव्यूह है (D) $\text{tr}(B^TA^T) = 4$

Ans. B

Sol. Compute $(AB)^T$ which is an identity matrix and $(AB)^T = B^TA^T$

6. If $A = \begin{bmatrix} \cos^2\alpha & \sin\alpha\cos\alpha \\ \sin\alpha\cos\alpha & \sin^2\alpha \end{bmatrix}; B = \begin{bmatrix} \cos^2\beta & \sin\beta\cos\beta \\ \sin\beta\cos\beta & \sin^2\beta \end{bmatrix}$ are such that, AB is a null matrix, then which of the following should necessarily be an odd integral multiple of $\frac{\pi}{2}$:

यदि $A=\begin{bmatrix} \cos^2\alpha & \sin\alpha\cos\alpha \\ \sin\alpha\cos\alpha & \sin^2\alpha \end{bmatrix}$; $B=\begin{bmatrix} \cos^2\beta & \sin\beta\cos\beta \\ \sin\beta\cos\beta & \sin^2\beta \end{bmatrix}$ इस प्रकार है, कि AB एक शून्य आव्यूह है तब

cr kb; sfuEu esa sdkSu vo' ; gh $\frac{\pi}{2}$ का विषम पूर्णांक गुणांक होगा :

(A) α (B) β (C) $\alpha - \beta$ (D) $\alpha + \beta$

Ans. C

Sol. $AB=\begin{pmatrix} \cos^2\alpha & \sin\alpha\cos\alpha \\ \sin\alpha\cos\alpha & \sin^2\alpha \end{pmatrix}\begin{pmatrix} \cos^2\beta & \sin\beta\cos\beta \\ \sin\beta\cos\beta & \sin^2\beta \end{pmatrix}$

$=\begin{pmatrix} \cos^2\alpha\cos^2\beta+\sin\alpha\cos\alpha\sin\beta\cos\beta & \cos^2\alpha\sin\beta\cos\beta+\sin\alpha\cos\alpha\sin^2\beta \\ \cos^2\beta\sin\alpha\cos\alpha+\sin^2\alpha\sin\beta\cos\beta & \sin\alpha\cos\alpha\sin\beta\cos\beta+\sin^2\alpha\sin^2\beta \end{pmatrix}$

$=\begin{pmatrix} \cos\alpha\cos\beta\cos(\alpha-\beta) & \cos\alpha\sin\beta\cos(\alpha-\beta) \\ \sin\alpha\cos\beta\cos(\alpha-\beta) & \sin\alpha\sin\beta\cos(\alpha-\beta) \end{pmatrix}$

$\Rightarrow$ $\alpha - \beta$ must be an odd integral multiple of $\pi/2$ $\Rightarrow$ (C)

7. Let A, B, C, D be (not necessarily square) real matrices such that $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ and $D^T = ABC$ for the matrix $S = ABCD$, consider the two statements.

I $S^3 = S$

II $S^2 = S^4$

(A) II is true but not I (B) I is true but not II

(C) both I and II are true (D) both I and II are false

माना A, B, C, D (जरूरी नहीं है कि वर्ग आव्यूह हो) वास्तविक आव्यूह है तथा $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ तथा $D^T = ABC$ आव्यूह $S = ABCD$, के लिए, दो कथनों पर विचार कीजिए :

I. $S^3 = S$

II. $S^2 = S^4$

(A) ppp सत्य है तथा I नहीं (B) p सत्य है तथा II नहीं

(C) p तथा II दोनों सत्य है (D) p तथा II दोनों असत्य है

Ans. C

Sol. $S = ABCD = A(BCD) = AA^T$ ....(1)

$S^3 = (ABCD)(ABCD)(ABCD)$

$= (ABC)(DAB)(CDA)(BCD)$

$= D^TC^TB^TA^T$

$= (BCD)^TA^T = AA^T$ ....(2)

from (1) and (2) $\boxed{S = S^3}$ $\Rightarrow$ I is correct

multiply both sides by S

$\boxed{S^2 = S^4}$ $\Rightarrow$ II is correct $\Rightarrow$ Both I and II are true Ans.

8. There are two possible values of A in the solution of the matrix equation $\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$ where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions, is :

यदि आव्यूह समीकरण $\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$ के हल में A के दो सम्भव मान है तथा A, B, C, D, E, F वास्तविक संख्याऐं हैं तो इन दोनों हलों के अन्तर का मापांक क्या होगा :

(A) $\frac{8}{3}$ (B) $\frac{11}{3}$ (C) $\frac{1}{3}$ (D) $\frac{19}{3}$

Ans. D

Sol. If $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj. } P = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

also $A^{-1} = \frac{\text{adj.A}}{\text{det.A}}$

$\therefore \quad \begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} = \frac{\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}}{2A^2+A-20}$

$\therefore \quad \frac{1}{2A^2+A-20} \begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$

The gives, $\frac{A^2+5A-10}{2A^2+A-20} = 14 \quad \Rightarrow A = 3 \text{ or } -\frac{10}{3}$

$\therefore \quad 3 + \frac{10}{3} = \frac{19}{3}$ Ans.

9. Number of real values of λ for which the matrix $A = \begin{bmatrix} \lambda-1 & \lambda & \lambda+1 \\ 2 & -1 & 3 \\ \lambda+3 & \lambda-2 & \lambda+7 \end{bmatrix}$ has no inverse :

(A) 0 (B) 1 (C) 2 (D) infinite

λ के कुल कितने मान होगें जब आव्यूह $A=\begin{bmatrix}\lambda-1 & \lambda & \lambda+1\\ 2 & -1 & 3\\ \lambda+3 & \lambda-2 & \lambda+7\end{bmatrix}$ व्युत्क्रमणीय नही होगा :

(A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. D

Sol. $|A|=0$ for all $\lambda \in R \Rightarrow A$ is singular. Hence inverse can not be found for any value of $\lambda \in R \Rightarrow$ (D)

Use $R_2 \to R_2 - \frac{1}{2}(R_3 - R_1)$

10. Let $A=\begin{bmatrix}l & m & n\\ p & q & r\\ 1 & 1 & 1\end{bmatrix}$ and $B=A^2$. If $(l-m)^2+(p-q)^2=9$, $(m-n)^2+(q-r)^2=16$, $(n-l)^2+(r-p)^2=25$, then the value of det. B equals :

माना $A=\begin{bmatrix}l & m & n\\ p & q & r\\ 1 & 1 & 1\end{bmatrix}$ तथा $B=A^2$ यदि $(l-m)^2+(p-q)^2=9$, $(m-n)^2+(q-r)^2=16$, $(n-l)^2+(r-p)^2=25$, तो det. B का मान होगा :

(A) 36 (B) 100 (C) 144 (D) 169

Ans. C

Sol. det. A is twice the area of the triangle with vertices (l, p), (m, q), (n, r) with sides 3, 4, 5.

$\Delta^2 = s(s-a)(s-b)(s-c)$

$\Delta^2 = 6(6-3)(6-4)(6-5)$

$\Delta^2 = 36 \quad \Rightarrow \quad D = 6$

Now $\det A = 2\Delta = 12 \quad \Rightarrow \quad \det. B = (\det A)^2 = 4\Delta^2 = 144$ Ans.

11. In a square matrix A of order 3 the elements, a_{ii}'s are the sum of the roots of the equation $x^2-(a+b)x+ab=0$; $a_{i,i+1}$'s are the product of the roots, $a_{i,i-1}$'s are all unity and the rest of the elements are all zero. The value of the det. (A) is equal to :

यदि कोटि 3 के वर्ग आव्यूह A के लिए, अवयवों a_{ii}'s समीकरण में $x^2-(a+b)x+ab=0$ के मूलों का योगफल है। $a_{i,i+1}$'s मूलों का xqkuQy gS$a_{i,i-1}$'s इकाई है तथा बचे हुए अवयव शून्य है तब det. (A) का मान बताओ :

(A) 0 (B) $(a+b)^3$ (C) a^3-b^3 (D) $(a^2+b^2)(a+b)$

Ans. D

Sol. Given $a_{11}=a_{22}=a_{33}=a+b$

$a_{12} = a_{23} = ab$; $a_{21} = 1 = a_{32}$, all others are zero.

$$\text{Det (A)} = \begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix} = (a^2 + b^2)(a + b)]$$

12. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$. If AB = I, where I is an identity matrix of order 3 then trace B has value equal to :

माना $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$ यदि AB = I, जहाँ I, 3 कोटि का इकाई आव्यूह है तो आव्यूह B का

vuqjs[kd ke ku gksxk:

(A) 0 (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) 5

Ans. B

Sol. We have AB = I, so

$$\begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5\alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10\alpha - 2 & 5\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (Given)}$$

$\Rightarrow 5\alpha = 1 \Rightarrow \alpha = \frac{1}{5}$

Now, trace (B) $= (-\alpha) + 1 + (-2\alpha) = 1 - 3\alpha = 1 - 3\left(\frac{1}{5}\right) = 1 - \frac{3}{5} = \frac{2}{5}$. Ans.

13. Let D_k is the k × k matrix with 0's in the main diagonal, unity as the element of 1st row and $\left(f(k)\right)^{th}$ column and k for all other entries. If $f(x) = x - \{x\}$ where $\{x\}$ denotes the fractional part function then the value of det. (D_2) + det. (D_3) equals :

(A) 32 (B) 34 (C) 36 (D) None

माना D_k, $k \times k$ का एक आव्यूह है, जिसके विकर्ण अवयव '0' है, प्रथम पंक्ति तथा $\left(f(k)\right)^{th}$ स्तम्भ में 1 है तथा अन्य प्रविष्टियाँ k है। ; fn $f(x) = x - \{x\}$ जहाँ पर $\{x\}$ अंश भाग फलन को व्यक्त करता है, तो सारणिक (D_2) तथा सारणिक (D_3) का योग होगा :

(A) 32 (B) 34 (C) 36 (D) कोई नहीं

Ans. B

Sol. Det. $(D_2) = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix}$ $(f(2) = 2 \Rightarrow 1^{st}$ and 2nd column $= 1)$

||||ly Det. $(D_3) = \begin{vmatrix} 0 & 3 & 1 \\ 3 & 0 & 3 \\ 3 & 3 & 0 \end{vmatrix}$ $(f(3) = 3 \Rightarrow a_{13} = 1)$

Det. $(D_2) = -2$; Det. $(D_3) = 36$

$\Rightarrow$ Det. (D_2) + Det. $(D_3) = 34$ Ans.

14. Consider a matrix $A(\theta) = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}$ then :

(A) $A(\theta)$ is symmetric (B) $A(\theta)$ is skew symmetric

(C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$

आव्यूह $A(\theta) = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}$ पर विचार कीजिए तब :

(A) ।;θ) सममित है (B) ।;θ) विषम सममित है

(C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$

Ans. C

Sol. As $A(\theta) = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}$

$A(\theta)$ is certainly neither symmetric nor skew symmetric

Further, $A(\pi - \theta) = \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

and $A(\theta) \cdot A(\pi - \theta) = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^{-1}(\theta) = A(\pi - \theta)$

$$A^2(\theta)=\begin{bmatrix}\sin\theta & \cos\theta\\ -\cos\theta & \sin\theta\end{bmatrix}\begin{bmatrix}\sin\theta & \cos\theta\\ -\cos\theta & \sin\theta\end{bmatrix}=\begin{bmatrix}-\cos 2\theta & \sin 2\theta\\ -\sin 2\theta & -\cos 2\theta\end{bmatrix}\neq a\left(\frac{\pi}{2}-2\theta\right)$$

15. Let A and B be 3×3 symmetric matrices such that $X = AB + BA$ and $Y = AB - BA$, then $(XY)^T$ is equal to : (Note : $(XY)^T$ is the transpose of the matrix XY)

माना A तथा B, 3×3 के सममित आव्यूह है जिनके लिये $X = AB + BA$ तथा $Y = AB - BA$, तो $(XY)^T$ बराबर होगा। (नोट : $(XY)^T$, XY का परिवर्त आव्यूह है)

(A) XY (B) YX (C) $-XY$ (D) $-YX$

Ans. D

Sol. We have $A = A^T$ and $B = B^T$

$X = AB + BA \Rightarrow X = X^T$ and $Y = AB - BA \Rightarrow Y = -Y^T$

Now, $(XY)^T = Y^T X^T = -YX$. Ans.

16. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}4 & 0\\ -1 & 1\end{bmatrix}$, then number of distinct values of α for which $A^2 = B$ equals :

यदि $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ तथा $B=\begin{bmatrix}4 & 0\\ -1 & 1\end{bmatrix}$, तो α के भिन्न मानो की संख्या जिसके के लिये $A^2 = B$ बराबर है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. We have

$A^2 = B$ (Given)

$$\Rightarrow\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}=\begin{bmatrix}4 & 0\\ -1 & 1\end{bmatrix}\Rightarrow\begin{bmatrix}\alpha^2 & 0\\ \alpha+1 & 1\end{bmatrix}=\begin{bmatrix}4 & 0\\ -1 & 1\end{bmatrix}$$

So, $\alpha^2 = 4$ and $\alpha + 1 = -1 \Rightarrow \alpha = \pm 2$ and $\alpha = -2$

$\therefore$ a $= -2$

Hence, only one possible value of $\alpha = -2$ exist. Ans.

17. If A is a diagonal matrix of order 3 such that $A^2 = A$, then number of possible matrices A, is $(A \neq O)$:

यदि A, तीन कोटि का विकर्ण आव्यूह $A^2 = A$ है, तो संभव आव्यूहों A की संख्या होगी $(A \neq O)$:

(A) 2 (B) 3 (C) 8 (D) 7

Ans. D

Sol. $A^2 = A$

$\Rightarrow$ $d_1, d_2, \ldots\ldots d_n$ be the diagonal matrix of A, then

$A^2 = A \Rightarrow d_1^2 = d_1, d_2^2 = d_2, \ldots\ldots, d_n^2 = d_n$

Now, $d_1^2 = d_1 \Rightarrow$ either $d_i = 0$ or $d_i = 1$

Number of ways in which diagonal elements can be filled up is 2^3, but all diagonal elements cannot be zero.

$\therefore$ Number of required matrices is $2^3 - 1 = 7$.Ans.

18. If $\omega \neq 1$ is the complex cube root of unity and matrix $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then H^{70} is equal to : **[AIEEE-2011]**

यदि $\omega \neq 1$ इकाई का एक सम्मिश्र घन मूल है तथा आव्यूह $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$ हैं, तो H^{70} बराबर है : **[AIEEE-2011]**

(A) H (B) 0 (C) –H (D) H^2

Ans. A

Sol. Since H is a diagonal matrix. We know that product of two diagonal matrix is always a diagonal matrix.

So $H^{70} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}\ldots\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$ 70 times

$= \begin{bmatrix} \omega^{70} & 0 \\ 0 & \omega^{70} \end{bmatrix} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = H$

19. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where each of a. b and c is either ω of ω^2. Then the number of distinct matrices in the set S is : **[JEE-2011]**

माना $\omega \neq 1$ इकाई का घनमूल है तथा S वह सभी व्युत्क्रमणीय आव्यूह $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$ का सम्मुचय है, जहाँ a. b तथा c ω या ω^2 में

l s, d eku y s sg\$r ks l Eeq; S में सम्भव भिन्न आव्यूहों की संख्या होगी : **[JEE-2011]**

(A) 2 (B) 6 (C) 4 (D) 8

Ans. A

Sol. $\begin{vmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{vmatrix}$

$=1-c\omega - a(\omega - \omega^2 c) = (1 - c\omega) - a\omega(1 - c\omega) = (1 - c\omega)(1 - a\omega)$

for non singular matrix

$$c \neq \frac{1}{\omega} \text{ and } a \neq \frac{1}{\omega}$$

$\Rightarrow$ $c \neq \omega^2$, $a \neq \omega^2$

$\Rightarrow$ a and c must be ω and b can be ω or ω^2

$\therefore$ total matrices = 2

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

20. If A and B are different matrices satisfying $A^3 = B^3$ and $A^2B = B^2A$, then :

(A) det $(A^2 + B^2)$ must be zero

(B) det $(A - B)$ must be zero

(C) det $(A^2 + B^2)$ as well as det $(A - B)$ must be zero

(D) At least one of det $(A^2 + B^2)$ or det $(A - B)$ must be zero

यदि A एवं B भिन्न–भिन्न आव्यूह समीकरणें $A^3 = B^3$ तथा $A^2B = B^2A$ को संतुष्ट करते है, तो :

(A) det $(A^2 + B^2) = 0$

(B) det $(A - B) = 0$

(C) कम्अज़ । $^2 + B^2)$ तथा det $(A - B)$ दोनों शून्य होगा

(D) कम्अज़ । $^2 + B^2)$ तथा det $(A - B)$ में कम से कम एक शून्य अवश्य होगा

Ans. AD

Sol. $A^3 = B^3$ .....(1)

and $A^2B = B^2A$ .....(2)

(1) – (2) gives, $A^3 - A^2B = B^3 - B^2A$

$A^2(A - B) = -B^2(A - B) \Rightarrow (A^2 + B^2)(A - B) = 0$

$\det (A^2 + B^2)(A - B) = 0$

$\det (A^2 + B^2) \cdot \det (A - B) = 0 \Rightarrow$ (D)

21. Given the matrices A and B as $A = \begin{bmatrix} 1 & -1 \\ 4 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$. The two matrices X and Y are such that XA = B and AY = B then which of the following hold(s) true?

दिये गये आव्यूह A तथा B जहाँ $A = \begin{bmatrix} 1 & -1 \\ 4 & -1 \end{bmatrix}$ और $B = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$ दो आव्यूह X और Y इस प्रकार हैं कि XA = B और AY = B तो निम्न में से कौन सा कथन सत्य है?

(A) $X = \frac{1}{3}\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$ (B) $Y = \frac{1}{3}\begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix}$ (C) det. X = det. Y (D) $3(X+Y) = \begin{bmatrix} 4 & -1 \\ 4 & 2 \end{bmatrix}$

Ans. CD

Sol. Note that A is non singular but B is singular hence only A^{-1} exists

Now $XA = B \Rightarrow X = BA^{-1}$ _____(1)

and $AY = B \Rightarrow Y = A^{-1}B$ _____(2)

Also $A^{-1} = \frac{1}{3}\begin{bmatrix} -1 & 1 \\ -4 & 1 \end{bmatrix}$

now verify

22. If there are three square matrix A, B, C of same order satisfying the equation $A^2 = A^{-1}$ and let $B = A^{2^n}$ and $C = A^{2^{(n-2)}}$ then which of the following statements are true?

(A) det. (B – C) = 0 (B) (B + C)(B – C) = 0

(C) B must be equal to C (D) None

यदि तीन वर्ग आव्यूह A, B, C समीकरण $A^2 = A^{-1}$ एवं माना $B = A^{2^n}$ तथा $C = A^{2^{(n-2)}}$ है, तो?

(A) det. (B – C) = 0 (B) (B + C)(B – C) = 0

(C) आव्यूह B, C के समान है (D) कोई नहीं

Ans. ABC

Sol. $B = A^{2^n} = A^{2\cdot 2^{n-1}} = (A^2)^{2^{n-1}} = (A^{-1})^{2^{n-1}} = \left(A^{2^{n-1}}\right)^{-1} = \left(A^{2\cdot 2^{n-2}}\right)^{-1} = \left((A^2)^{2^{n-2}}\right)^{-1}$

$= \left((A^{-1})^{-1}\right)^{2^{n-2}} = A^{2^{(n-2)}}$

so B = C

(B – C) = null $\Rightarrow$ (B + C)(B – C) = 0 $\Rightarrow$ det. (B – C) = 0

$\Rightarrow$ A, B, C are answers

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. $A_{3\times 3}$ is a matrix such that $|A| = a$, $B = (\text{adj } A)$ such that $|B| = b$. Find the value of $(ab^2 + a^2b + 1)S$ where $\frac{1}{2}S = \frac{a}{b} + \frac{a^2}{b^3} + \frac{a^3}{b^5} + \ldots\ldots$ up to ∞, and $a = 3$.

$I_{3\times 3}$ एक आव्यूह इस प्रकार है कि $|A| = a$, $B = (\text{adj } A)$ इस प्रकार है कि $|B| = b$ है, तो $(ab^2 + a^2b + 1)S$ का मान ज्ञात कीजिये।

जहा $\frac{1}{2}S = \frac{a}{b} + \frac{a^2}{b^3} + \frac{a^3}{b^5} + \ldots\ldots$ ∞ तक तथा $a = 3$ है।

Ans. 225

Sol. $|A| = a = 3$, $|B| = b = |\text{adj. } A| = |A|^2 = 9$

$$\frac{1}{2}S = \frac{\frac{a}{b}}{1-\frac{a}{b^2}} = \frac{9}{26}; S = \frac{9}{13}$$

$(ab^2 + a^2b + 1)S = 225$

2. Given the matrix $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ and X be the solution set of the equation $A^x = A$, where $x \in N - \{1\}$.

Evaluate $\prod\left(\frac{x^3+1}{x^3-1}\right)$ where the continued product extends $\forall\, x \in X$.

माना आव्यूह $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ दिया गया है तथा X समीकरण $A^x = A$ के हलों का समुच्चय है, जहाँ $x \in N - \{1\}$ है , तो

$\prod\left(\frac{x^3+1}{x^3-1}\right)$ का मान ज्ञात कीजिए। जहाँ $\forall\, x \in X$ के लिए सतत गुणनफल है।

Ans. $\frac{3}{2}$

Sol. $A^2 = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}\begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

$A^2 = A^3 = A^4 = \ldots\ldots\ldots\ldots = A$

$\Rightarrow x = 2,3,4,5,\ldots\ldots\ldots$

$$\Pi\left(\frac{x^3+1}{x^3-1}\right) = \left(\Pi\left(\frac{x+1}{x-1}\right)\right)\left(\Pi\left(\frac{x^2-x+1}{x^2+x+1}\right)\right)$$

$$\Rightarrow \lim_{n\to\infty}\left(\prod_{x=2}^{n}\frac{x+1}{x-1}\right)\left(\prod_{x=2}^{n}\frac{x^2-x+1}{x^2+x+1}\right)$$

$$\lim_{n\to\infty}\left(\frac{3\cdot4\cdot5\ldots\ldots n+1}{1\cdot2\cdot3\ldots\ldots n-1}\right)\left(\frac{3\cdot7\cdot\ldots\ldots\left(n^2-n+1\right)}{7\cdot13\cdot\ldots\ldots\left(n^2+n+1\right)}\right)$$

$$\Rightarrow \lim_{n\to\infty}\frac{n(n+1)}{2}\times\frac{3}{n^2+n+1}\Rightarrow\frac{3}{2}$$

3. If A is a skew symmetric matrix and I + A is non singular, then prove that the matrix $B = (I - A)(I + A)^{-1}$ is an orthogonal matrix. Use this to find a matrix B given $A = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}$.

यदि A एक विषम सममित आव्यूह है तथा I + A व्युत्क्रमणीय है, तो सिद्ध कीजिये आव्यूह $B = (I - A)(I + A)^{-1}$ एक लम्ब कोणीय vkO;wg gksxkA bl dk mi;ksx djrs gq, vkO;wg B ज्ञात कीजिए यदि दिया गया है कि $A = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}$

Ans. $\frac{1}{13}\begin{bmatrix} -12 & -5 \\ 5 & -12 \end{bmatrix}$

Sol. $B = (I - A)(I + A)^{-1}$

$B^T = ((I + A)^{-1})^T (I - A)^T$

$B^T = (I + A^T)^{-1} (I - A^T)$

$BT = (I - A)^{-1} (I + A)$

$\Rightarrow B^TB = (I - A)^{-1} (I + A) (I - A) (I + A)^{-1}$

$= (I - A)^{-1} (I - A) (I + A) (I + A)^{-1}$

$\Rightarrow I$

$\Rightarrow$ B is an orthogonal matrix.

$$B = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}\right)\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}\right)^{-1}$$

$$= \begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix}\begin{bmatrix} 1 & 5 \\ -5 & 1 \end{bmatrix}^{-1}$$

$$\Rightarrow \frac{1}{26}\begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix}\begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix} \Rightarrow \frac{1}{26}\begin{bmatrix} -24 & -10 \\ 10 & -24 \end{bmatrix}$$

4. Given that $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 10 \\ 13 \\ 9 \end{bmatrix}$ and that CB = D. Solve the matrix equation Ax = B.

दिया गया है कि $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 10 \\ 13 \\ 9 \end{bmatrix}$ तथा CB = D आव्यूह समीकरण Ax = B को हल कीजिये।

Ans. $x_1 = 1, x_2 = -1, x_3 = 1$

Sol. CB = D

AX = B

CAX = CB = D

$X = (CA)^{-1} D$

$$CA = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 5 & 10 \\ 7 & 7 & 13 \\ 4 & 3 & 8 \end{bmatrix}$$

$$(CA)^{-1} = \begin{bmatrix} -\frac{17}{5} & 2 & 1 \\ \frac{4}{5} & 0 & -1 \\ \frac{7}{5} & -1 & 0 \end{bmatrix}$$

$$X = (CA)^{-1} D = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

5. If $A = \begin{bmatrix} k & m \\ \ell & n \end{bmatrix}$ and $kn \neq \ell m$; then show that $A^2 - (k + n)A + (kn - \ell m) I = O$. Hence find A^{-1}.

यदि $A = \begin{bmatrix} k & m \\ \ell & n \end{bmatrix}$ तथा $kn \neq \ell m$ है, तो सिद्ध कीजिये $A^2 - (k + n)A + (kn - \ell m) I = O$ तत्पश्चात A^{-1} ज्ञात कीजिये।

Ans. $\frac{1}{kn - \ell m} \begin{bmatrix} n & -m \\ -\ell & k \end{bmatrix}$

Sol. $\begin{vmatrix} k - \lambda & m \\ \ell & n - \lambda \end{vmatrix} = 0$

$\lambda^2 - \lambda(k + n) + (kn - \ell m) = 0$

Put $\lambda = A$

6. Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. Find all possible matrix X in the following cases.

दिया गया है कि $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$ जहाँ I दो कोटि का इकाई आव्यूह है । सभी संभव आव्यूह X ज्ञात कीजिये यदि

(i) $AX = A$ (ii) $XA = I$ (iii) $XB = O$ but $BX \neq O$

Ans. (i) $X = \begin{bmatrix} a & b \\ 2 - 2a & 1 - 2b \end{bmatrix}$ for $a, b \in R$; (ii) X does not exist;

(iii) $X = \begin{bmatrix} a & -3a \\ c & -3c \end{bmatrix}$ a, c $\in$ R and $3a + c \neq 0$; $3b + d \neq 0$

Sol. (i) $AX = A$

let $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2a+c & 2b+d \\ 2a+c & 2b+d \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$2a + c = 2$, $c = 2(1 - a)$, $a \in R$

$2b + d = 1$, $d = 1 - 2b$, $b \in R$

(ii) $XA = I$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2(a+b) & a+b \\ 2(c+d) & c+d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$ which is not possible

(iii) $XB = 0$ but $BX \neq 0$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3(3a+b) & 3a+b \\ 3(3c+d) & 3c+d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$3a + b = 0$ and $3c + d = 0$

$b = -3a$ & $d = -3c$, a, c $\in$ R

but $BX \neq 0$

$$\begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3(3a+c) & 3(3b+d) \\ 3a+c & 3b+d \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$3a + c \neq 0,\ 3b + d \neq 0$

7. Find the product of two matrices A and B, where $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and use it to solve the following system of linear equations $x + y + 2z = 1;\ 3x + 2y + z = 7;\ 2x + y + 3z = 2$.

दो आव्यूह A तथा B का गुणनफल ज्ञात कीजिये। जहाँ $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ तथा गुणनफल का प्रयोग कर निम्न समीकरण निकाय का हल ज्ञात कीजिये।

$x + y + 2z = 1;\ 3x + 2y + z = 7;\ 2x + y + 3z = 2$

Ans. $x = 2,\ y = 1,\ z = -1$

Sol. $$AB = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4I$$

$$B\left(\frac{1}{4}A\right) = I$$

$$B^{-1} = \frac{1}{4}A$$

Given system of equation : $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix}$

$BX = C$

$X = B^{-1}C$

$$X = \frac{1}{4}\begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

8. If $A=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$ then, find a non-zero square matrix X of order 2 such that AX = O. Is XA = O. If $A=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$, is it possible to find a square matrix X such that AX = O. Give reasons for it.

यदि $A=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$ है, तो दो कोटि का अशून्य वर्ग आव्यूह X ज्ञात कीजिये इस प्रकार की AX = O। क्या XA = O होगा। यदि $A=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$ क्या एक वर्ग आव्यूह X, जो AX = O को संतुष्ट करे, ज्ञात करना संभव है ? कारण बताइये।

Ans. $X=\begin{bmatrix}-2c & -2d\\ c & d\end{bmatrix}$, where $c, d \in R-\{0\}$, NO

Sol. Let $X=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

$AX = 0$

$$\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}\begin{bmatrix}a & b\\ c & d\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$a + 2c = 0, b + 2d = 0$

$$X=\begin{bmatrix}-2c & -2d\\ c & d\end{bmatrix}, c, d \in R-\{0\}$$

$$XA=\begin{bmatrix}-2c & -2d\\ c & d\end{bmatrix}\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}\neq 0$$

9. Determine the values of a and b for which the system $\begin{bmatrix}3 & -2 & 1\\ 5 & -8 & 9\\ 2 & 1 & a\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}b\\ 3\\ -1\end{bmatrix}$

(i) has a unique solution; (ii) has no solution and

(iii) has infinitely many solutions

तथा b का मान ज्ञात कीजिये, जिसके लिए निकाय $\begin{bmatrix}3 & -2 & 1\\ 5 & -8 & 9\\ 2 & 1 & a\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}b\\ 3\\ -1\end{bmatrix}$ के

(i) अद्वितीय हल है (ii) कोई हल नहीं है और

(iii) अनन्त हल है

Ans. (i) $a \neq -3$, $b \in R$; (ii) $a = -3$ and $b \neq 1/3$; (iii) $a = -3$, $b = 1/3$

Sol. $\begin{vmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{vmatrix} = 3(-8a-9)+2(5a-18)+5+16$

$\Rightarrow -14a - 42$

(i) For unique solution : $-14a - 42 \neq 0$

$a \neq -3, b \in R$

(ii) For no solution : $a = -3$

$$\begin{vmatrix} b & -2 & 1 \\ 3 & -8 & 9 \\ -1 & 1 & -3 \end{vmatrix} \neq 0$$

$b(24 - 9) + 2(-9 + 9) + (3 - 8) \neq 0$

$15b \neq 5$

$$b \neq \frac{1}{3}$$

(iii) for infinitely many solutions

$$a = -3,\ b = \frac{1}{3}$$

10. $\text{If} A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}; B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}; C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ then solve the following matrix equation.

यदि $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}; B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}; C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ तथा $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ है, तो निम्न आव्यूह समीकरण को हल कीजिये।

(a) $AX = B - I$ (b) $(B - I)X = IC$ (c) $CX = A$

Ans. (a) $X = \begin{bmatrix} -3 & -3 \\ \frac{5}{2} & 2 \end{bmatrix}$, (b) $X = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, (c) no solution

Sol. (a) $X = A^{-1}(B - I)$

$$X = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 6 & 6 \\ -5 & -4 \end{bmatrix}$$

(b) $X = (B - I)^{-1}C$

$$= -\frac{1}{3}\begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = -\frac{1}{3}\begin{bmatrix} -3 & -6 \\ 3 & 6 \end{bmatrix}$$

(c) $CX = A$

$X = C^{-1}A$

Inverse is not possible as $|C| = 0$

11. If A is an orthogonal matrix and B = AP where P is a non singular matrix then show that the matrix PB^{-1} is also orthogonal.

यदि A लाम्बिक आव्यूह है तथा B = AP है, जहाँ P व्युत्क्रमणीय आव्यूह हो, तो सिद्ध कीजिए कि PB^{-1} भी लाम्बिक आव्यूह है।

Ans.

Sol. $AA^T = I$, $B = AP$, $|B| \neq 0$

$\Rightarrow (PB^{-1})(PB^{-1})^T = P(AP)^{-1}(B^{-1})^T P^T$

$\Rightarrow PP^{-1}A^{-1}(P^{-1}A^{-1})^T P^T$

$\Rightarrow A^{-1}(A^{-1})^T (P^{-1})^T P^T$

$\Rightarrow (A^TA)^{-1}(P^T)^{-1} P^T$

$\Rightarrow I\, I = I$

12. $A = \begin{bmatrix} a & 1 & 0 \\ 1 & b & d \\ 1 & b & c \end{bmatrix}$, $B = \begin{bmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{bmatrix}$, $U = \begin{bmatrix} f \\ g \\ h \end{bmatrix}$, $V = \begin{bmatrix} a^2 \\ 0 \\ 0 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

If AX = U has infinitely many solution, then prove that BX = V cannot have a unique solution. If further $afd \neq 0$, then prove that BX = V has no solution.

$A=\begin{bmatrix} a & 1 & 0 \\ 1 & b & d \\ 1 & b & c \end{bmatrix}, B=\begin{bmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{bmatrix}, U=\begin{bmatrix} f \\ g \\ h \end{bmatrix}, V=\begin{bmatrix} a^2 \\ 0 \\ 0 \end{bmatrix}$ तथा X स्तम्भ सदिश है अर्थात $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ है। यदि $AX = U$ के अनन्त हल है, तो सिद्ध कीजिये कि $BX = V$ का कोई अद्वितीय हल नहीं हो सकता है। यदि $afd \neq 0$ है, तो सिद्ध कीजिये कि $BX = V$ का कोई हल नहीं है।

Ans.

Sol. $AX = U$ has infinite solutions

$\Rightarrow |A| = 0$

$$\begin{vmatrix} a & 1 & 0 \\ 1 & b & d \\ 1 & b & c \end{vmatrix} = 0 \Rightarrow ab = 1 \text{ or } c = d$$

$$\& \quad |A_1| = \begin{vmatrix} a & 1 & f \\ 1 & b & g \\ 1 & b & h \end{vmatrix} = 0 \Rightarrow -g(ab-1)+h(ab-1)=0$$

$\Rightarrow g = h$

$$|A_2| = \begin{vmatrix} a & f & 0 \\ 1 & g & d \\ 1 & h & c \end{vmatrix} = 0 \Rightarrow g = h$$

$$|A_3| = \begin{vmatrix} f & 1 & 0 \\ g & b & d \\ h & b & c \end{vmatrix} = 0 \Rightarrow -d(bf-h)+c(bf-g)=0$$

$\Rightarrow g = h, c = d$

Now, $BX = V$ has no unique solution.

$$|B| = \begin{vmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{vmatrix} = 0$$, C_2 & C_3 are identical.

$\Rightarrow BX = V$ has no unique solution.

$$|B_1| = \begin{vmatrix} a^2 & 1 & 1 \\ 0 & d & c \\ 0 & g & h \end{vmatrix} = 0$$

$$|B_2| = \begin{vmatrix} a & a^2 & 1 \\ 0 & 0 & c \\ f & 0 & h \end{vmatrix} = a^2cf = a^2df$$

$$|B_3| = \begin{vmatrix} a & 1 & a^2 \\ 0 & d & 0 \\ f & g & 0 \end{vmatrix} = a^2df$$

If $adf \neq 0$, then $|B_2| = |B_3| \neq 0$.

Hence, no solution exists.

13. Find the number of 2×2 matrix satisfying :

2×2 कोटि की मेट्रिक्स की संख्या ज्ञात कीजिये जो निम्न को संतुष्ट कीजिए :

(i) a_{ij} is 1 or –1; (ii) $a_{11}^2 + a_{12}^2 = a_{21}^2 + a_{22}^2 = 2$; (iii) $a_{11}\, a_{21} + a_{12}\, a_{22} = 0$

Ans. 8

Sol. $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$a_{ij} = 1$ or -1

$a_{11}a_{21} + a_{12}a_{22} = 0$

$$\begin{array}{cccc} a_{11} & a_{21} & a_{12} & a_{22} \\ 1 & -1 & (-1) & (-1) \\ & & (1) & (1) \\ -1 & 1 & (-1) & (-1) \\ & & 1 & 1 \end{array} \Bigg\} 4 \times 2 = 8 \text{ ways}$$

14. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Such that $AB = B$ and $a + d = 5050$. Find the value of $(ad - bc)$.

माना $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ तथा $B=\begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ इस प्रकार है कि AB = B और a + d = 5050 है, तो (ad – bc) को मान ज्ञात कीजिये।

Ans. 5049

Sol. AB = B

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} p \\ q \end{bmatrix}=\begin{bmatrix} p \\ q \end{bmatrix}$$

$\Rightarrow ap + bq = p \Rightarrow bq = p(1 - a)$ _____(1)

$cp + dq = q \Rightarrow (1 - d)q = cp$ _____(2)

(1) ÷ (2)

$$\frac{b}{1-d}=\frac{1-a}{c} \Rightarrow ad - bc = 5049$$

15. If the matrices $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (a, b, c, d not all simultaneously zero) commute, find the value of $\frac{d-b}{a+c-b}$. Also show that the matrix which commutes with A is of the form $\begin{bmatrix} \alpha-\beta & 2\beta/3 \\ \beta & \alpha \end{bmatrix}$.

यदि आव्यूह $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ तथा $B=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ क्रमविनिमय है। जहाँ (a, b, c, d एक साथ शून्य नहीं है) तो $\frac{d-b}{a+c-b}$ का मान ज्ञात कीजिये। यह भी सिद्ध कीजिए कि आव्यूह जो A के साथ क्रमविनिमय है $\begin{bmatrix} \alpha-\beta & 2\beta/3 \\ \beta & \alpha \end{bmatrix}$ के रूप में है।

Ans. 1

Sol. $$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}=\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a+2c & b+2d \\ 3a+4c & 3b+4d \end{bmatrix}=\begin{bmatrix} a+3b & 2a+4b \\ c+3d & 2c+4d \end{bmatrix}$$

On equating we get

$2c = 3b$ ... (1)

$2d = 2a + 3b$ ... (2)

$c = d - a$ ... (3)

$$\frac{d-b}{a+c-b}=\frac{d-b}{d-b}=1$$

Let $d=\alpha, c=\beta$ then $A=\begin{bmatrix}\alpha-\beta & 2\beta/3\\ \beta & \alpha\end{bmatrix}$

16. Let k be a positive real number and let $A=\begin{bmatrix}2k-1 & 2\sqrt{k} & 2\sqrt{k}\\ 2\sqrt{k} & 1 & -2k\\ -2\sqrt{k} & 2k & -1\end{bmatrix}$ and $B=\begin{bmatrix}0 & 2k-1 & \sqrt{k}\\ 1-2k & 0 & 2\sqrt{k}\\ -\sqrt{k} & -2\sqrt{k} & 0\end{bmatrix}$.

If det (adj A) + det (adj B) = 10^6, then [k] is equal to [Note: adj M denotes the adjoint of a square matrix M and [k] denotes the largest integer less than or equal to k]. **[JEE-2010]**

माना k धनात्मक वास्तविक संख्या है तथा माना $A=\begin{bmatrix}2k-1 & 2\sqrt{k} & 2\sqrt{k}\\ 2\sqrt{k} & 1 & -2k\\ -2\sqrt{k} & 2k & -1\end{bmatrix}$ तथा $B=\begin{bmatrix}0 & 2k-1 & \sqrt{k}\\ 1-2k & 0 & 2\sqrt{k}\\ -\sqrt{k} & -2\sqrt{k} & 0\end{bmatrix}$ है। यदि

det (adj A) + det (adj B) = 10^6 है, तो [k] का मान होगा [नोट: adj M वर्ग आव्यूह M का सहखण्डज है तथा [k] महत्तम i wkd d ksn' kZkg$t ksk के बराबर या k से छोटा है] **[JEE-2010]**

Ans. 4

Sol. $\det(A)=\begin{vmatrix}2k-1 & 2\sqrt{k} & 2\sqrt{k}\\ 2\sqrt{k} & 1 & -2k\\ -2\sqrt{k} & 2k & -1\end{vmatrix}$ $C_2\to C_2-C_3$

$=\begin{vmatrix}2k-1 & 0 & 2\sqrt{k}\\ 2\sqrt{k} & 1+2k & -2k\\ -2\sqrt{k} & 2k+1 & -1\end{vmatrix}$ $R_2\to R_2-R_3$

$=\begin{vmatrix}2k-1 & 0 & 2\sqrt{k}\\ 4\sqrt{k} & 0 & 1-2k\\ -2\sqrt{k} & 2k+1 & -1\end{vmatrix}=(2k+1)^3$

$\because$ B is a skew-symmetric matrix of odd order therefore det(B) = 0

Now $\det(adjA)+\det(adjB)=10^6$

$\Rightarrow$ $\{(2k+1)^3\}^2+0=10^6\Rightarrow$ $2k+1=10$, as $k>0$ $\Rightarrow$ $k=4.5$ $\Rightarrow$ $[k]=4$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 17 to 19

Consider $N = \begin{vmatrix} 77 & 44 \\ 7 & 30 \end{vmatrix} = \prod_{i=1}^{4} (a_i)^{b_i}$ where b_i's $\in N$ and a_i's are prime numbers and $a_i < a_{i+1}\ \forall$ i. Let two circles be defined as

$C_1 : \begin{bmatrix} x^2 & y^2 \end{bmatrix} \begin{bmatrix} a_1 \\ b_1 + b_2 \end{bmatrix} = [8b_1]$ and

$C_2 : \begin{bmatrix} (x+y)^2 & (x-y)^2 \end{bmatrix} \begin{bmatrix} b_3 \\ b_4 \end{bmatrix} = [2b_2]$

माना $N = \begin{vmatrix} 77 & 44 \\ 7 & 30 \end{vmatrix} = \prod_{i=1}^{4} (a_i)^{b_i}$ जहाँ b_i's $\in N$ तथा a_i's अभाज्य संख्यायें है तथा $a_i < a_{i+1}\ \forall$ i. माना दो वृत्त निम्न तरीके से परिभाषित है।

$C_1 : \begin{bmatrix} x^2 & y^2 \end{bmatrix} \begin{bmatrix} a_1 \\ b_1 + b_2 \end{bmatrix} = [8b_1]$ तथा

$C_2 : \begin{bmatrix} (x+y)^2 & (x-y)^2 \end{bmatrix} \begin{bmatrix} b_3 \\ b_4 \end{bmatrix} = [2b_2]$

17. The value $\sum_{i=1}^{4} (a_i + b_i)$ is equal to :

$\sum_{i=1}^{4} (a_i + b_i)$ का मान होगा :

(A) 23 (B) 33 (C) 25 (D) 37

Ans. D

Sol. We have $N = \begin{vmatrix} 77 & 44 \\ 7 & 30 \end{vmatrix} = 2310 - 308 = 2002 = 2 \times 7 \times 11 \times 13$

So, $b_1 = b_2 = b_3 = b_4 = 1$

and $a_1 = 2, a_2 = 7, a_3 = 11, a_4 = 13$

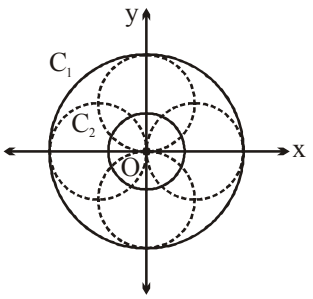
(i) Clearly, $\sum_{i=1}^{4} (a_i + b_i) = 37$

18. Radius of the circle which passes through centre of the circle C_2 and touches the circle C_1, is :

वृत्त C_2 के केन्द्र से गुजरने वाले तथा वृत्त C_1 को स्पर्श करने वाले वृत्त की त्रिज्या होगी :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{2}}$

Ans. A



Sol. C_1 : $[x^2 \quad y^2]\begin{bmatrix}2\\2\end{bmatrix} = [8] \Rightarrow C_1 : x^2 + y^2 = 4$

and C_2 : $\left[(x+y)^2 \quad (x-y)^2\right]\begin{bmatrix}1\\1\end{bmatrix} = [2]$

$\Rightarrow C_2 : (x + y)^2 + (x - y)^2 = 2 \Rightarrow C_2 : x^2 + y^2 = 1.$

Circle which passes through centre of C_2 and touches the circle C_1 has its radius equal to 1.

19. Iφ τηε λνε ξ + λy = 1 + kλ always passes through a fixed point on the circle C_1 for all values of λ then length of the tangent from the point $\left(1, \sqrt{3}k\right)$ to the circle C_2, is :

यदि रेखा $x + \lambda y = 1 + k\lambda$ सदैव वृत्त C_1 के नियत बिन्दु से गुजरती है, λ के सभी मानो के लिये तो बिन्दु $\left(1, \sqrt{3}k\right)$ से वृत्त C_2 पर स्पर्शी d hy Eckbzgkbh:

(A) k (B) k^2 (C) 2k (D) $2k^2$

Ans. B

Sol. We have $\xi + \lambda y = 1 + k\lambda$

Þ $x - 1 + \lambda (y - k) = 0 \Rightarrow L_1 + \lambda L_2 = 0$

So, $x = 1, y = k$. Hence (1, k) lies on C_1 (Given)

So, $1 + k^2 = 4 \Rightarrow k = \pm \sqrt{3}$

Hence, length of the tangent from $(1, \pm 3)$ to the circle C_2 is $= \sqrt{1+9-1} = 3 = k^2$. Ans.

Paragraph for Question nos. 20 to 22

Let A be the set of all 3 × 3 symmetric matrices all of whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0.

माना A कोटि 3 × 3 के सभी सममित आव्यूह का समूह है जिसके सभी अवयव 0 या 1 है। इनमें से पाँच अवयव 1 तथा चार अवयव 0 है।

20. The number of matrices in A is : **[JEE-2009]**

A में आव्यूहों की संख्या होगी : **[JEE-2009]**

(A) 12 (B) 6 (C) 9 (D) 3

Ans. A

Sol. Case I : All three diagonal elements are 1

No. of matrices = $^3C_1 = 3$

Case II : Two diagonal elements are zero & one element is one

No. of matrices = $^3C_1 . {}^3C_1 = 9$

Total matrices = 3 + 9 = 12

21. The number of matrices A in A for which the system of linear equations

$$A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

has a unique solution, is : **[JEE-2009]**

(A) less than 4 (B) at least 4 but less than 7

(C) at least 7 but less than 10 (D) at least 10

समूह A में आव्यूहों A की संख्या, जिनके लिए रेखिक समीकरणों के निकाय $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ के अद्वतीय हल हो, है : **[JEE-2009]**

(A) 4 से काम (B) कम से कम 4 तथा 7 से कम

(C) कम से कम 7 लेकिन 10 से कम (D) कम से कम 10

Ans. B

Sol. $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

For unique solution det (A) $\neq 0$

Case I

$$\det(A) = \begin{vmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{vmatrix} = 1 - a^2 - b^2 - c^2 + 2abc \neq 0$$

Here a, b, c is selected from 1, 0, 0. (No case is possible)

Case II

(i) $\det(A) = \begin{vmatrix} 1 & a & b \\ a & 0 & c \\ b & c & 0 \end{vmatrix} = 2abc - c^2 \neq 0$

Here a, b, c are selected from 1, 1, 0. (2 cases are possible)

(ii) $\det(A) = \begin{vmatrix} 1 & a & b \\ a & 0 & c \\ b & c & 0 \end{vmatrix} = 2abc - b^2 \neq 0$

Here a, b, c are selected from 1, 1, 0. (2 cases are possible)

(iii) $\det(A) = \begin{vmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 1 \end{vmatrix} = 2abc - a^2 \neq 0$

Here a, b, c are selected from 1, 1, 0. (2 cases are possible)

Hence there are exactly 6 matrices for unique solution . Hence option B is correct

22. The number of matrices A in A for which the system of linear equations $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is inconsistent, is :

[JEE-2009]

(A) 0 (B) more than 2 (C) 2 (D) 1

समूह A में विद्यमान आव्यूहों A की संख्या जिसके लिए रेखीय समीकरण निकाय $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ के हल असंगत हो, है :

[JEE-2009]

(A) 0 (B) 2 से अधिक (C) 2 (D) 1

Ans. B

Sol. $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Case I :

$\begin{bmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ a, b, c are selected from 1, 0, 0

$\Rightarrow$ $x + ay + bz = 1$

$ax + y + cz = 0$
$bx + cy + z = 0$

(i) If $a = 1$, $b = c = 0$
then $x + y = 1$ Inconsitent system of equation
$x + y = 0$

(ii) If $a = 0 = c$, $b = 1$
then $x + z = 1$
$y = 0$ Inconsitent system of equation
$x + z = 0$

(iii) If $c = 1$, $a = b = 0$
then $x = 1$, $z = 0$, $y = 0$

Case II :

(i) $\begin{bmatrix} 1 & a & b \\ a & 0 & c \\ b & c & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ a, b, c are selected from 1, 1, 0

$\Rightarrow$ $x + ay + bz = 1$
$ax + cz = 0$
$bx + cy = 0$ Clearly, In all three cases, solutions are possible so system is consistent.

(ii) $\begin{bmatrix} 0 & a & b \\ a & 1 & c \\ b & c & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow$ $ay + bz = 1$
$ax + y + cz = 0$
$bx + cy = 0$
Clearly , $b = 0$, $a = c = 1$ gives $y = 1$
$x + y + z = 0$ Inconsistent system $y = 0$
More than 2 matrices are possible. Hence option B is correct

Paragraph for Questions 23 to 25

Let p be an odd prime number and T_p be the following set of 2×2 matrices :

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, ..., p-1\} \right\}$$

माना कि p एक विषम अभाज्य संख्या (odd prime number) है तथा T_p , 2×2 आव्यूहों का निम्न समुच्चय है :

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, ..., p-1\} \right\}$$

23. The number of A in T_p such that A is either symmetric or skew-symmetric or both, and det (A) divisible by p is : **[JEE-2010]**

T_p में आव्यूह A की संख्या, जो इस प्रकार है कि A या तो विषम सममित या दोनों है तथा det (A), p से भाज्य है, होगी : **[JEE-2010]**

(A) $(p-1)^2$ (B) $2(p-1)$ (C) $(p-1)^2+1$ (D) $2p-1$

Ans. D

Sol. (i) If A is symmetric, $A^T = A$

$$\Rightarrow \begin{bmatrix} a & b \\ c & a \end{bmatrix} = \begin{bmatrix} a & c \\ b & a \end{bmatrix}$$

$\Rightarrow b = c$

If A is skew symmetric, $A^T = -A$

$$\Rightarrow \begin{bmatrix} a & b \\ c & a \end{bmatrix} = \begin{bmatrix} -a & -c \\ -b & -a \end{bmatrix}$$

$\Rightarrow a = 0, b + c = 0$

$\because b, c \geq 0 \Rightarrow a = 0, b = 0, c = 0$

Now, $\det(A) = a^2 - bc$

$= a^2 - b^2$ ($\because$ b = c for A being symmetric or skew symmetric or both)

$= (a-b)(a+b)$ is divisible by p.

Let $(a-b)(a+b) = \lambda p, \lambda \in I$

Range of (a + b) is 0 to 2p – 2 which includes only one multiple of p i.e. p

$\therefore a + b = p$ and $a - b \in I$

$\Rightarrow$ possible number of pairs of a and b will be p – 1.

Also, range of (a – b) is 1 – p to p – 1 which includes only one multiple of p i.e. 0

$\therefore a - b = 0$ & $a + b \in I$

$\Rightarrow$ Possible number of pairs of a & b will be p.

Hence total number of A in T_p will be

$p + p - 1 = 2p - 1$

24. The number of A in T_p such that the trace of A is not divisible by p but det (A) is divisible by p is : [Note : The trance of a matrix is the sum of its diagonal entries.] **[JEE-2010]**

T_p में आव्यूह A की संख्या जिसके लिए A का ट्रेस (अनुरेख) p से भाज्य नहीं है परन्तु det (A), p से विभाजित है, होगी। [नोटः आव्यूह d kv uqjs[kh; km l sfod .kZ d hi fo"V; ksx g\$ **[JEE-2010]**

(A) $(p-1)(p^2-p+1)$ (B) $p^3-(p-1)^2$ (C) $(p-1)^2$ (D) $(p-1)(p^2-2)$

Ans. C

Sol. $a^2 - bc \div p \quad \Rightarrow \quad$ a can be chosen in $p - 1$ ways $(a \neq 0)$

Let a be 4 & $p = 5 \quad \Rightarrow \quad$ so $a^2 = 16$ and hence bc should be chosen such that $a^2 - bc \div p$

Now b can be chosen in $p - 1$ ways and c in only 1 (one be is chosen)

Explanation : If $b = 1 \Rightarrow c = 1$; $\quad b = 2 \Rightarrow c = 3$

$b = 3 \Rightarrow c = 2$; $\quad b = 4 \Rightarrow c = 4$

Hence a can be chosen in $p - 1$ ways

and then b can be chosen in $p - 1$ ways

c can be chosen in 1 ways

so $(p - 1)^2$

25. The number of Δ in T_p such that det (A) is not divisible by p is : **[JEE-2010]**

T_p में आव्यूह A की संख्या जो इस प्रकार है कि det (A), p से विभाजित नहीं है, होगी : **[JEE-2010]**

(A) $2p^2$ (B) $p^3 - 5p$ (C) $p^3 - 3p$ (D) $p^3 - p^2$

Ans. D

Sol. Total number of A in $T_p = p^3$

when $a \neq 0$ & det(A) is divisible by p, then number of A will be $(p-1)^2$

When a = 0 & det(A) is divisible by p, then number of A will be 2p–1.

So, total number of A for which det(A) is divisible by p

$= (p - 1)^2 + 2p - 1$

$= p^2$

So number of A for which det(A) is not divisible by p

$= p^3 - p^2$

Paragraph for Question Nos. 26 to 28

Let a, b and c three real numberes satisfying

माना a, b और c तीन वास्तविक संख्याएँ संतुष्ट करती है

$$[a\ b\ c]\begin{bmatrix}1 & 9 & 7\\ 8 & 2 & 7\\ 7 & 3 & 7\end{bmatrix} = [0\ 0\ 0] \quad \text{.....(E)}$$

26. If the point P(a, b, c) with reference to (E), lies on the plane $2x + y + z = 1$, then the value of $7a + b + c$ is : **[JEE-2011]**

यदि समीकरण (E) के संदर्भ में, बिन्दु P(a, b, c) समतल $2x + y + z = 1$ पर स्थित है, तो $7a + b + c$ का मान है : **[JEE-2011]**

(A) 0 (B) 12 (C) 7 (D) 6

Ans. D

Sol. $a + 8b + 7c = 0 \quad$ ..... (i)

$9a + 2b + 3c = 0$ ..... (ii)

$a + b + c = 0$ ..... (iii)

$\Delta = \begin{vmatrix} 1 & 8 & 7 \\ 9 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = 1.(-1) - 8(6) + 7(7) = 0$

$\Lambda\varepsilon\tau\,\chi = \lambda$

$\therefore\ a + 8b = -7\lambda \quad \Rightarrow \quad a + b = -\lambda \quad \Rightarrow \quad b = \frac{-6}{7}\lambda \ \&\ a = \frac{-\lambda}{7}$

$\therefore\ (a, b, c) \equiv \left(\frac{-\lambda}{7}, \frac{-6\lambda}{7}, \lambda\right)$ where $\lambda \in R$

P(a, b, c) lies on the plane $2x + y + z = 1$

$\therefore\ \frac{-2\lambda}{7} - \frac{6\lambda}{7} + \lambda = 1 \quad \Rightarrow \frac{-\lambda}{7} = 1 \quad \Rightarrow\ \lambda = -7$

$\therefore\ 7a + b + c = 7 + 6 - 7 = 6$

27. Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with b and c satisfying(E), then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to : **[JEE-2011]**

माना कि ω समीकरण $x^3 - 1 = 0$ का हल है, जहाँ $\text{Im}(\omega) > 0$ है। यदि $a = 2$ और b एवं c समीकरण (E) को संतुष्ट करते हैं, तो $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ का मान है : **[JEE-2011]**

(A) –2 (B) 2 (C) 3 (D) –3

Ans. A

Sol. $a = 2 \quad \Rightarrow \quad \lambda = -14$

$\therefore\ b = 12 \ \&\ c = -14$

Now $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c} = \frac{3}{\omega^2} + \frac{1}{\omega^{12}} + 3.\omega^{14} = 3\omega + 1 + 3\omega^2 = 3(\omega + \omega^2) + 1 = -2$

28. Let $b = 6$, with a and c satisfying (E). If α and β are the roors of the quadratic equation $ax^2 + bx + c = 0$, then $\sum_{n=0}^{\infty}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^n$ is : **[JEE-2011]**

माना $b = 6$ और संख्याऐं a एवं c समीकरण (E) को संतुष्ट करती है। यदि α, β द्विघात समीकरण $ax^2 + bx + c = 0$ के मूल है। तो $\sum_{n=0}^{\infty}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^n$ का मान है : **[JEE-2011]**

(A) 6 (B) 7 (C) $\frac{6}{7}$ (D) ∞

Ans. B

Sol. $b = 6 \Rightarrow \lambda = -7$

$\Rightarrow a = 1$ & $c = -7$

now $ax^2 + bx + c = 0 \quad \Rightarrow \quad x^2 + 6x - 7 = 0 \Rightarrow \quad x = -7, 1$

$$\therefore \sum_{n=0}^{\infty}\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)^n = \sum_{n=0}^{\infty}\left(\frac{6}{7}\right)^n = 1 + \frac{6}{7} + \left(\frac{6}{7}\right)^2 + \ldots\ldots \infty = \frac{1}{1-\frac{6}{7}} = 7$$

MATCH THE COLUMN TYPE QUESTION(S)

29. Consider the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$ and let P be any orthogonal matrix and $Q = PAP^T$ and $R = P^TQ^KP$ also $S = PBP^T$ and $T = P^TS^KP$:

Column I	**Column II**
(A) If we vary K from 1 to n then the first row first column elements at R will form	(P) G.P. with common ratio a
(B) If we vary K from 1 to n then the 2nd row 2nd column elements at R will form	(Q) A.P. with common difference 2
(C) If we vary K from 1 to n then the first row first column elements of T will form	(R) G.P. with common ratio b
(D) If we vary K from 3 to n then the first row 2nd column elements of T will represent the sum of	(S) A.P. with common difference– 2.

आव्यूह $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ तथा $B = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$ पर विचार कीजिये तथा माना P कोई लाम्बिक आव्यूह है तथा $Q = PAP^T$ तथा $R = P^TQ^KP$ तथा $S = PBP^T$ तथा $T = P^TS^KP$ है :

Column I	**Column II**
(A) यदि हम K को 1 से n तक परिवर्तित करते है, तो आव्यूह R की पहली पंक्ति और पहला स्तम्भ के अवयव बनाते है	(P) गुणोतर श्रेणी जिसका सार्वअनुपात a है
(B) यदि हम K को 1 से n तक परिवर्तित करते है, तो आव्यूह R की दूसरी पंक्ति तथा दूसरे स्तम्भ के अवयव बनाते है	(Q) समान्तर श्रेणी जिसका सार्वअन्तर 2 है
(C) यदि हम K को 1 से n तक परिवर्तित करते है, तो आव्यूह T की प्रथम पंक्ति तथा पहले स्तम्भ के अवयव बनाते है	(R) गुणोतर श्रेणी जिसका सार्वअनुपात b है
(D) यदि हम K को 3 से n तक परिवर्तित करते है, तो आव्यूह T की प्रथम पंक्ति दूसरे स्तम्भ के अवयव किसके योगफल को निरूपित करते है	(S) समान्तर श्रेणी जिसका सार्वअनुपात– 2 है

Ans. (A) Q (B) S (C) P (D) P

Sol. $PP^T = P^TP = I$

$Q = PAP^T \Rightarrow Q^2 = (PAP^T)(PAP^T) = PA^2P^T$

$R = P^TQ^kP = P^T(PA^kP^T)P = A^k$

Similarly, $T = B^k$

$$A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 7 & -12 \\ 3 & -5 \end{bmatrix}$$

$A \to Q, B \to S$

$$B^2 = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a^2 & ab+b \\ 0 & 1 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} a^2 & ab+b \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a^3 & a^2b+ab+b \\ 0 & 1 \end{bmatrix}$$

$(C) \to P, (D) \to P$

30. Consider a square matrix A of order 2 which has its elements as 0,1,2 and 4. Let N denote the number of such matrices :

Column - A	**Column - B**
(A) Possible non-negative value of det(A) is	(P) 2
(B) Sum of values of determinants corresponding to N matrices is	(Q) 4
(C) If absolute value of (det(A)) is least, then possible value of |adj(adj(adj A))|	(R) –2
(D) If det (A) is algebraically least, then possible value of det($4A^{-1}$) is	(S) 0
	(T) 8

2 कोटि के वर्ग आव्यूह A जिसके अवयव 0,1,2 तथा 4 है तथा माना N ऐसे आव्यूहों की संख्या को निरूपित करता है जिनके सभी अवयव fHkUu&fHkUu gS

Column - A	**Column - B**
(A) कम्मज़ । द्धका सम्भवतः अऋणात्मक मान है	(P) 2
(B) छ आव्यूहों के संगत सारिणकों के मान योग है	(Q) 4
(C) यदि (det(A)) का निरपेक्ष मान न्यूनतम हो तो |adj(adj(adj A))| का सम्भवतः मान होगा	(R) –2
(D) यदि det (A) बीज गणितीज रूप से न्यूनतम है, तो det($4A^{-1}$) का सम्भवतः मान होगा	(S) 0
	(T) 8

Ans. (A) PQT (B) S (C) PR (D) R

Sol. (A) $\begin{vmatrix}2 & 4\\ 0 & 1\end{vmatrix}=2;\begin{vmatrix}4 & 2\\ 0 & 1\end{vmatrix}=4;\begin{vmatrix}4 & 1\\ 0 & 2\end{vmatrix}=8$

(B) Possible values of determinants, $\pm2, \pm4, \pm8$

sum = 0

(C) Absolute least value of $|A| = \pm2$

$|\text{adj. (adj. (A))}| = |A|^{(2-1)^2} = \pm2$

(D) $|A| = -8$

$|4A^{-1}| = 16\times\frac{1}{|A|} = -2$

31. Match the statements/Expression in Column-I with the statements/Expressions in Column-II and indicate your answer : **[JEE-2008]**

Column-I	**Column-II**
(A) The minimum value of $\frac{x^2+2x+4}{x+2}$ is $(x>0)$	(P) 0
(B) Let A and B be 3×3 matrices of real numbers, where A is symmetric, B is skew-symmetric, and $(A+B)(A-B)=(A-B)(A+B)$. If $(AB)^t=(-1)^kAB$, where $(AB)^t$ is the transpose of the matrix AB, then the possible values of k are :	(Q) 1
(C) Let $a=\log_3\log_32$. An integer k satisfying $1<2^{(-k+3^{-a})}<2$, must be less than	(R) 2
(D) If $\sin\theta=\cos\phi$, then the possible values of $\frac{1}{\pi}\left(\theta\pm\phi-\frac{\pi}{2}\right)$ are	(S) 3

स्तम्भ -I में कथनों/व्यजकों को स्तम्भ -II में कथनों/व्यजकों से मिलाईये : **[JEE-2008]**

Column-I	**Column-II**
(A) $\frac{x^2+2x+4}{x+2}$ का न्यूनतम मान है $(x>0)$	(P) 0
(B) माना A तथा B कोटि 3×3 का वास्तविक संख्याओं के आव्यूह है, जहाँ A सममित है तथा B विषम सममित है। तथा $(A+B)(A-B)=(A-B)(A+B)$ यदि $(AB)^t=(-1)^kAB$, जहाँ $(AB)^t$ आव्यूह AB का परिवर्त आव्यूह हो, तो k के संम्भव मान होगें	(Q) 1

(C) माना $a = \log_3 \log_3 2$ तथा k का पूर्णांक मान जो $1 < 2^{(-k+3^{-a})} < 2$, को संतुष्ट करता है, किससे छोटा होगा। (R) 2

(D) यदि $\sin\theta = \cos\phi$ है तो $\frac{1}{\pi}\left(\theta \pm \phi - \frac{\pi}{2}\right)$ के सम्भावित मान है (S) 3

Ans. $A \to R; B \to Q,S; C \to R,S; D \to P,R$

Sol. (A) $f'(x) = \frac{(x+2)(2x+2) - (x^2+2x+4)}{(x+2)^2} = 0$

$\Rightarrow 2x^2 + 6x + 4 - x^2 - 2x - 4 = 0$

$x^2 + 4x = 0 \Rightarrow x = 0, -4$

$f(x) = 2, -6$

(B) $A^T = A, B^T = -B$

$(A + B)(A - B) = (A - B)(A + B) \Rightarrow AB = BA$

$(AB)^T = B^T A^T \Rightarrow -(BA) = -(AB)$

$k = 1, 3$

(C) $3^{-9} = 3^{\log_3 \log_2 3} = \log_2 3$

$1 < 2^{-k} \cdot 2^{\log_2 3} < 2$

$\frac{1}{3} < 2^{-k} < \frac{2}{3}$

(D) $\cos\left(\frac{\pi}{2} - \theta\right) = \cos\phi$

$\Rightarrow \frac{\pi}{2} - \theta = 2n\pi \pm \phi$

$\theta \pm \phi = \frac{\pi}{2} + 2n\pi$

$\frac{1}{\pi}\left(\theta \pm \phi - \frac{\pi}{2}\right) = 2n$

(D) P, R